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SCHOOL OF LIFE SCIENCES**



Master project in Life Science Engineering

STUDYING MOTIFS IN CONNECTOME MODELS

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Abstract

In the present work we analyze the connectivity graph of a computational model of rat somatosensory cortex (Markram et al., 2015) looking for Robust Motifs (Curto et al., 2019) and showing that there exist a particular Robust Motif, the bisimplex. We devise an algorithm to find such motif and show that it is overrepresented in the connectivity graph. We show that bisimplices concentrate in some specific neuron morphological types. We analyze the bidirectional edge distribution inside simplices and compare it to new control models. We show that there is a concentration of bidirectional edges inside simplices and that the formation of reciprocal connections inside simplices is non-random. We show that the presence of bidirectional edges in simplices influences the network activity.

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1 Introduction

Thanks to incredible efforts, the Blue Brain Project has developed the first computational model of a rat's cortical column, based on extensive literary research and experimental data (Markram et al., 2015, Reimann et al., 2015). This is a particularly valuable model, as experimental observation of the activity or even the morphology of big ensembles of neurons is an impossible task. Data on connectivity from this model was summarized in a directed graph where nodes are neurons. In this graph, edges between pairs of nodes represent synaptic connections. The edge is directed from the presynaptic neuron to the postsynaptic one.

The connectivity graph of the computational model has been studied, revealing that there exist motifs that are overrepresented and play an important role in the network's activity. These motifs are called directed cliques (Reimann et al., 2017).

A directed clique on $n + 1$ nodes is a directed graph with edges $\{(i, j) | i, j \in (0, \dots, n) \text{ and } i < j\}$.

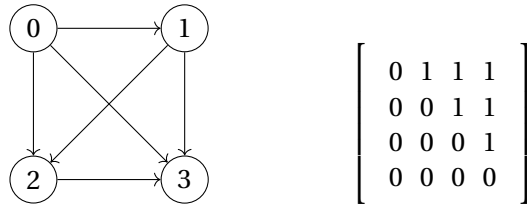


Figure 1.1 – A directed clique and the corresponding connectivity matrix.

A set of neurons represented by a directed clique is a very peculiar network. In the clique, all pairs of neurons are connected, but neurons can be ordered such if i comes before j , then there is an edge from i to j . Then, a directed clique is a small feed-forward network.

In the present work we investigate other motifs in the model. Threshold-linear network models are firing rate models that represent interactions in a network of neurons. Let $x = (x_0, \dots, x_n)$

be the vector of the firing rates of the $n + 1$ neurons in the network. A TLN follows a simple system of differential equations:

$$\frac{dx}{dt} = -x + [Wx + b]_+$$

where $[\cdot]_+$ is the positive part function, and W is the synaptic weight matrix. The weight matrix W can be encoded in a graph by thresholding the value of the synaptic weights.

A particular class of graphs, called robust motifs, was introduced in Curto et al., 2019. When these graphs represent the connectivity of a TLN model, the fixed point supports of the system do not change when perturbing the synaptic weights. In the TLN setting, robust motifs have the dynamics of feed-forward networks. Directed cliques themselves are part of the robust motif family, so it's natural to ask whether other robust motifs are relevant in the computational model.

Among these motifs, we identify two that are computationally tractable, bisimplices and extended simplices.

The **bisimplex** of dimension n is the graph G with vertex set

$$V = 0, \dots, n$$

and edge set

$$E = \{(i, j) \mid (i < j)\} \cup \{(n, n-1)\}.$$

The **extended simplex** of dimension n is the graph G with vertex set

$$V = 0, \dots, n$$

and edge set

$$E = \{(i, j) \mid (i < j) \text{ and } i, j \in [n-1]\} \cup \{(n, n-1), (n-1, n)\}$$

We develop an algorithm to find these motifs in the connectivity graph of the matrix based on the flagser algorithm (Luetgehetmann et al., 2019), and we prove that bisimplices are overrepresented in the model, while extended simplices are not, even when controlling for potential confounders. We also show that they tend to concentrate in particular morphological types.

Since reciprocal connections play a crucial role in forming the motifs we consider, we also examine whether the distribution of bidirectional edges in the model explains our findings

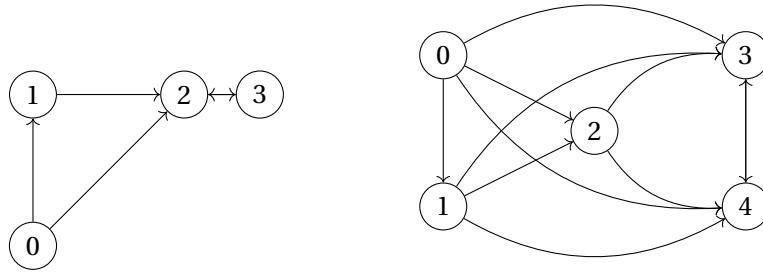


Figure 1.2 – Examples of an extended simplex on 4 nodes and a bisimplex on 5 nodes.

on bisimplices. We introduce control models that have a comparable amount of simplices of the examined graph but have random bidirectional connections. We show that bisimplices overrepresentation is due to an increase in concentration of bidirectional edges in simplices, but that there is no preferential bidirectional edge formation at the end of a simplex (which is the difference between a simplex and a bisimplex). We also show that the overall distribution of bidirectional edges in simplices differs from random controls.

We perform correlation analysis on spike trains of the computational model. We show that neurons in bidirectional edges inside simplices have a higher correlation than their directional counterpart, and that those involved in a bidirectional connection at the end of a simplex show an higher correlation than neuron pairs in other positions.

The thesis is structured in the following way.

In Chapter 2, we introduce basic definitions and notations related to directed graphs and directed simplicial complexes.

In Chapter 3, we introduce basic definitions related to threshold-linear networks, robust motifs and the results obtained in Curto et al., 2019. We define the two motifs we selected for study and the algorithm employed.

Chapter 4 shows that bisimplices are overrepresented in the computational model and that they concentrate in specific morphological type.

In Chapter 5, we study the overall distribution of bidirectional edges in the model. We introduce two new control models. We show that bidirectional edges tend to concentrate in simplices and that they have a non-random distribution inside them.

In chapter 6, we analyze spike train activity and show that bidirectional edges at the end of simplices have increased correlation.

2 Basic definitions and notation

This chapter introduces some basic notations and definition on directed graphs, simplices and simplicial complexes which will be used in the remainder of the work.

2.1 Directed graphs

A directed graph G is a pair (V, E) , where V is the set of vertices of the graph and the set $E \subseteq V \times V$ is the set of edges of the graph. Directed graphs are commonly drawn using dots to represent vertices and arrows to represent edges. Arrows start at the first element of the edge and point to the second element. Alternatively, graphs can also be represented by their connectivity matrix. The connectivity matrix $A(G)$ of the graph is defined as:

$$A_{ij} = \begin{cases} 1 & \text{if } (i, j) \in E \\ 0 & \text{otherwise} \end{cases}$$

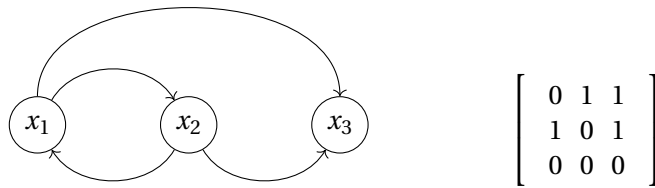


Figure 2.1 – A graph on $V = \{x_1, x_2, x_3\}$ and the corresponding connectivity matrix.

The **in-degree** of a node is the number of its incoming connections. The **out-degree** of a node is the number of its outgoing connections. The **bi-degree** of a node is the number of reciprocal

connections it has.

The **directionality** of a graph is the quantity:

$$\sum_{v \in V} (\text{in-degree}(v) - \text{out-degree}(v))^2 \quad (2.1)$$

In the present work, we will restrict ourselves to directed graphs with no self loops, i.e. edges of the type (v, v) for some $v \in V$. Such graphs are generally referred to as simple directed graphs.

A **path** is a collection of vertices $(v_0, \dots, v_n)$ such that $(v_i, v_{i+1}) \in E \forall i \in 0, \dots, n$. A **cycle** is a path $(v_0, \dots, v_n = v_0)$. A Directed Acyclic Graph (in short, DAG) is a graph with no cycle.

2.2 Directed cliques

A directed clique $\Delta[n]$ on $n + 1$ vertices is the graph $G = (V, E)$ with $V = \{v_0, v_1, \dots, v_n\}$ and $E = \{(v_i, v_j) \in V \times V \mid i < j\}$. The ordered set $(v_0, v_1, \dots, v_n)$ is called the **spine** of the clique. A clique is not entirely determined by its vertex set, while it's entirely determined by its spine. For example, the cliques with spines $(1, 2, 3)$ and $(1, 3, 2)$ share the same vertex set.

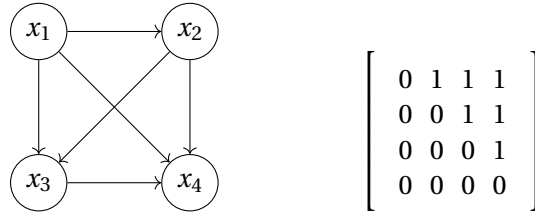


Figure 2.2 – A directed clique with spine on (x_1, x_2, x_3, x_4) and the corresponding connectivity matrix.

Directed cliques on n nodes have the highest directionality among graphs with the same number of nodes (Reimann et al., 2017).

2.3 Directed simplicial complex

The set of natural numbers lower or equal than n will be denoted by $[n] = \{0, \dots, n\}$. A directed simplicial complex $\mathbf{K}$ on a finite set of vertices V is a collection of ordered subsets of V such that, if S is in $\mathbf{K}$, then all its subsets, with the order inherited from S , are in $\mathbf{K}$. For example,

$$\{(0, 1, 2), (0, 1), (1, 2), (0, 2), (0), (1), (2)\}$$

is a simplicial complex on $[2]$, but

$$\{(0, 1, 2), (0, 1), (0, 2), (0), (1), (2)\}$$

is not, because $(1, 2)$, which is a subset of $(0, 1, 2)$ with its inherited order, is missing. The elements of the directed simplicial complex are called simplices.

The *dimension* of a simplex S is $|S| - 1$. A simplex with $n + 1$ elements is thus called an simplex of dimension n , or an n -simplex. The set of all n -simplices of $\mathbf{K}$ is denoted by $\mathbf{K}_n$.

The **faces** of an n -simplex $S = (v_0, \dots, v_n)$ are the $n + 1$ ordered sets:

$$(v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_n) \text{ with } i \in [n].$$

The faces of a n -simplex are the $n - 1$ -simplices obtained by deleting a vertex from it.

The **geometric n -simplex** $\Delta^n \subseteq R^{n+1}$ is the set

$$\Delta^n = \{(x_0, \dots, x_n) \mid x_i \geq 0 \forall i \in [n] \text{ and } \sum_{i \in [n]} x_i = 1\}.$$

See Figure 2.3 for examples of geometric simplices.

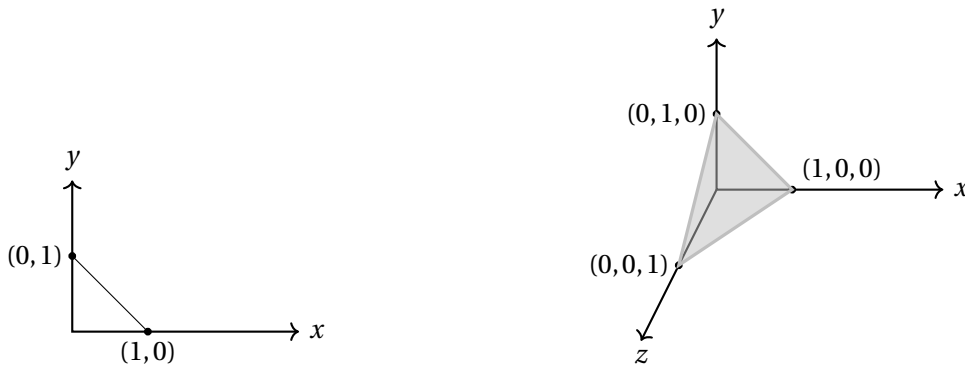


Figure 2.3 – Two geometric simplices. You can see a 1-simplex (left) and a 2-simplex (right)

The **faces** of Δ^n are the sets

$$\{(x_0, \dots, x_{j-1}, x_{j+1}, \dots, x_n) \mid x_i \geq 0 \forall i \in [n] \text{ and } \sum_{i \in [n], i \neq j} x_i = 1\}.$$

Every directed simplicial complex K has a geometric realization $|K|$. For each maximal n -

simplex in K , consider the corresponding geometric n -simplex. The geometric realization of the simplicial complex is the topological space made by 'gluing' together (via quotienting) all geometric simplices along corresponding faces. Figure 2.4 shows the geometric realization of the directed simplicial complex

$$\{(0, 1, 2), (0, 1), (1, 2), (0, 2), (1, 3), (2, 3), (3, 4), (0), (1), (2), (3), (4)\}.$$

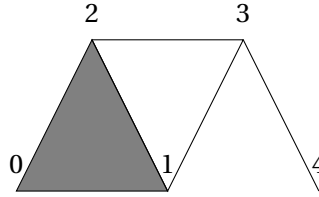


Figure 2.4 – The geometric realization of a directed simplicial complex.

2.4 Directed flag complex

Given a directed graph, it's possible to associate a directed simplicial complex to it. We say that S is a subgraph of G if the vertices and edges of S are contained in those of G , and we say that S is in G . Let V be the vertex set of G . The directed flag complex of G , denoted by $\mathcal{F}(G)$, is the simplicial complex with vertices V consisting of all the spines of directed cliques in G . In this case, $\mathcal{F}(G)_0$ is the vertex set of G , while $\mathcal{F}(G)_1 = E$. The n -simplices of the complex, $\mathcal{F}(G)_n$, are spines of length $n + 1$.

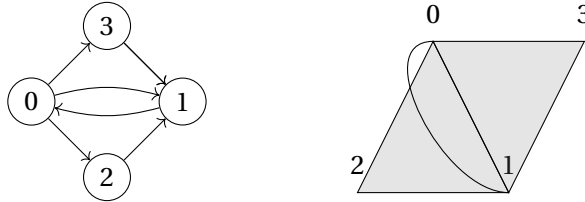


Figure 2.5 – A graph and the geometric realization of its directed flag complex.

2.5 Other simplicial complexes

While it's natural to associate a directed flag complex to a directed graph, there are alternative simplicial complexes that can be built out of it. These complexes can represent different phenomena in the graph.

The two graphs in Figure 2.6 have different flag complexes: but these flag complexes have

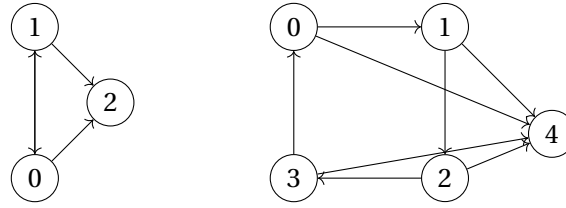


Figure 2.6 – Two graphs with isomorphic flag complex geometric realization.

homeomorphic geometric realization. However, it might be that both graphs represent connectivity in a biological network of neurons, and while in the first graph the coupled neurons tend to fire together, in the second network maybe neurons tend to fire sequentially in a repeating pattern. In this case, we would be better off treating the reciprocally connected neurons as a single neuron: no cycle would appear in the first graph, while it would still be present in the second.

Bidirectional edges in the connectivity graph have a central role in this work, as shown in Chapter 5. Being able to build graphs or simplicial complexes that expose their role in the computational model would be an interesting further advancement.

Here we propose two different models that can be studied in the future.

PROPOSAL 1

```

Start with a graph G
While there are reciprocal connections in the G
  For each pair of bidirectionally connected neurons i,j
    Generate a new graph G' where j is removed,
    where i,k is an edge if i,k or j,k is an edge in G and
    where k,i is an edge if k,i or k,j is an edge in G
    Consider G = G'
Take the directed flag complex of G

```

In this case, the simplicial complex of the graph at the left would have no cycle as desired, while the one to the right would retain it.

Better again, one could try and keep the information on multiple connections and merged vertices on weights on the graph.

PROPOSAL 2

```

Start with a graph G
Assign weight 1 to every vertex
Assign weight 1 to every edge
Consider missing edges with weight 0.

```

```

While there are reciprocal connections in the G
  For each pair of bidirectionally connected neurons i,j
    Generate a new graph G' where j is removed,
    where i,k is an edge if i,k or j,k is an edge in G
      with weight  $w(i,k) + w(j,k)$ 
    where k,i is an edge if k,i or k,j is an edge in G
      with weight  $w(k,i) + w(k,j)$ 
    assign weight  $w(i) + w(j)$  to i
  Consider  $G = G'$ 
Take a directed complex of G that takes into account edge/node weight.

```

In this case, the resulting graph will retain the information that the node is originated from a bidirectional connection. Multiple edges will be removed, and their presence will still be represented by the edge weight.

3 Robust motifs

This chapter introduces some basic notions on TLN models and their fixed points, introducing the idea of robust motifs and the results on their classification. It concludes presenting the motifs we are conducting this project on. All the material in this chapter, except for Section 3.5, is based on Curto et al., 2018, Curto et al., 2019. Our notation will differ slightly for consistency with chapter 2.

3.1 Threshold-linear networks

A threshold-linear network model (TLN for short) is a firing rate model of a network of neurons. Given a set of neurons $[n] = 0, \dots, n$, the vector $x = (x_0(t), \dots, x_n(t))$ is the vector of instantaneous firing rates of the neurons. In a TLN, firing rates obey the following differential equations:

$$\frac{dx}{dt} = -x + [Wx + b]_+ \quad (3.1)$$

where $b = (b_0, \dots, b_n)$ represents the activation threshold of neurons, W is the matrix representing synaptic weights, and $[\cdot]_+$ is the positive part function. A TLN is **competitive** when all the matrix entries are negative, $W_{ij} \leq 0$, and the elements on the diagonal are zero, $W_{ii} = 0 \forall i \in [n]$. In this case, increasing activity in a neuron inhibits the activity in all others.

3.2 Fixed points of TLNs

A point $(x_0^*, \dots, x_n^*)$ is a fixed point of the TLN if it is a fixed point of the corresponding dynamical system. The fixed point then satisfies the following equations:

$$x_i^* = \left[\sum_{j=1}^n W_{ij} x_j^* + b \right]_+ \quad \text{for } i = 0, \dots, n \quad (3.2)$$

The support of a fixed point is the set of neurons that have non-zero activity at the fixed point, that is,

$$\text{Supp}(x^*) = \{i \in V \mid x_i^* > 0\} \quad (3.3)$$

The support of the fixed point in the TLN contains all the information we need about the fixed point. In fact, each fixed point support corresponds to at most one fixed point, meaning that we can retrieve the value of all fixed points from equation 3.2 by knowing the set of supports. In the remainder of the chapter, we care about the fixed point support of a TLN:

$$FP(W, b) := \{\sigma \subset [n] \mid \sigma \text{ is a fixed point support.}\} \quad (3.4)$$

3.3 Compatible matrices

Given the nature of TLNs and the object they represent, it's tempting to try and associate a graph to them that represent the mutual influence of neurons. We will restrict ourselves to competitive TLNs.

In Curto et al., 2019, the authors associate a graph to the weight matrix in the following way: $e = (i, j)$ is an edge in G_W if $W_{ij} > -1$. Then G_W is the graph of synaptic connections such that the overall (direct and indirect) stimulation inhibits the postsynaptic neuron less than the presynaptic neuron self inhibition.

A possible scenario that competitive TLN could represent is an excitatory network regulated by an extra network of inhibitory neurons. The idea is that our model only represents a part of a population of neurons. While the network portrays the activity of population A, there is an additional inhibitory population, population B, that regulates the overall activity of A. Neuron activity in A excites population B, which has inhibitory feedback on A, so that neurons in A effectively reduce each other's activity. Every neuron in A always indirectly inhibits other neurons in A more than it inhibits itself.

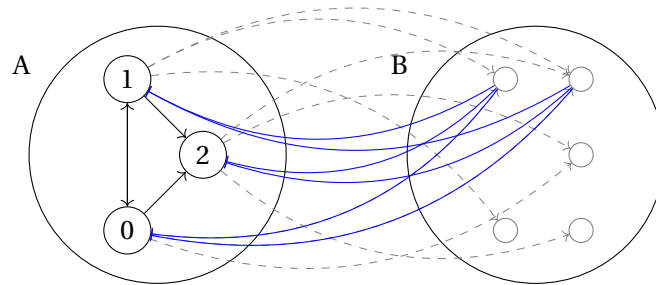


Figure 3.1 – The TLN captures the activity in population A, regulated by an inhibitory hidden population B.

Some neurons in A might also be connected by excitatory synapses. This way, the overall effect of a single neuron activity might influence differently other neurons in A . In our set of equations 3.1, we suppose the overall effect of inhibition of neuron i to itself is -1 . Neuron i inhibits other neurons j a bit more, $W_{ij} = -1 - N_{ij}$. However, it also excites those neurons k it's directly connected to, with weight $-1 - N_{ik} + E_{ik}$. Supposing $E_{ij} > N_{ij}$, one could distinguish excitatory connections between neurons from the weight of the connection matrix:

$$i \longrightarrow j \quad \begin{cases} \text{is present} & \text{if } W_{ij} > -1 \\ \text{is not present} & \text{if } W_{ij} \leq -1 \end{cases}$$

For example, suppose we have the following network of excitatory connections:

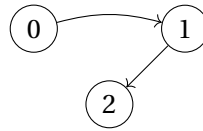


Figure 3.2 – A network of neurons.

In this network, there is an excitatory connection between neuron 0 and 1 and between neuron 1 and 2, but there is no excitatory connection between 0 and 2. We assign $W_{01} = S_{01} - 1$, $W_{12} = S_{12} - 1$, while $W_{02} = -N_{02} - 1$, $W_{20} = -N_{20} - 1$, $W_{10} = -N_{10} - 1$. Then the graph in figure 3.2 actually somehow captures the connectivity and activity of the network.

As anticipated at the beginning of the section, this motivates the following definition. Given a dynamics matrix W , we can associate a connectivity graph G_W as follows:

$$j \longrightarrow i \text{ in } G_W \iff W_{ji} > -1$$

A matrix W is said to be compatible with a graph G if $G_W = G$.

3.4 Robust motifs

We say that a graph is a robust motif if $FP(W)$ is the same for all compatible W . This means that for these motifs, even for great perturbations of the weights W_{ij} , their set of fixed point supports does not change. While it is not clear that robust motifs should exist, the authors of Curto et al., 2019 have been able to show that they indeed do exist and managed to break down the family of robust motifs to these three categories:

- *Small robust motifs* (Figure 3.3)

- *The DAG1 family.* This is the family of all directed graphs that can be decomposed as a DAG G_1 plus a node that is a target of G_1 with no edges from it (Figure 3.4).
- *The DAG2 family.* This is the family of all directed graphs that can be decomposed as a DAG G_1 plus a 2-cycle G_2 , such that one of the nodes of the 2-cycle is a target and there are no edges back from the 2-cycle to the graph (Figure 3.4).

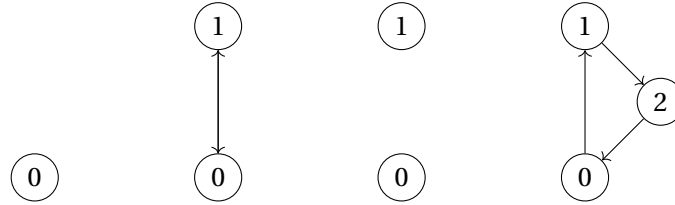


Figure 3.3 – All robust motifs not part of DAG1 or DAG2.

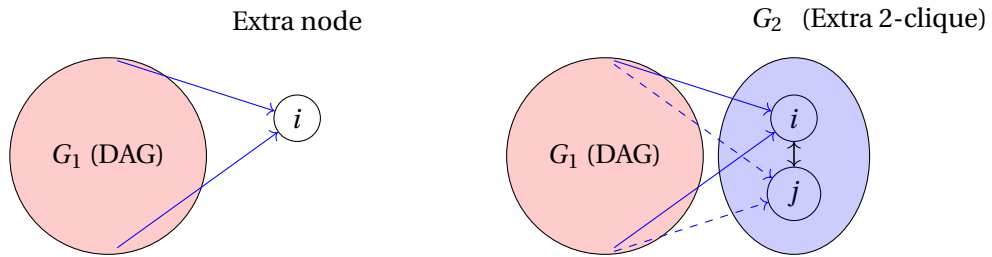


Figure 3.4 – A representation of the DAG1 (left) and DAG2 (right) families. Dashed edges are optional.

Notice that a directed clique is a robust motif since it belongs to the DAG1 family. Interestingly, robust motifs of the DAG1 and DAG2 family have either the target or the 2-clique as the only fixed point support.

Robust motifs are motifs with generally high directionality due to the presence of a target, thus further motivating the comparison with directed cliques, since these are the motif with the highest directionality on graphs with the same number of nodes.

3.5 Bisimplices and extended simplices

Among robust motifs, two classes of graphs were particularly interesting, both for their relationship with directed cliques and their computational tractability.

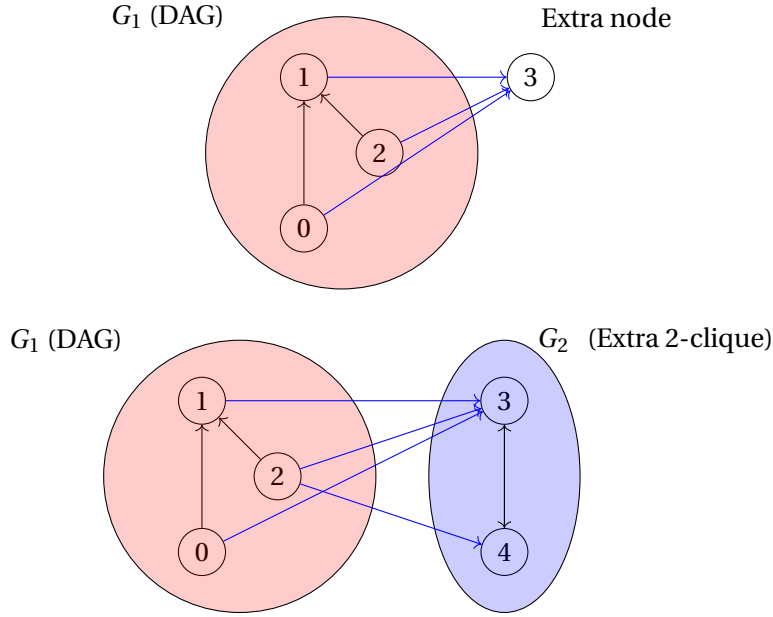


Figure 3.5 – Examples of robust motifs of the DAG1 (above) and DAG2 (below) families.

Definition 3.5.1 The **bisimplex** of dimension n is the graph G with:

$$V = [n]$$

$$E = \{(i, j) \mid (i < j)\} \cup \{(n, n-1)\}.$$

The bisimplex is the DAG2 robust motif such that both members of the 2-cliques are target of the directional part of the motif (G_1 as shown in figure 3.4).

A bisimplex contains two simplices of dimension n , $(0, \dots, n-1, n)$ and $(0, \dots, n, n-1)$. These two simplices are completely identical, except for the position of the extra node and the sink. These two nodes are completely interchangeable in the motif. This is the reason why we decided to name the motif bisimplex.

Definition 3.5.2 The **extended simplex** of dimension n is the graph G with:

$$V = [n]$$

$$E = \{(i, j) \mid (i < j) \text{ and } i, j \in [n-1]\} \cup \{(n, n-1), (n-1, n)\}$$

An extended simplex of dimension n is the DAG2 motif on $n+1$ nodes such that one of the members of the 2-clique doesn't have connections to G_1 . Bisimplices contain two extended

simplices of dimension n each.

We call the target of the DAG part of the motif (G_1 as in Figure 3.4) the **sink** of the motif. The other member of the 2-clique is the **extra node**.

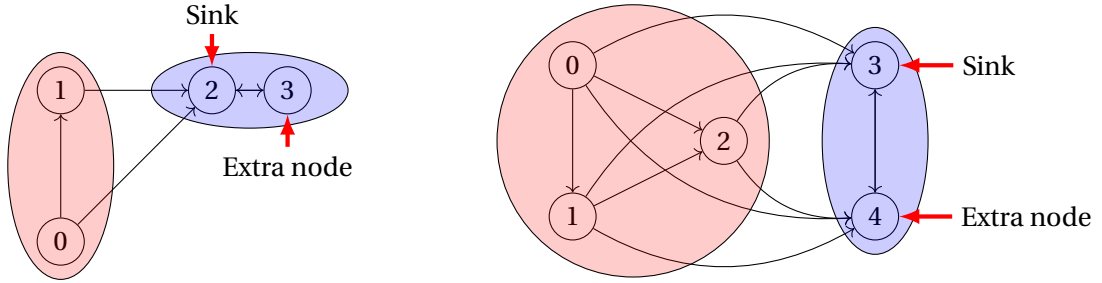


Figure 3.6 – Examples of an extended simplex on 4 nodes and a bisimplex on 5 nodes. G_1 in red, G_2 in blue.

3.5.1 Finding bisimplices and extended simplices in graphs

The ease of computability of these motifs comes from the fact that there are very efficient tools to obtain full lists of simplices (Reimann et al., 2017, Luetgehetmann et al., 2019). Starting from the full list of simplices of the graph, one can obtain the full list of the bisimplices or extended simplices using the simple following algorithms.

```
Finding extended simplices in a network from its simplex list
Input:
  L: dimension n simplex array, an M x (n+1) matrix containing
    the list of simplices and their spines
  B: connectivity matrix containing bidirectional edges only
  max_dimension: maximum simplex dimension of the graph
Output:
  extra_nodes_pointers: positions of extra_nodes indicating where
    the Extended Simplices that contain a simplex start.
  extra_nodes: extra nodes that make a simplex an Extended simplex.
for dimension in 1,...,max_dimension:
  extra_node_counter = 0
  for simplex in L:
    simplex_sink = simplex[-1]
    for nonzero_element in B[simplex_sink]:
      if nonzero_element not in simplex:
        add nonzero_element to extra_nodes
        add 1 to extra_node_counter
    end
    add extra_node_counter to extra_nodes_pointers
  end
end
```

```

Finding bisimplices in a network from its simplex list
Input:
  L: dimension n simplex array, an M x (n+1) matrix containing
    the list of simplices and their spines
  A: connectivity matrix
  B: connectivity matrix containing bidirectional edge only
  max_dimension: maximum simplex dimension of the graph
Output:
  extra_nodes_pointers: positions of extra_nodes indicating where
    the Extended Simplices that contain a simplex start.
  extra_nodes: extra nodes that make a simplex an Extended simplex.
for dimension in 1,...,max_dimension:
  extra_node_counter = 0
  for simplex in L:
    simplex_sink = simplex[-1]
    for nonzero_element in B[simplex_sink]:
      if nonzero_element not in simplex:
        bs = 1
        for element in simplex:
          if A[nonzero_element, element] is 0:
            bs = 0
        if bs == 1:
          add nonzero_element to extra_nodes
          add 1 to extra_node_counter
        end
      add extra_node_counter to extra_nodes_pointers
    end
  end
end

```

For bisimplices only, the following algorithm is more efficient, and has the advantage that it doesn't repeat bisimplices, which instead happens in the above algorithm:

```

Finding bisimplices in a network from its simplex list
Input:
  L: dimension n simplex array, an M x (n+1) matrix containing the
    list of simplices and their spines
  A: connectivity matrix
  B: connectivity matrix containing bidirectional edge only
  max_dimension: maximum simplex dimension of the graph
Output:
  is_bisimplex: a boolean array of dimension M containing whether
    the corresponding n-dimensional simplex is also a bisimplex.
for dimension in 1,...,max_dimension:
  extra_node_counter = 0
  for i in 1, ..., len(L):
    if A[L[i][-2], L[i][-1]] == 1:
      is_bisimplex[i] = 1
    else:
      is_bisimplex[i] = 0
  end
end

```

However, since the two previous algorithms were more similar in implementation, and are easily extendible to look for other classes of Robust motifs, we used them anyway.

You can find the algorithm along with the code used to analyze the results at the following GitHub repository: https://github.com/matsantoro/counting_motifs

3.5.2 Finding other robust motifs

In the next chapter, we show that bisimplices are overrepresented in a computational model of a rat somatosensory cortex, while extended simplices are underrepresented. This suggests that bisimplices are a special type of robust motif. However, it would still be possible that other robust motifs are overrepresented in the cortex, and that they share some characteristic with bisimplices. While we didn't have time to explore this possibility, we propose an algorithm that could help in this regard.

Definition 3.5.3 Let $G = (V, E)$ be a DAG2 on $n + 1$ vertices robust motif, and let G_1 be its DAG part (as shown in Figure 3.4). Let $n - 1$ be the target of the Robust Motif and n the extra node as in Figure 3.6. The signature of G is the vector $v(G) \in \mathbb{Z}_2^{n-2}$ defined as:

$$v(G)_i = \begin{cases} 1 & \text{if } (i, n) \in E \forall i \in [n-2] \\ 0 & \text{otherwise} \end{cases}$$

The following algorithm finds all DAG2 robust motifs such that G_1 is a simplex and classifies them based on their signature.

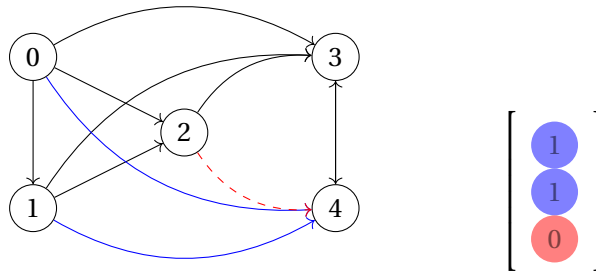


Figure 3.7 – The signature of a simplex-containing DAG2 motif.

Finding simplex-containing DAG2's in a network from its simplex list

Input:

L: dimension n simplex array, an $M \times (n+1)$ matrix containing the list of simplices and their spines


```

A: connectivity matrix
B: connectivity matrix containing bidirectional edge only
max_dimension: maximum simplex dimension of the graph
Output:
  extra_nodes_pointers: positions of extra_nodes indicating
  where the Extended Simplices that contain a simplex start.
  extra_nodes: extra nodes that make a simplex an Extended simplex.
  signatures: integer representing the robust motif type.
for dimension in 1,...,max_dimension:
  extra_node_counter = 0
  for simplex in L:
    simplex_sink = simplex[-1]
    for nonzero_element in B[simplex_sink]:
      if nonzero_element not in simplex:
        add nonzero_element to extra_nodes
        add 1 to extra_node_counter
        signature = A[simplex][extra_node]
        add signature to signatures
      end
    add extra_node_counter to extra_nodes_pointers
  end
end
end

```

In this case, signatures will contain only the maximal (in terms of inclusion) simplex-containing robust motif signature. While counting all robust motifs with a certain signature, one needs to count it from the signature of the maximal motif. For example, when counting all motifs with signature (0, 1, 1), one should include motifs with signature (1, 1, 1) in the count.

4 Bisimplices and extended simplices in the rat somatosensory cortex model

In this chapter we discuss our findings regarding overrepresentation of bisimplices in the Blue Brain Project rat somatosensory cortex model. We begin by briefly exposing the characteristics of the computational model in section 4.1, and we recap the findings of Reimann et al., 2017 which inspired the present work in section 4.2. We then present the control models we compare the computational model with in section 4.3. Finally, we expose our results on the overrepresentations of bisimplices in the model.

4.1 The somatosensory cortex model

The BBP computational model we investigate here is a full reconstruction of a rat somatosensory cortex column in its morphological and physiological characteristics, based on literature and experimental observation. The computational model is fully described in Markram et al., 2015, Reimann et al., 2015. We only recall the most prominent characteristics. Each instance of the model is $\approx 0.30\text{mm}^2$ of neuronal tissue, containing around 33000 neurons and 8 million connections. Neurons from 2-week-old Wistar rat were analyzed both in their morphology and physiology.

Randomized reconstructions of their soma and arborization were placed in a volume according to statistical parameters inferred from in-vitro measurements, like relative density of a certain m-type at a certain layer. The morphological randomizations were obtained from jittering of imaged neuronal morphologies.

The connectivity of the model was inferred using the algorithm exposed in Reimann et al., 2015. The algorithm was apposition based, as synapses between neurons were placed when their arborizations touched. Potential synapses were then pruned to preserve the distribution of multiple connections between different m-types.

The electrophysiological model of each morpho-electrical type combination was obtained by fitting ion-channel parameters of an Hodgkin-Huxley model to the different neuronal sections, optimizing for salient electrical features such as spike frequency, spike amplitude and width.

The work described in Markram et al., 2015 produced 42 different instances of a rat somatosensory cortex. Statistical measurements of 5 different rats gave rise to 7 statistical instances each. Another 7 instances were generated based on averaged measurements (we call this “average rat” instances).

We refer to these instances as the “microcircuit”.

From the microcircuit, we summarize the connectivity information in a directed graph whose vertex set is the set of all neurons and where two neurons are connected only if there is a synapse between them. The direction of the edge is given by which of the neurons is the presynaptic one.

4.2 Directed cliques in the computational model

The application of topological analysis to the above model yielded very interesting results Reimann et al., 2017.

Directed clique counts in the directed graph were compared per dimension against different graph models (Erdős–Rényi random graph, a Peters’ Rule based pruning model, and a distance-based connectivity model). It was found that in all dimensions, directed cliques were significantly more abundant than the controls.

Simplices tend to reside in layer 5 and of the model, with layer 5 having the highest simplex/neuron count.

Activity data from simulated thalamic inputs was analyzed, and it was shown that

- the correlation of the activity of a connected neuron pair increased with the number of simplices the edge belonged to.
- the correlation of the activity of a connected neuron pair increased with simplex dimension the neuron pair belonged to
- the correlation of the activity of a connected neuron pair changed with their position within the simplex. In particular, it was higher when the edge was the connection between the sink and the previous neuron in the spine.

This demonstrated that simplices not only were particularly expressed in the cortex, but that

they also played a significant role in shaping its activity.

Other interesting results regarding the homology of the microcircuit were obtained in Reimann et al., 2017, but they are outside the scope of this work.

4.3 Control models

To compare Bisimplex and Extended Simplex counts, we initially generated three control models from the microcircuit.

- *Erdős–Rényi* The first model is a simple Erdős–Rényi model with the same number of connections that the microcircuit has.
- *Adjusted ER* The second model, which we call Adjusted Erdős–Rényi, has the same number of connections and reciprocal connections that the microcircuit has. It was built in the following way:
 - Generate an Erdős–Rényi random graph with the same number of connections as the microcircuit.
 - C_m = number of bidirectional connections of the ER graph.
 - C_c = number of bidirectional connections of the microcircuit.
 - Pick $C_c - C_m$ connections at random from the ER graph and remove them.
 - Pick $C_c - C_m$ of the remaining ER graph non-reciprocal connections at random and make them bidirectional. We used this model to control for the total amount of bidirectional edges in the connectivity graph, which could increase the total count of motifs we analyzed.
- *Pathway SBM (Stochastic Block Model)* For the third model, we divided the connectivity matrix into blocks according to the presynaptic and postsynaptic morphological type, and we shuffled the connections in these blocks (without forming self loops).

The in-degree, out-degree and bi-degree distribution of these models can be seen in Figures 4.1, 4.2, 4.3. These distributions are very far from the microcircuit, and are far in standard graph metrics. More refined control models will be presented in Section 5.1. It must be noted, however, that generating control models that have similar simplex counts or even joint degree distributions is still an open problem. The controls in Reimann et al., 2017 were an Erdős–Rényi random graph, a Peters’ Rule based pruning model, and a distance-based connectivity model, which are not much more refined than the ones we present.

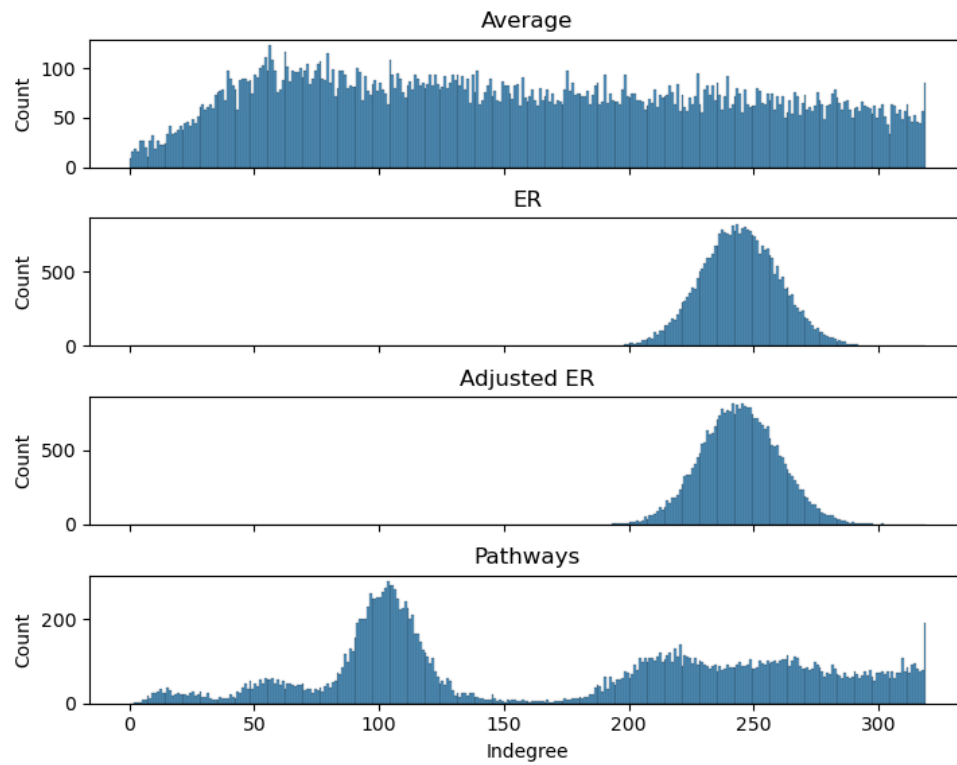


Figure 4.1 – In-degree distributions for the column and controls. From top to bottom, you can see the microcircuit, Erdős–Rényi, Adjusted ER and Pathway SBM. Models are described in Section 4.3

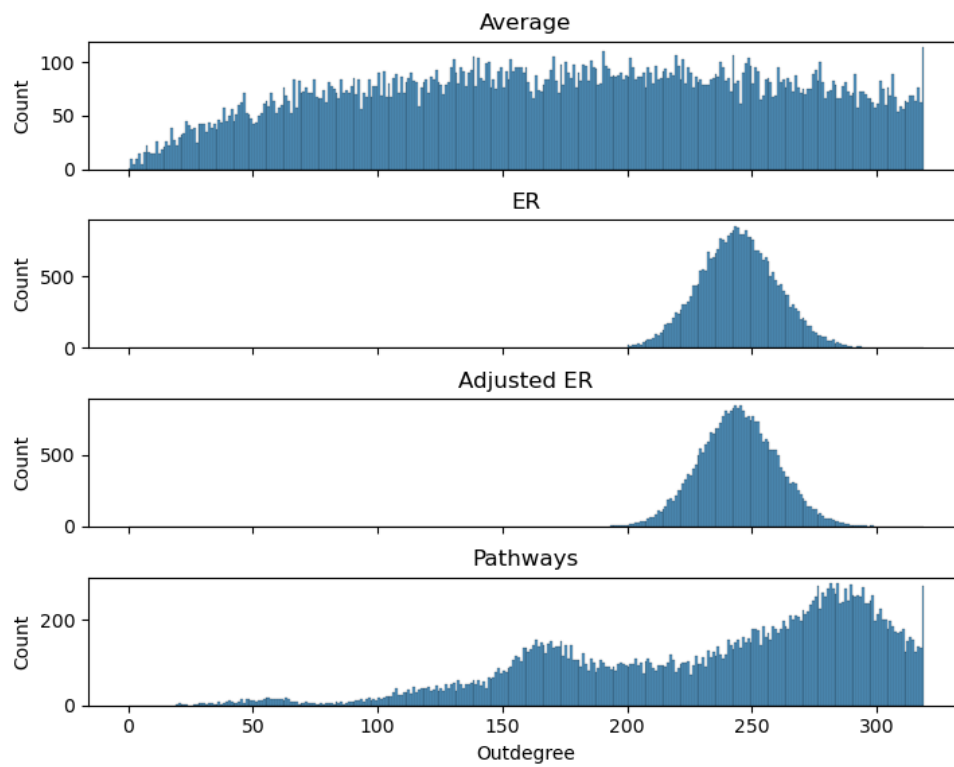


Figure 4.2 – Out-degree distributions for the column and controls. From top to bottom, you can see the microcircuit, Erdős–Rényi, Adjusted ER and Pathway SBM. Models are described in Section 4.3

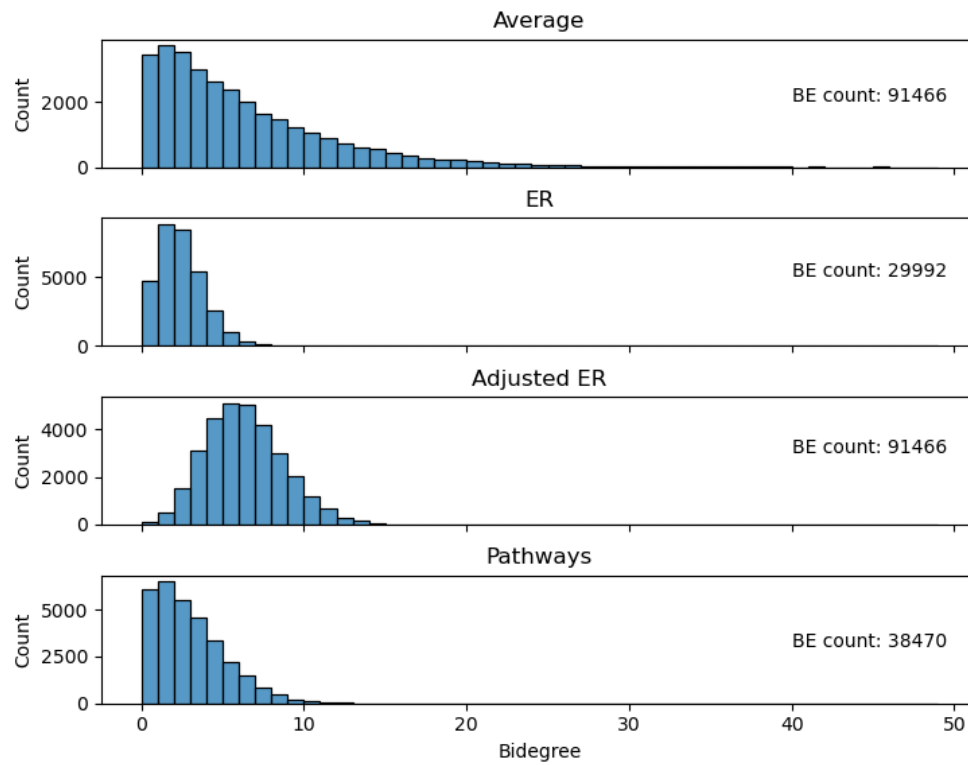


Figure 4.3 – Bi-degree distributions for the column and controls. From top to bottom, you can see the microcircuit, Erdős–Rényi, Adjusted ER and Pathway SBM. Models are described in Section 4.3

4.4 Bisimplices are overrepresented

To ascertain whether the two examined classes of robust motifs were over represented, we compared the absolute count of extended simplices and bisimplices per dimension in the average rat instances with controls generated from it.

Note that bisimplices and extended simplices in a generic random graph must be correlated with the total number of bidirectional edge present. We recall that this is the reason why we generated the Adjusted ER models.

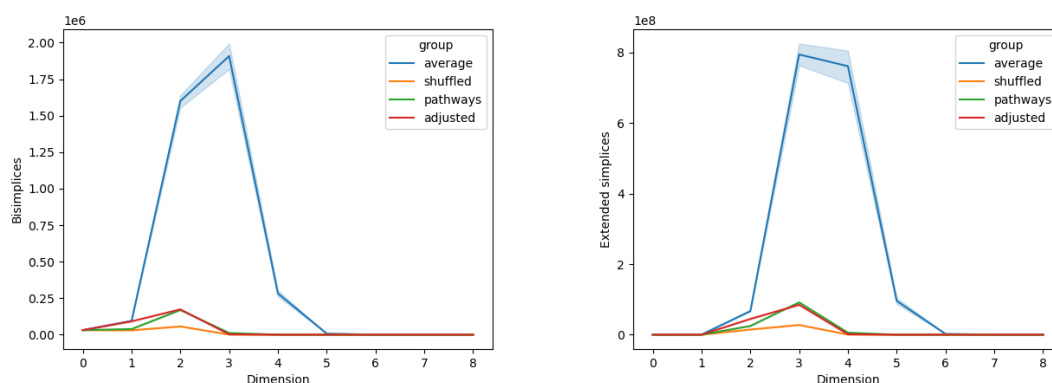


Figure 4.4 – Bisimplices and extended simplices in the model and controls described in section 4.3. Blue line is the microcircuit, other lines are ER (orange), adjusted ER (red) and Pathway SBM (green). Shaded area is 95% CI. 7 instances of the average rat microcircuit against 5 instances for each control model. Control model instances are generated from the same average rat instance.

Of course, the presence of these motifs might be due to the abundance of directed cliques in the graph. Since every bisimplex and extended simplex includes a directed clique, an increased number of directed cliques must increase the other motifs counts. We control for that by plotting the ratio of the motifs per simplex of the same dimension, which can be seen in Figure 4.5.

From figure 4.5, we can see that:

- The ratio bisimplices/simplices follows an almost-constant trend in the shuffled and adjusted controls, while it's increasing (linearly) in the microcircuit.
- The ratio extended simplices /simplices doesn't increase as fast as it does in the control models.

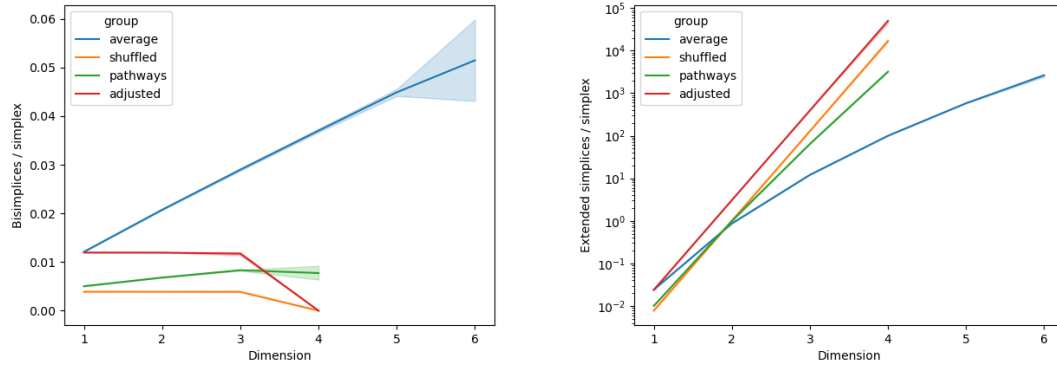


Figure 4.5 – Ratio of Bisimplices and Extended Simplices over simplices in the model and controls described in section 4.3. The lack of points of the controls is because there are no simplices in those dimensions. Blue line is the microcircuit, other lines are ER (orange), adjusted ER (red) and Pathway SBM (green). Shaded area is 95% CI. 7 instances of the average rat microcircuit against 5 instances for each control model. Control model instances are generated from the same average rat instance.

This shows that bisimplices are overrepresented in the cortical microcircuit, while extended simplices are not as represented, increasingly so in higher dimensions. This means that bisimplices do play an important role in the overall microcircuit structure.

There is an increase in variance at higher dimensions for bisimplices / extended simplices, which is due to the lower number of simplices present in that dimension.

These findings were not limited to the average rat instance. We repeated the counts for all rat instances in Figures 4.6, 4.7. For each rat, bisimplices are overrepresented, as shown in 4.7. Moreover, the total number of bisimplices per dimension changes between rat, as was found in Reimann et al., 2017 for directed cliques.

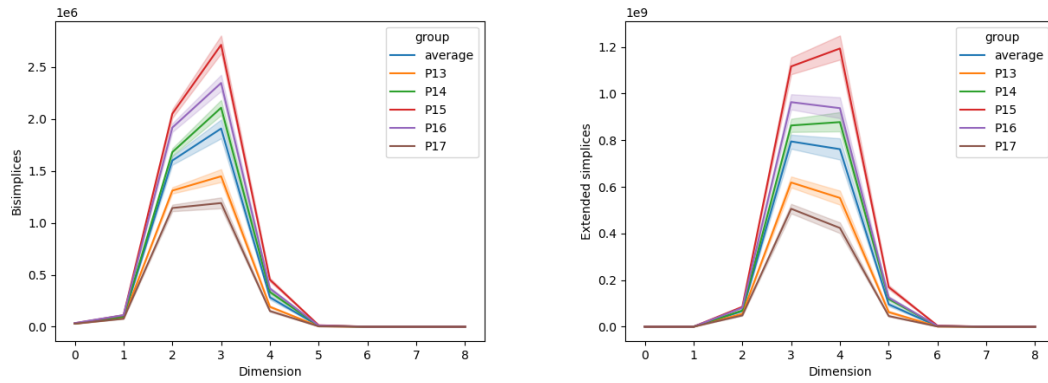


Figure 4.6 – Count of Bisimplices and Extended Simplices in the individual rat instances. Shaded area is 95% CI. 7 instances for each rat type.

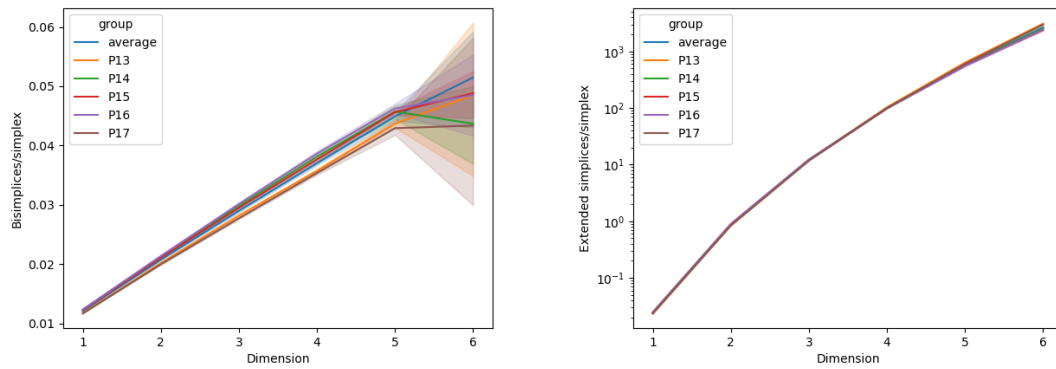


Figure 4.7 – Count of Bisimplices and Extended Simplices over simplices in the individual rat instances. Shaded area is 95% CI. 7 instances for each rat type.

4.5 M-type and layers of bisimplices

Reinforcing the idea that bisimplices are not just a random byproduct of connectivity, we tried to identify those areas where they happened the most. We checked where in the model bisimplices or extended simplices tended to have the sink and the extra edge (as shown in figure 3.5). Layer counts for dimension 3 are in figures 4.8, 4.9, while morphological type counts for the same dimension are in figures 4.10, 4.11. Full counts for all dimension in the appendix.

The count matrix for bisimplices is symmetrical because there is no difference between the sink of the simplex and the extra neuron.

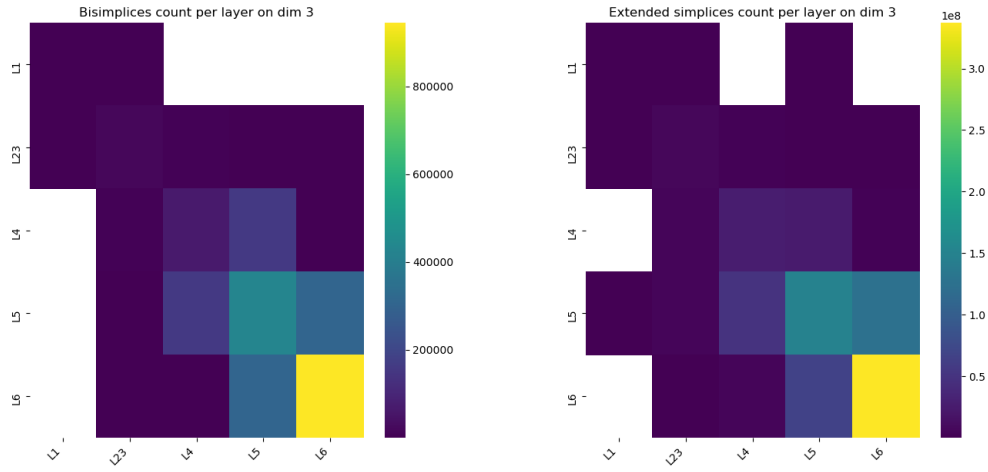


Figure 4.8 – Count of Bisimplices and Extended Simplices per sink - extra edge layer combination. Single instance.

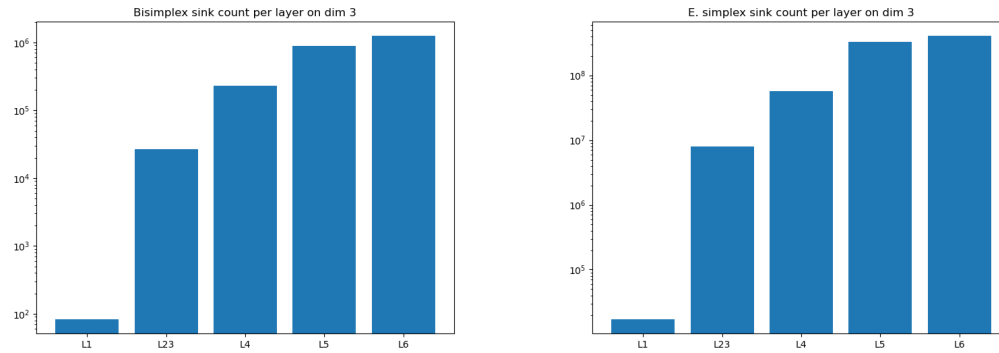


Figure 4.9 – Count of Bisimplices and Extended Simplices sink layer. Single instance.

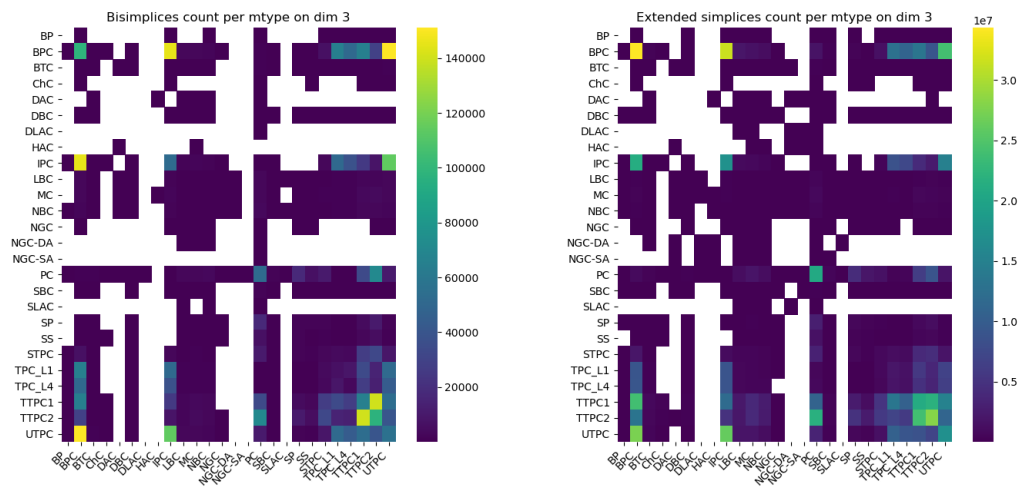


Figure 4.10 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination. Single instance.

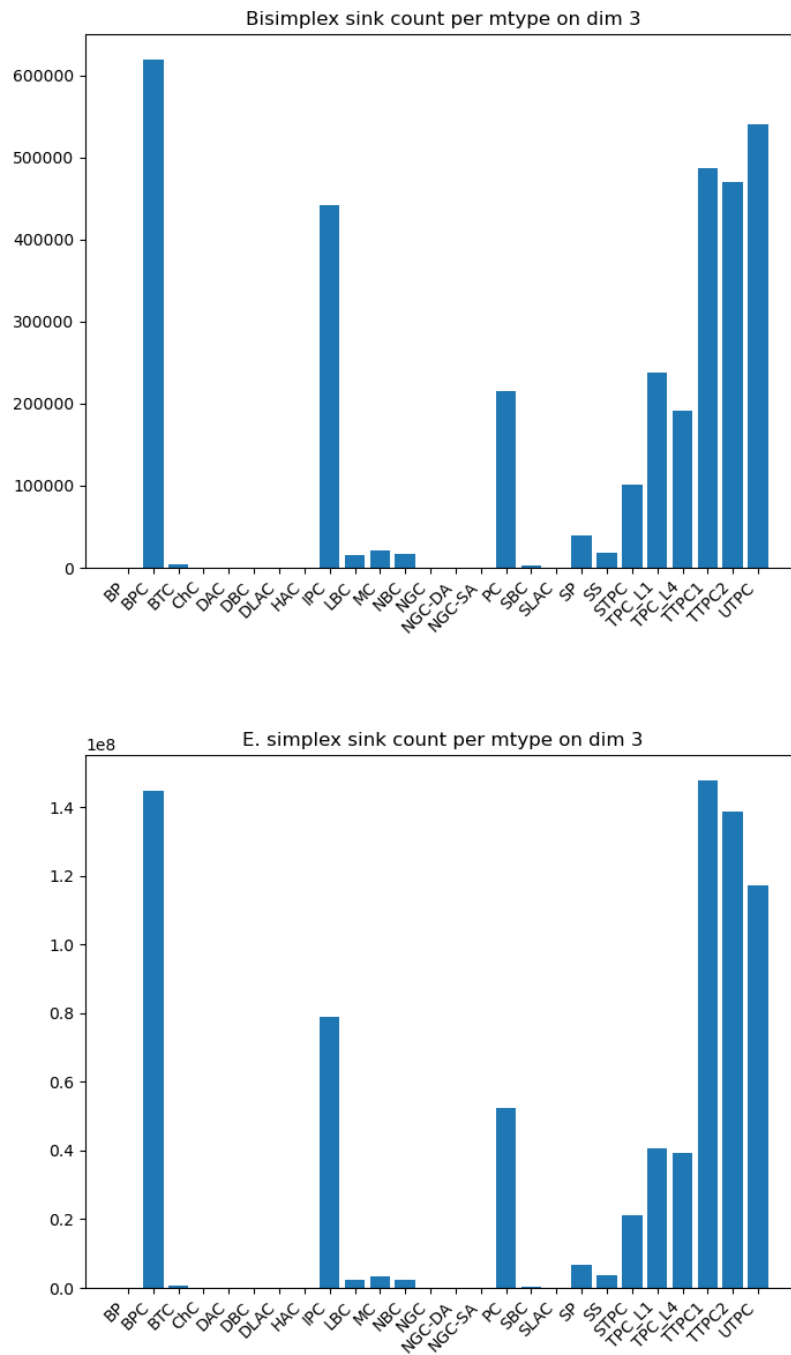


Figure 4.11 – Count of Bisimplices and Extended Simplices sink mtype. Single instance.

It is however not clear whether this concentration could be due to the presence of bidirectional edges, so we controlled by bidirectional edge presence by showing the motif/bidirectional edge ratio. In particular, motif counts in 2d histograms are compared to bidirectional edge counts between neurons with the same combination in Figures 4.12, 4.14. In 1d histograms, the motif count per sink type is compared to the number of bidirectional edges that include at least one neuron of that type in Figures 4.13, 4.15. Some areas had higher bisimplex/bidirectional edge concentration than others, supporting the idea that bisimplices are not a random byproduct of the model generation algorithm.

As shown in Figures 4.12 and 4.13, layer 5 seem to show a lot more of these motifs than the other layers.

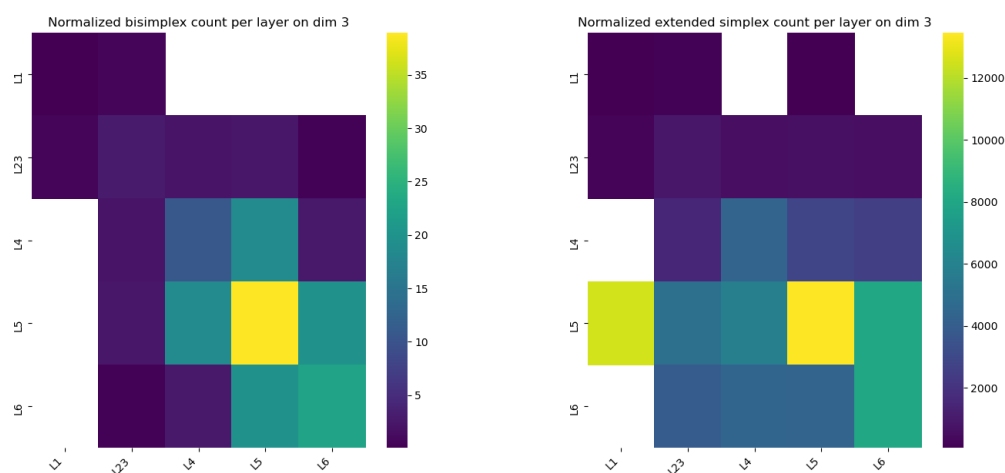


Figure 4.12 – Count of bisimplices and extended simplices per sink - extra edge layer combination over bidirectional edges on neurons with the same combination. Single instance.

Unfortunately, we lacked the time to control for the presence of simplices in a particular layer or morphology. In fact, simplices do tend to concentrate in proportion in layer 5, to which all the overrepresented morphologies belong. A motif count normalized on number of simplices with the last two terminal neurons characterization would shed light on the randomness of location of bisimplices as well. It's worth noticing, however, that the motif ratio with bidirectional edges distribution differs between extended simplices and bisimplices, so that it's likely that the effect is not simply reducible to the presence of simplices.

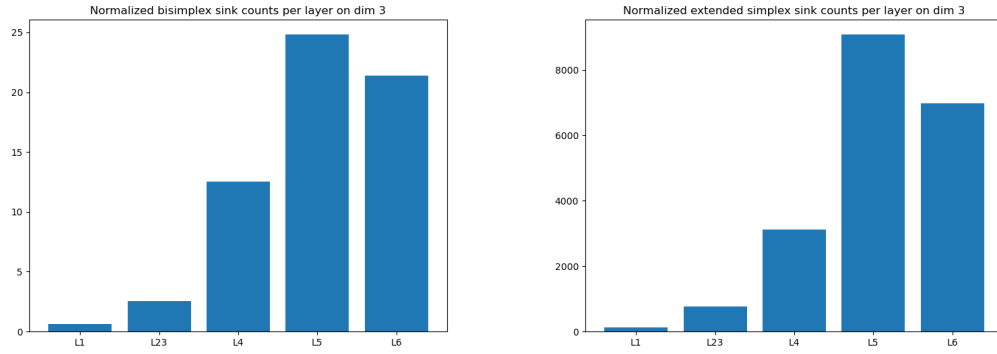


Figure 4.13 – The motif count per layer type is compared to the number of bidirectional edges that include at least one neuron of that layer. Single instance.

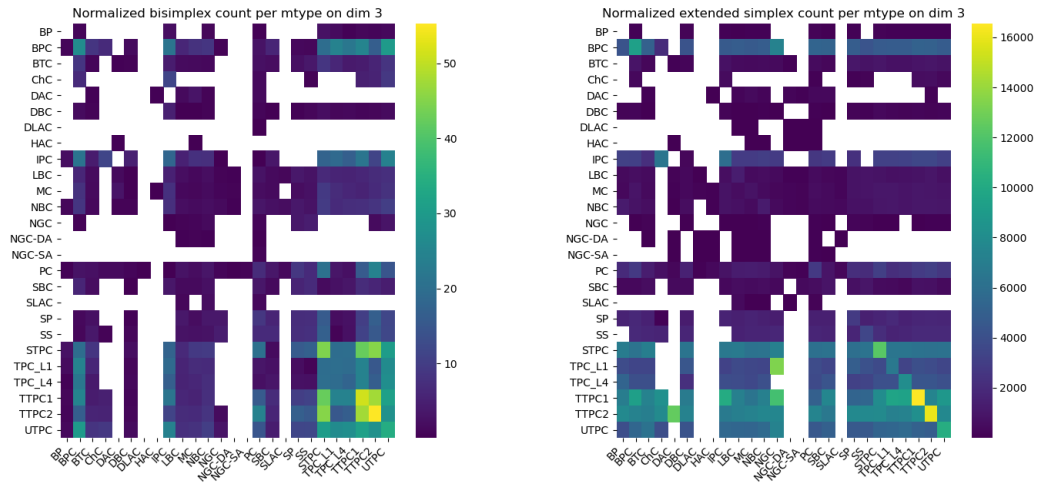


Figure 4.14 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination over the number of bidirectional edges with that combination. Single instance.

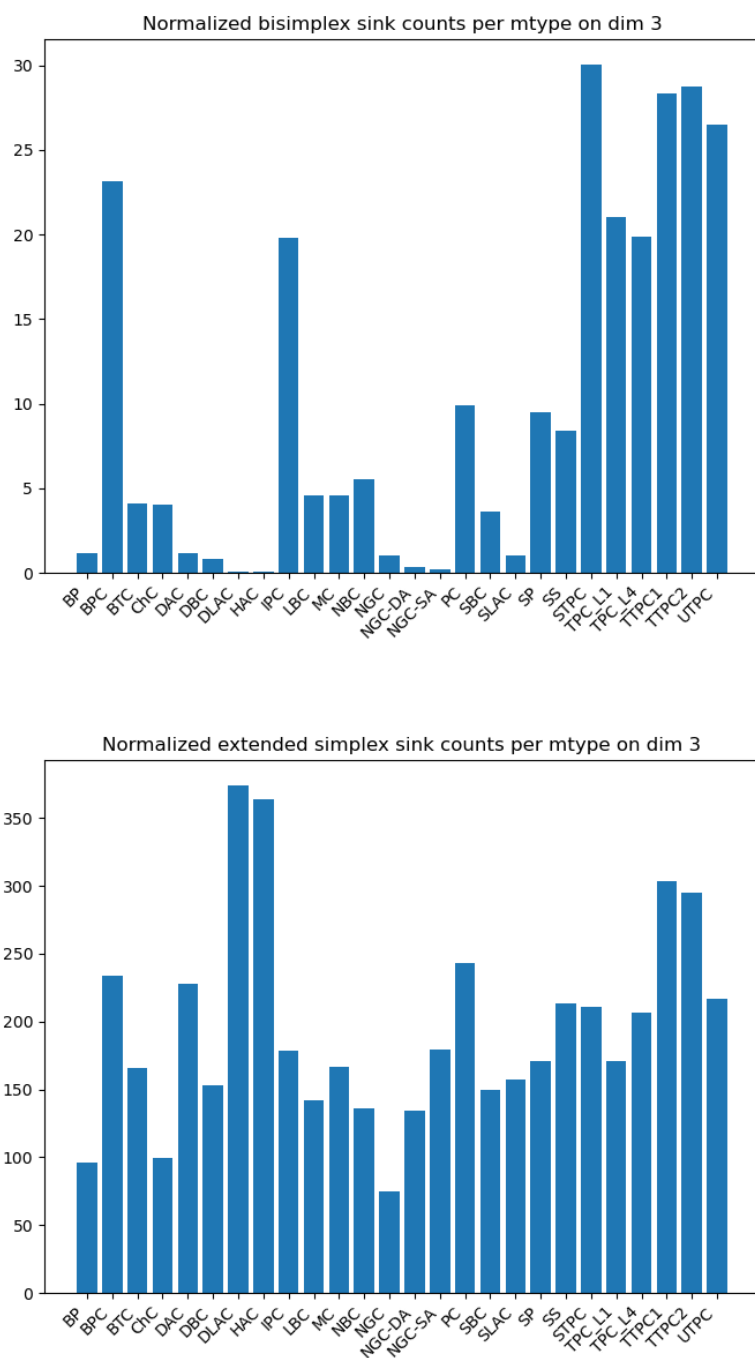


Figure 4.15 – The motif count per m-type is compared to the number of bidirectional edges that include at least one neuron of m-type. Single instance.

5 Bidirectional edges in simplices

In this chapter we will show that the preferential formation of bisimplices, that was shown in the previous chapter, is actually not due to the concentration of bidirectional edges at the end of simplices, but instead is due to the higher concentration of bidirectional edges inside simplices in general. Moreover, we show that the distribution of bidirectional edges in simplices is different from random controls.

We start by exposing new models that allow us to control for the number of simplices in the graph in Section 5.1. We show that bidirectional edges concentrate in simplices in Section 5.2. We briefly expose the problems associated with determining the preferential formation of bidirectional edges in the various positions of a directed clique in section 5.3 and introduce a new tool, the bidirectional edge matrix, that allows us to examine the position of bidirectional edges. Finally, we show that the bidirectional edge distribution inside simplices is not random.

5.1 Controlling for number of simplices

In Section 4.3, we introduced the Adjusted ER control model to exclude the confounding factor of number of bidirectional edges from the model. We have shown in Section 4.4 that bisimplices are overrepresented in the model. The natural question that arises is whether this overrepresentation of bisimplices is due to a preferential formation of bidirectional edges between nodes $n - 1$ and n of the simplex. However, none of the models introduced in 4.3 is adequate for testing, since it doesn't present simplex counts that are at least similar to the microcircuit simplex counts.

For this reason, we introduce two new control models that allow us to compare bidirectional edge formation in simplices. These models have simplex counts that are similar to the microcircuit.

- *Bishuffled* This model is generated by turning all bidirectional connections in the microcircuit into directional connections at random (each direction with 50% chance). We then pick connections in the resulting graph at random and turn them into bidirectional connections, so that the number of bidirectional connections in this control and the microcircuit is the same.
- *Underlying* This model, which we refer to as the 'underlying' model, is generated by turning the graph into a DAG by adding the transpose of the connectivity matrix to itself and then taking its upper triangular part. We then treat it as the connectivity matrix of a directed graph. This way the resulting graph is a DAG. We finally add bidirectional connections at random in the resulting graph so that the number of bidirectional connections in this control and the microcircuit is the same.

As these models are derived directly from the computational model, their degree distributions are much more similar to the microcircuit distribution as shown in Figures 5.1, 5.2, 5.3 compared to those in Section 4.3.

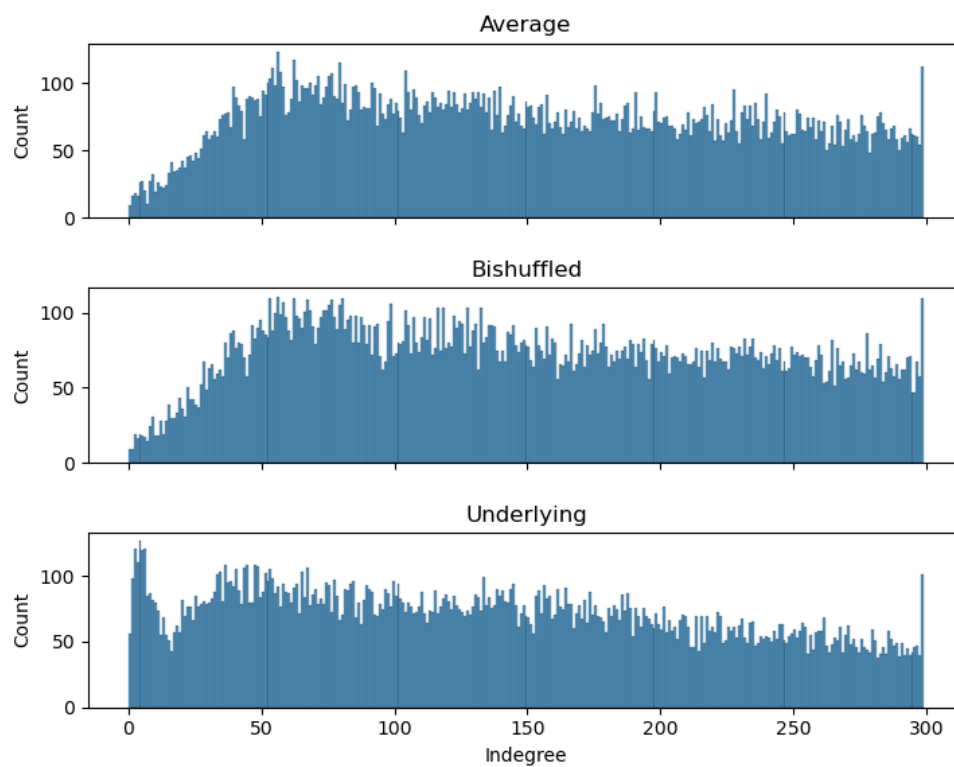


Figure 5.1 – In-degree distributions for the column and controls. From top to bottom, you can see the microcircuit, the bishuffled model, and the underlying model. Models are described in section 5.1

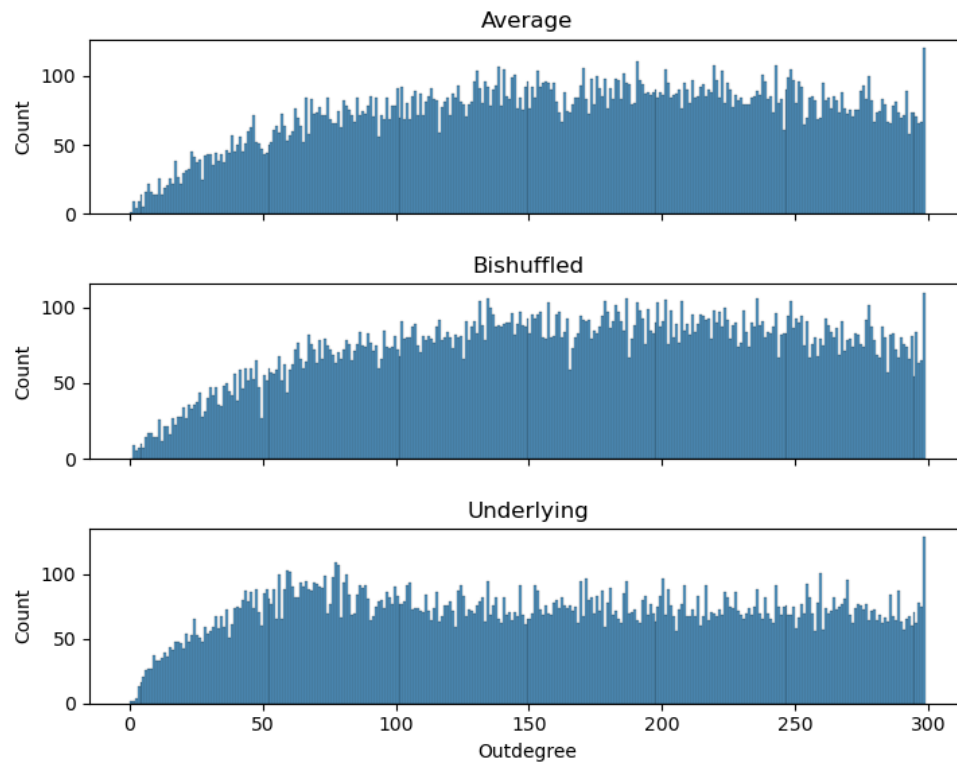


Figure 5.2 – Out-degree distributions for the column and controls. From top to bottom, you can see the column, the bishuffled model, and the underlying model. Models are described in section 5.1

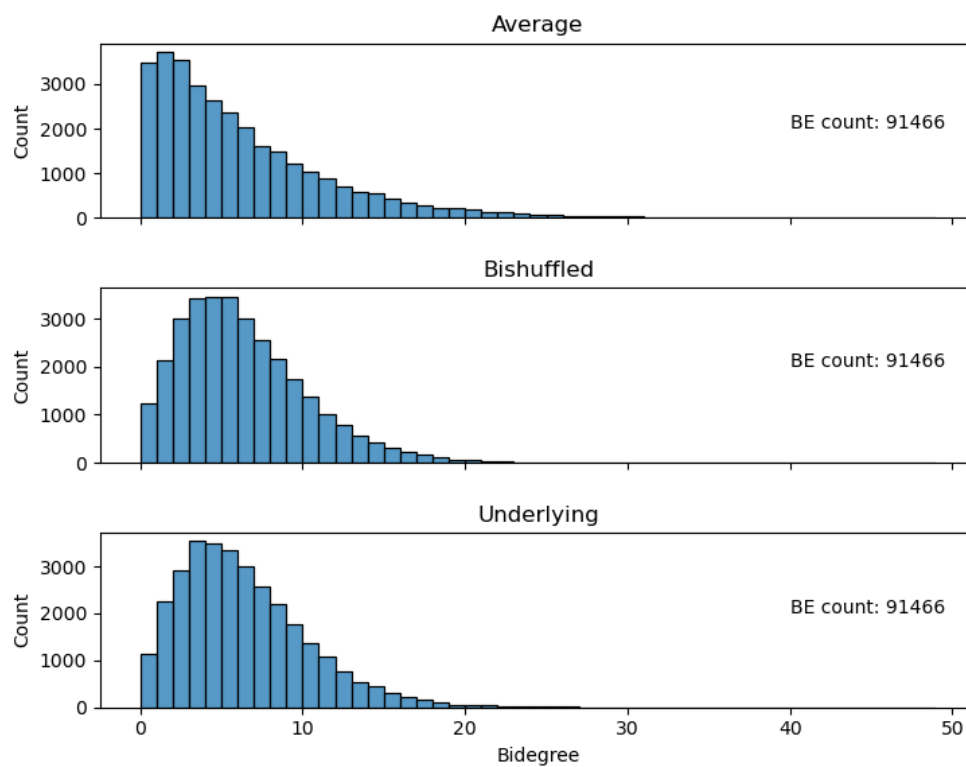


Figure 5.3 – Bi-degree distributions for the column and controls. From top to bottom, you can see the column, the bishuffled model, and the underlying model. Models are described in section 5.1

5.2 Bidirectional edges preferentially form in simplices

As mentioned, these models have a simplex counts comparable to those of the microcircuit, as shown in Figure 5.4. However, bisimplex overrepresentation is still present when comparing the column with these computational models, as shown in Figure 5.5. The overrepresentation is less prominent, but it is much more significant, as the simplex counts are comparable. In the case of the underlying model, simplex counts are even greater than in the microcircuit.

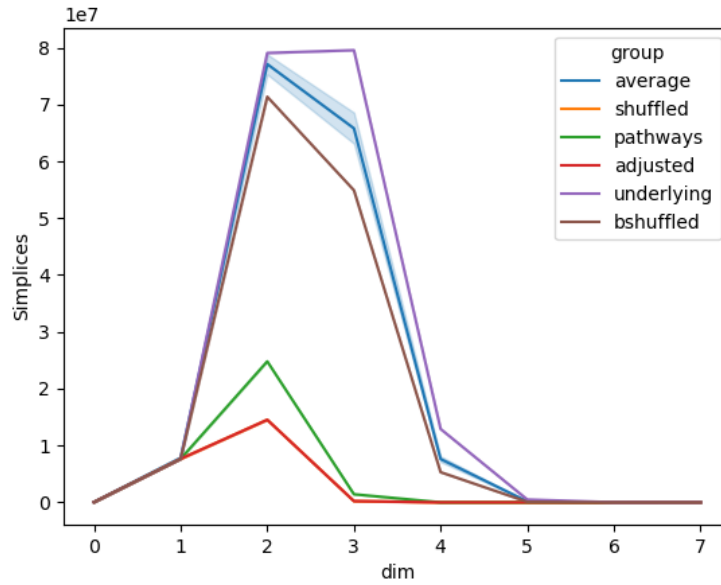


Figure 5.4 – Simplex counts for all models presented until now. The microcircuit is in blue, while other colors are the control models. ER in orange, pathway SBM in green, Adjusted ER in red, Underlying in purple and Bishuffled in brown. 7 instances of the average rat microcircuit against 5 instances for each control model. Control model instances are generated from the same average rat instance. Models are described in sections 4.3 and 5.1

Interestingly, the ratios of these two new models against number of simplices is constant along dimension, as can be seen in Figure 5.6. This is a feature we had already seen in the Adjusted ER model (4.5). The ratio of bisimplices to simplices in the microcircuit increases with dimension.

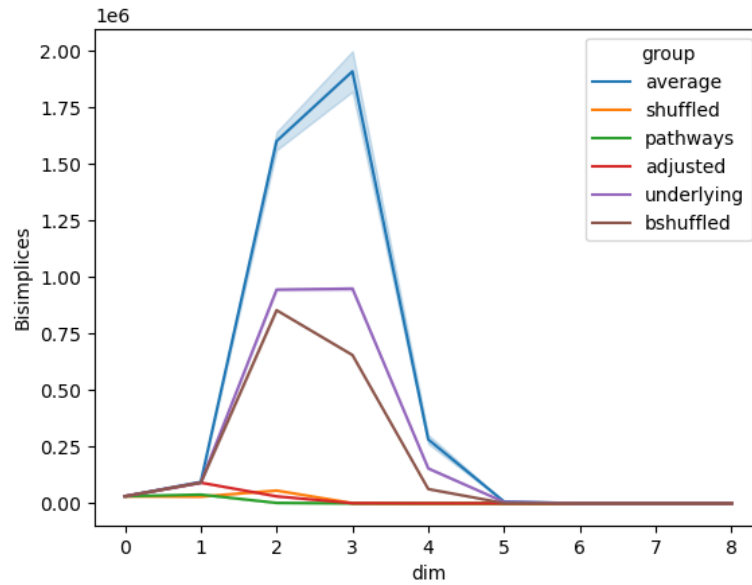


Figure 5.5 – Bisimplex counts for all models presented until now. The microcircuit is in blue, while other colors are the control models. ER in orange, pathway SBM in green, Adjusted ER in red, Underlying in purple and Bishuffled in brown. 7 instances of the average rat microcircuit against 5 instances for each control model. Control model instances are generated from the same average rat instance. Models are described in sections 4.3 and 5.1

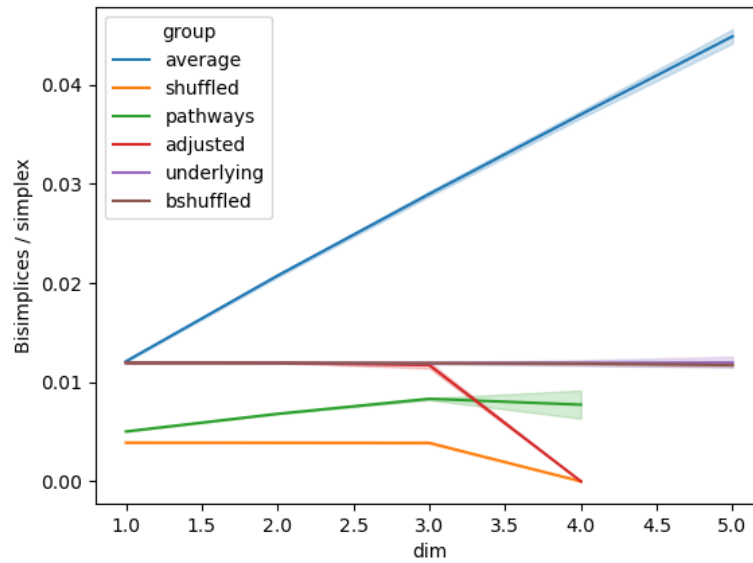


Figure 5.6 – Bisimplex counts over simplices for all models presented until now. The microcircuit is in blue, while other colors are the control models. ER in orange, pathway SBM in green, Adjusted ER in red, Underlying in purple and Bishuffled in brown. Models are described in sections 4.3 and 5.1. 7 instances of the average rat microcircuit against 5 instances for each control model. Control model instances are generated from the same average rat instance. Lines for bishuffled and underlying are overlapped.

This happens because bidirectional edges tend to concentrate in simplices. We show in Figures 5.7, 5.8 the bidirectional edge count and normalized bidirectional edge count in simplices of a given dimension for the column and various models. Normalized edge counts are obtained by dividing by the total number of simplices in a given dimension. This is done considering, for each simplex, the subgraph spanned by its vertices and counting the number of bidirectional connections in it.

In Figure 5.5 we can clearly see that the total absolute count of bidirectional edges inside simplices is much greater than the control models. Interestingly, in Figure 5.6 we can see an increasing trend on the ratio of bidirectional edges on the total number of simplices, and a definite increase with respect to the control models.

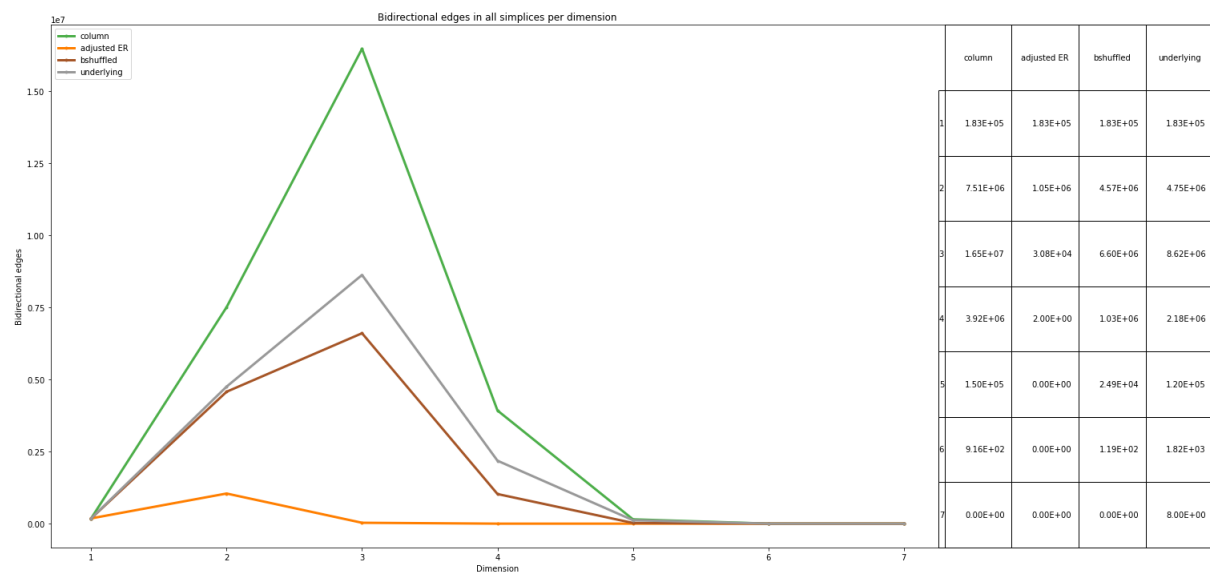


Figure 5.7 – Bidirectional edge count inside simplices for some models. The microcircuit is in green, Underlying in gray, Bishuffled in brown and Adjusted ER in orange. Models are described in sections 4.3 and 5.1. A single instance for each curve.

Similar results are obtained when maximal simplices only are considered, as shown in Figures 5.9, 5.10. This excludes influence from the overlapping of simplices across dimensions.

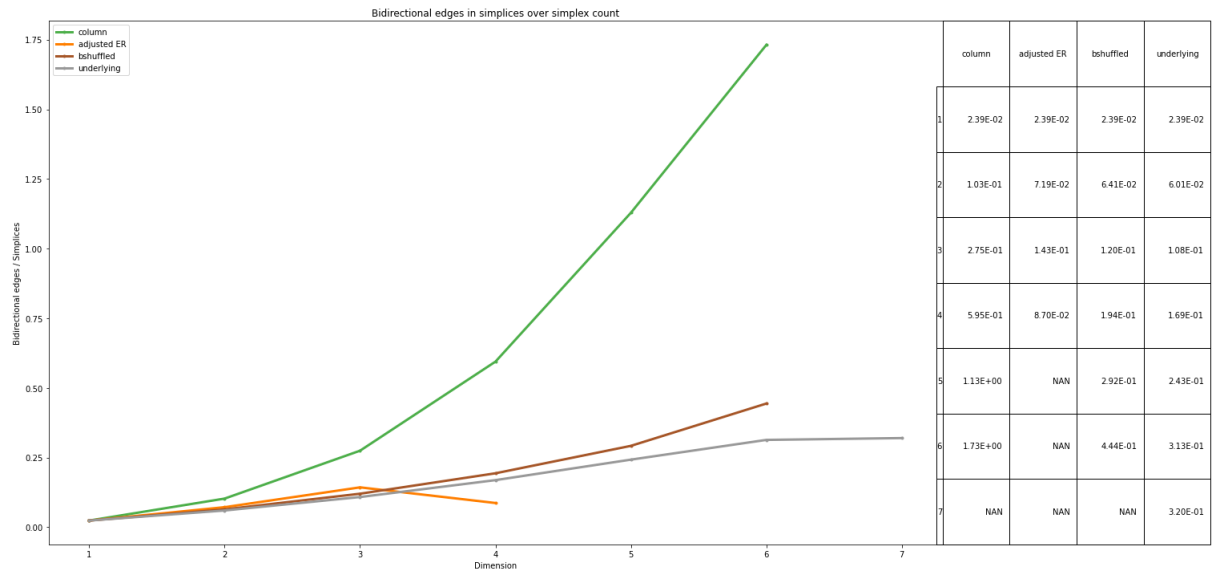


Figure 5.8 – Normalized bidirectional edge count inside simplices per dimension. Normalization is over the number of simplices in a given dimension. The microcircuit is in green, Underlying in gray, Bishuffled in brown and Adjusted ER in orange. Models are described in sections 4.3 and 5.1. A single instance for each curve.

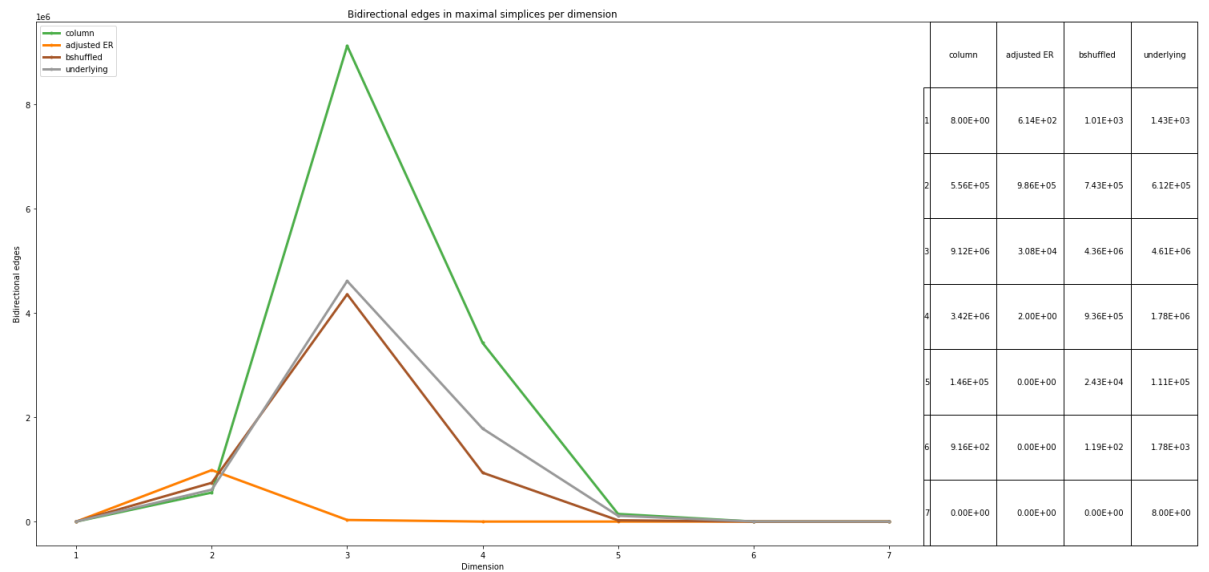


Figure 5.9 – Bidirectional edge count inside maximal simplices for some models. The microcircuit is in green, Underlying in gray, Bishuffled in brown and Adjusted ER in orange. Models are described in sections 4.3 and 5.1. A single instance for each curve.

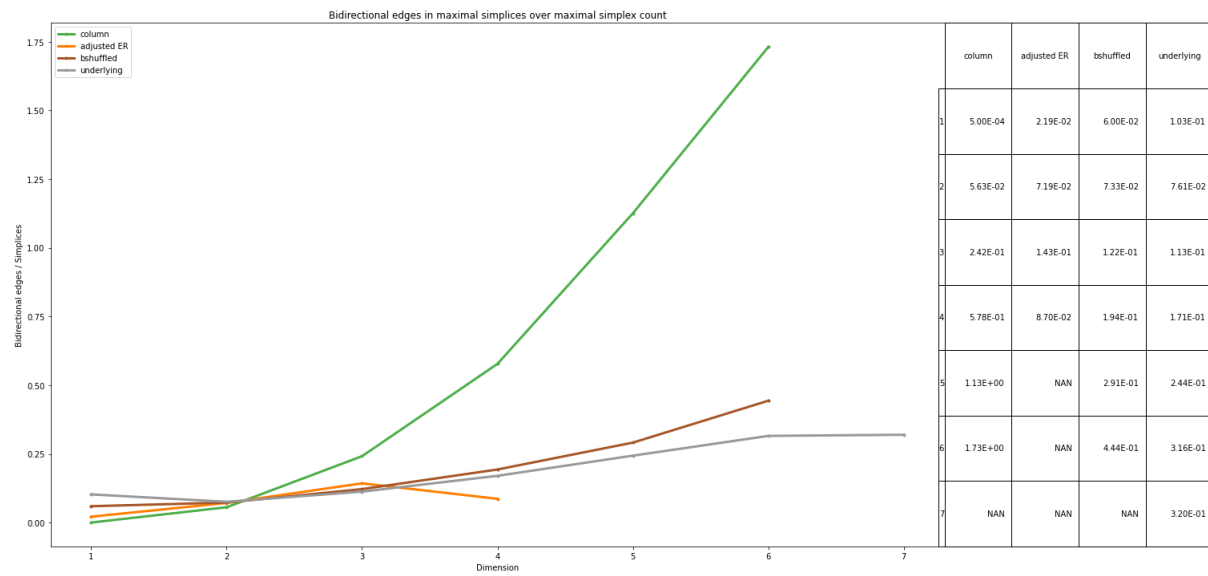


Figure 5.10 – Normalized bidirectional edge count inside maximal simplices per dimension. Normalization is over the number of maximal simplices in a given dimension The microcircuit is in green, Underlying in gray, Bishuffled in brown and Adjusted ER in orange. Models are described in sections 4.3 and 5.1. A single instance for each curve.

5.3 The position of bidirectional edges in simplices

We have shown in Section 5.2 that bidirectional edges preferentially form inside simplices. Since bisimplex count was high compared to control models, we wondered whether this was due to a preference for bidirectional edges to be found in position $n, n-1$ inside a simplex. We decided to perform the analysis for all positions inside simplices.

The main tool we use is a matrix that counts the number of bidirectional edge occurrences of all simplices in a graph, distinguishing them by position. We first define it for a single simplex, and then for the whole graph.

Definition 5.3.1 Let $S = (v_0, \dots, v_n)$ be a directed clique spine inside a graph G . Consider the subgraph G_S on $\{v_0, \dots, v_n\}$ that retains all the edges between elements in S . We call the **bidirectional edge matrix** of S , and we denote it by B_S , the matrix defined as:

$$B_{Sij} = \begin{cases} 1 & \text{if there is a bidirectional edge between } i \text{ and } j \\ 0 & \text{otherwise} \end{cases}$$

We're interested in the matrices of all counts of bidirectional edge position in simplices of a given dimension. Given a graph G , we define the **bidirectional edge matrix** of dimension n of G as:

$$B_{Gn} = \sum_{S \in \mathcal{F}(G)_n} B_S,$$

where the sum is over $\mathcal{F}(G)_n$, i.e. the directed n -cliques G . This matrix has the counts of bidirectional edge positions inside the simplices. You can see an example of it in Figure 5.11.



Figure 5.11 – A 4-bisimplex and its bidirectional edge matrix of dimension 4. Note that there is a 2 in position 4, 3 because the graph has two simplices, $(0, 1, 2, 3, 4)$ and $(0, 1, 2, 4, 3)$, and both simplices have a bidirectional edge in that position.

When comparing results with models, we will show bidirectional edge matrices of slightly different types.

- *Absolute count*

Absolute count bidirectional edge matrices are those defined in section 5.3.

- *Percentage*

Percentage bidirectional edge matrices show the relative percentage of times a bidirectional edge appears in a given position in all simplices. They are obtained from absolute matrices by dividing every element by the total number of bidirectional edges found in simplices of a given dimension.

- *Normalized*

Normalized bidirectional edge matrices show the number of bidirectional edges per simplex of the graph. They are obtained from absolute matrices by dividing every element by the total number of simplices in a given dimension.

However, we must be careful when considering these matrices to infer preferences in bidirectional edge positions. Consider the graph in Figure 5.11. This graph has exactly two simplices of dimension 4 in it: $(0, 1, 2, 3, 4)$ and $(0, 1, 2, 4, 3)$. When building B_{G_n} , the bidirectional edge in G is counted twice, once per simplex.

Remark 5.3.1 *Consider a directed clique $\Delta[n]$. Adding the edge $(i + 1, i)$, $i \in [n - 1]$ generates a new simplex $(0, \dots, i + 1, i, \dots, n)$. Adding an edge (i, j) with $j \neq i - 1$ doesn't generate any new simplex.*

As per Remark 5.3.1, the same happens whenever we add a bidirectional edge to the spine of a clique, but doesn't happen when we add it to any other position. This means that somehow positions outside the spine are underrepresented in the bidirectional edge matrix.

In fact, consider now a random graph G_R built from a directed clique on $n + 1$ nodes with exactly one bidirectional edge added to it at random, in any position. While every edge has the same probability of becoming bidirectional, the expected value of B_{G_R} is not equal in all positions, because of this “simplex duplication” problem. In fact, positions in the subdiagonal will have exactly double the expected value of the other positions.

A similar problem happens when we try to retrieve all the bidirectional edge positions in the microcircuit. Bidirectional edges in particular positions will still generate extra simplices and thus be more represented than bidirectional edges in other positions. This might give the false impression that bidirectional edges tend to concentrate along the spine of simplices, for example.

In figure 5.12 we show the average bidirectional edge percent matrix of a 5-simplex to which we add 9 bidirectional edges at random, obtained experimentally. The percentage gradient

from edge (5, 0) to the spine might suggest preferential bidirectional edge formation the closer to the spine we get, but to obtain the matrix we added connections at random.

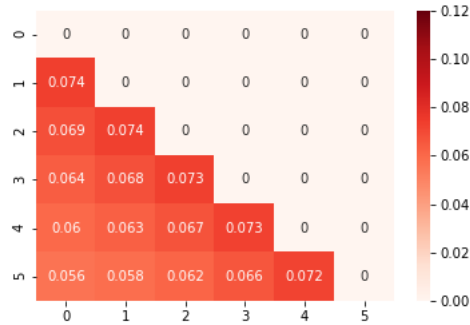


Figure 5.12 – Percent bidirectional edge count for a 5-simplex with 9 extra edges.

For this reason, comparison with control models with a similar amount of simplices is crucial.

5.4 Position of bidirectional edges in the microcircuit

The bisimplex was overrepresented in the microcircuit, but it is possible that other motifs with bidirectional edges are present as well. In fact, preferential bidirectional edge formation at the end of the simplex would mean that the bisimplex is a very particular motif that is present in the connectivity graph. However, this was not the case.

5.4.1 Bidirectional edge matrices

We plot and compare the bidirectional edge matrices, defined in section 5.2, of the Adjusted ER, Underlying, and Bishuffled models (models are defined in Sections 4.3 and 5.1). We show results for dimensions 4 and 5, and show plots for all dimensions in the appendix.

The bidirectional edges absolute count is always higher in the computational model than in control models, as shown in Figures 5.13, 5.14, and as anticipated in Section 5.2. This is not true for the highest dimensions, where the microcircuit doesn't have as much simplices as the underlying model.

The bidirectional edge concentration is even more evident in the normalized matrices of the same dimensions (figures 5.15, 5.16).

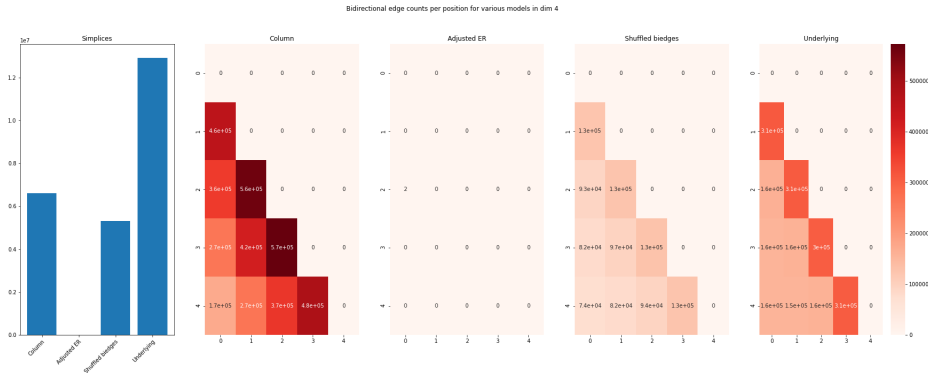


Figure 5.13 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 4. To the left, a simplex count histogram for dimension 4. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

We compared percentage matrices with the control models. We have seen that bidirectional edges do not really tend to form at the end of the simplex. In fact, the chance of this position having a bidirectional edge is even lower than in the underlying control model. There is also a lower tendency to form bidirectional edges between node 0 and node n . Figures 5.17, 5.18 compare the results for dimension 4 and 5. Figure 5.19 shows difference in percentages with

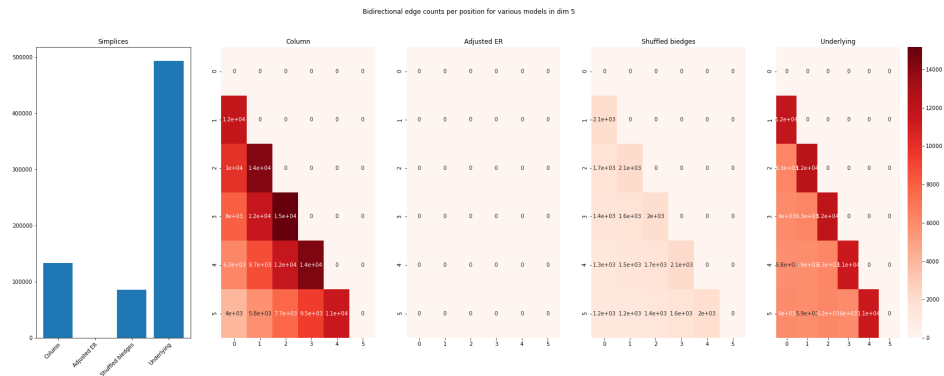


Figure 5.14 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 5. To the left, a simplex count histogram for dimension 5. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

the microcircuit.

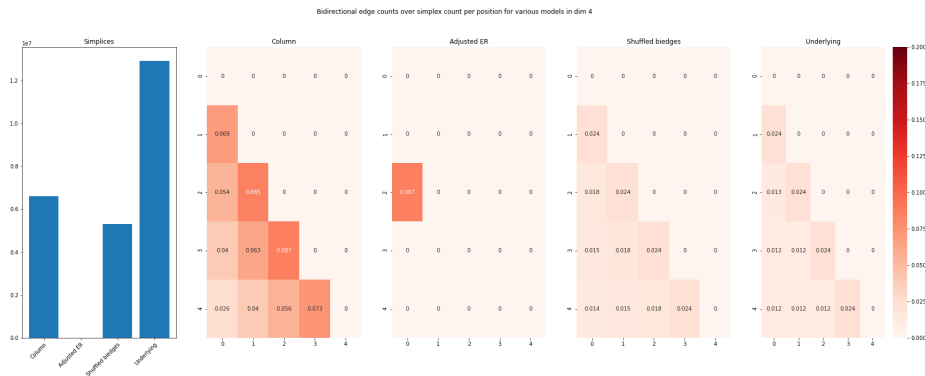


Figure 5.15 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 4. To the left, a simplex count histogram for dimension 4. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

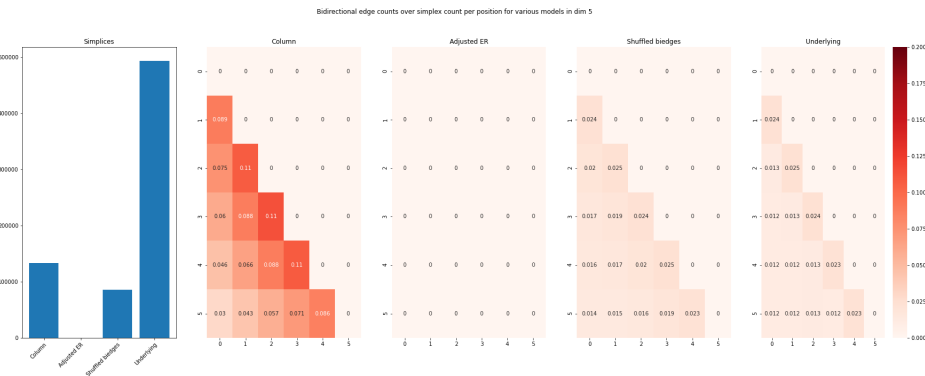


Figure 5.16 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 5. To the left, a simplex count histogram for dimension 5. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

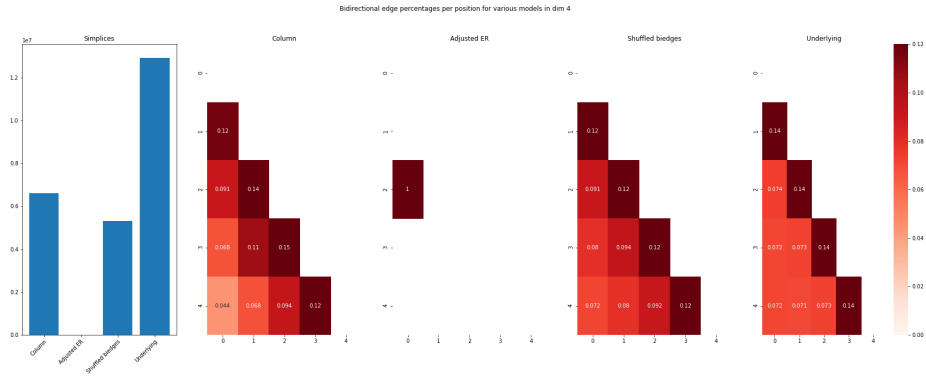


Figure 5.17 – Percent bidirectional edge count for the model and the controls in dimension 4. To the left, a simplex count histogram for dimension 4. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

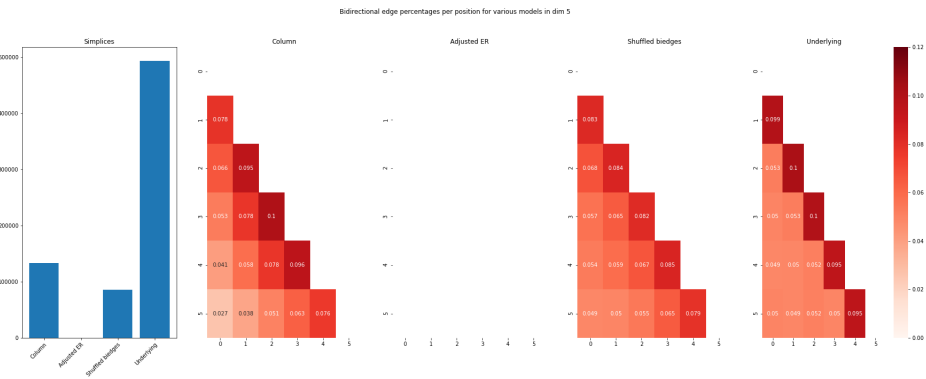


Figure 5.18 – Percent bidirectional edge count for the model and the controls in dimension 5. To the left, a simplex count histogram for dimension 5. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

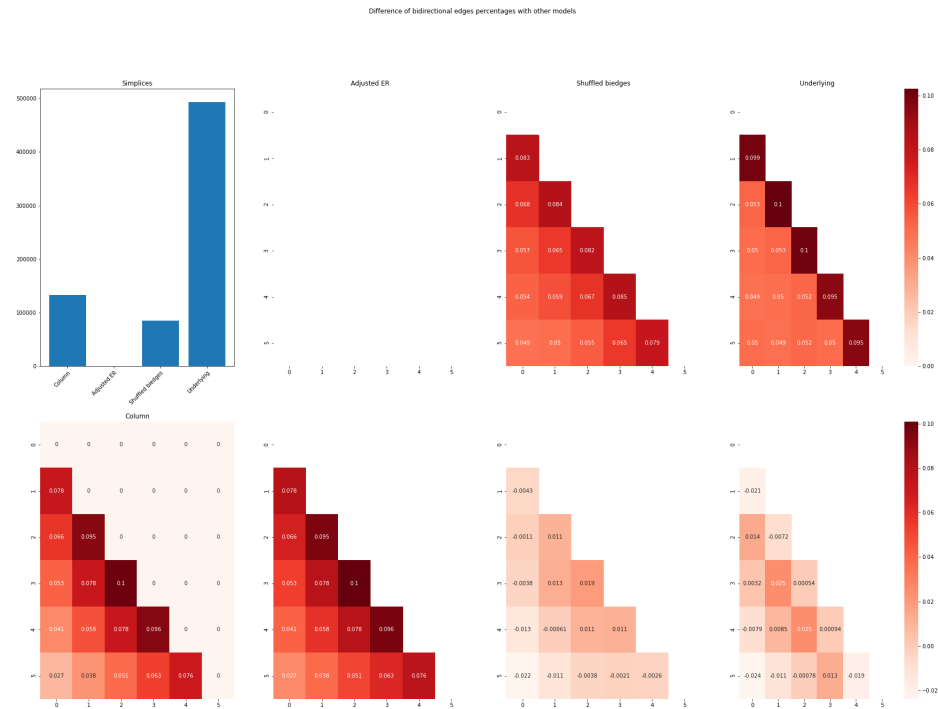


Figure 5.19 – Percent bidirectional edge count difference for the model and the controls in dimension 5. Upper row is controls, the microcircuit counts are at the left. Difference in counts between microcircuit and each model is shown in the lower row. A single instance for each matrix.

To quantify this different tendency, we plotted the curves of normalized bidirectional edge counts per position $((0, 1), (0, n), (n, n-1))$ over the number of simplices for the microcircuit and models 4 and 5 in Figure 5.20. As mentioned, the microcircuit shows a different behaviour. The normalized count is bigger in absolute terms, and the difference among positions $(n, n-1)$ and $(0, n)$ increases along dimension. This reinforces the fact that bidirectional edge distribution in the microcircuit is not random.

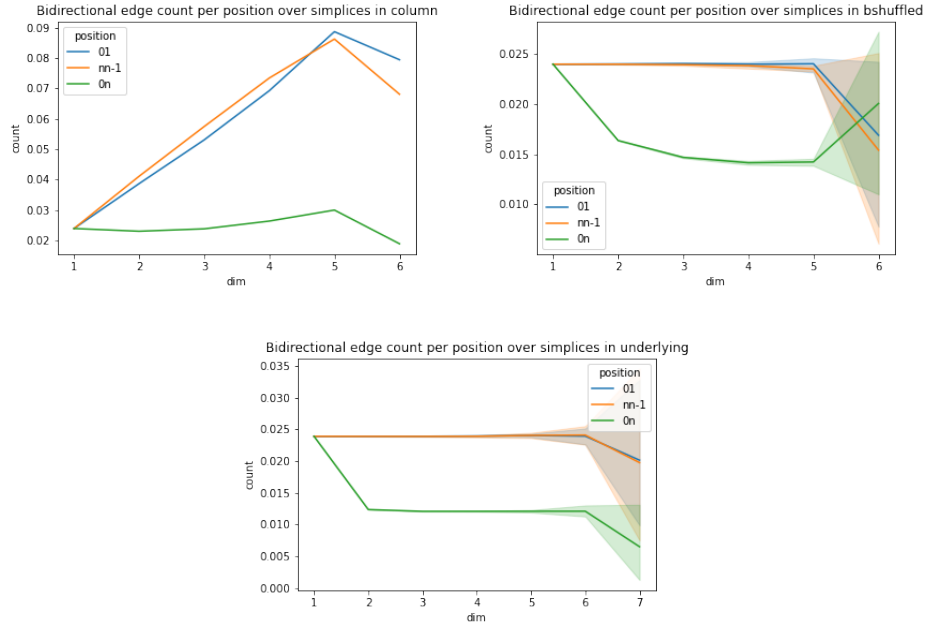


Figure 5.20 – Normalized bidirectional edge count per position in the column (top left), bishuffled (top right) and underlying (bottom) models. Models are described in 5.1. Notice the different trend in the column with respect to the controls, and the difference in scale. 5 instances for the control models.

5.4.2 T-tests for percentage matrices

To demonstrate the statistical significance of the differences between the matrix and the underlying model, we performed a one-sided t-test per position over 30 different instances of the underlying model. While this number of instances is not enough to ensure the validity of the t-test, we weren't able to perform more in the allotted time.

In Figure 5.22 we show the obtained t-test significances for dimensions 4 and 5. Red squares represent the microcircuit having lower occurrence percentage than the underlying model. Blue squares are the other way around. This shows that the bidirectional edge distribution inside simplices is different between the microcircuit and the underlying model.

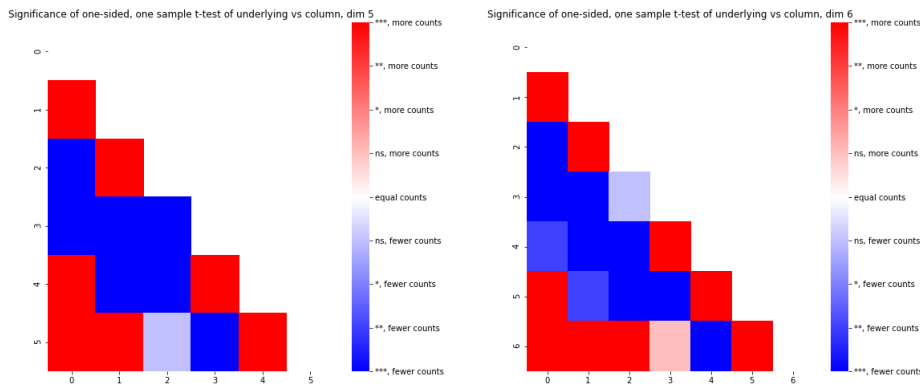


Figure 5.21 – Significance of 1-sided t-tests of bidirectional edge counts in different position in computational model versus the underlying model. Blue squares mean that the original model has more bidirectional edges in percentage, red squares mean that the underlying model has more bidirectional edges in percentage. 30 instances of the control model.

We have identified two groups whose relative count is consistently above or below the underlying model.

- *Group 1*, positions corresponding to the first subdiagonal of the matrix. There are more bidirectional edges in the original model than in the underlying model.
- *Group 2*, positions $((0, 1), (0, n), (n, n - 1))$. There are less bidirectional edges in the original model than in the underlying model.

The trend is not respected in lower dimensions, where the two groups are not defined (dimension 1) or overlap (dimension 2). We thus show only dimensions starting from 3.

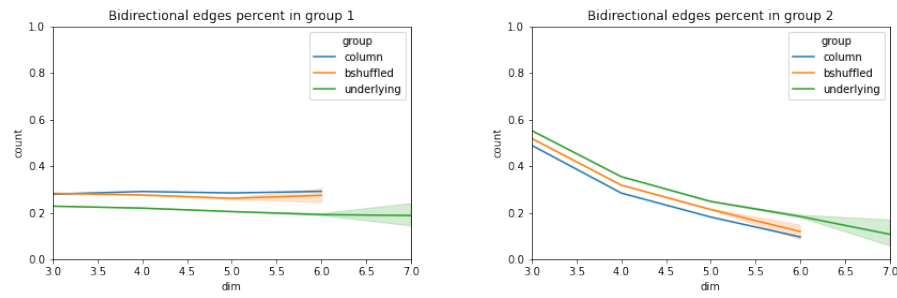


Figure 5.22 – Percent count of bidirectional edges in group 1 and group 2 in the original model against bishuffled and underlying model. 5 instances for the control models.

The bishuffled model was much more similar to the original model than the underlying model. However, we lacked the time to perform similar tests on it.

6 Structure and function in simplices

In this chapter, we explore the relationship between the structure of the graph and its motifs and their activity. In particular, we analyze the spike train correlation in connections inside simplices, distinguishing bidirectional connections from the others. We show that neurons involved in bidirectional edges have higher correlation when the edge is at the end of a simplex rather than other positions.

6.1 Spike train correlation in simplices

We have shown in Chapter 4 that bisimplices really are an important motif in the microcircuit and that they are overrepresented in the connectivity graph. We now want to analyze whether this overrepresentation also comes with an effect in the network activity.

In line with experiments done in Reimann et al., 2017, we analyzed spike trains to see how neuron correlation relates to position inside a maximal simplex. However, differently from Reimann et al., 2017, we distinguished correlation between bidirectional and directed edges.

Spike trains are obtained from ongoing plasticity simulations. They differ from those used in Reimann et al., 2017, as only simulation time bins of above-average activity were considered. We thank András Ecker for performing the simulations in the computational model and extracting all spike trains.

Figures in this chapter only show dimension from 2 to 5. The thresholding and the lower number of maximal simplices in these dimensions increased the noise in dimension 1 and 6, which were therefore excluded. Full plots on all dimensions are shown in the appendix.

The measure of correlation we use is Pearson Correlation.

Figure 6 shows the average correlation of bidirectional edges inside maximal simplices of

a given dimension. The dotted lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Correlations follow the same increasing trend shown in Reimann et al., 2017. Bidirectional edges tend to show more correlation than directed edges as shown in Figure 6.1. Bidirectional edges as well show an increase in correlation with dimension.

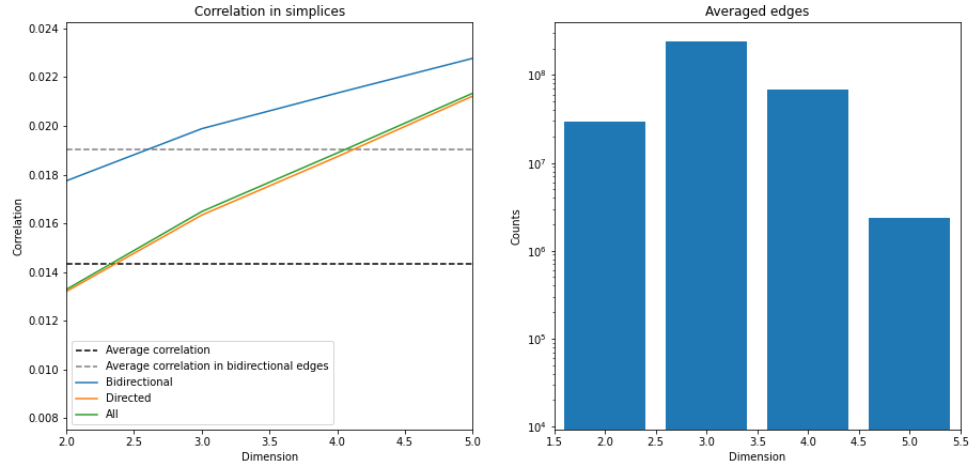


Figure 6.1 – Average correlation of spike trains of neurons in edges in any position of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the edge type - blue for bidirectional, orange for non-bidirectional edges and green for all edges. To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

6.2 Spike train correlation per position

We distinguished activity of three different position groups of the simplex:

- *Group 1* Any edge inside the simplex.
- *Group 2* Edges in the spine of the simplex.
- *Group 3* The edge $(n, n - 1)$ of the simplex. We refer to this as the end of the simplex. Note that when a bidirectional edge is present in this position, there is a bisimplex in the graph.

In line with the results in Reimann et al., 2017, the correlation between neurons increases

when the bidirectional edge was at the end of the simplex as shown in Figure 6.2. However, all edges presented this increase in correlation, as seen in Figure 6.3.

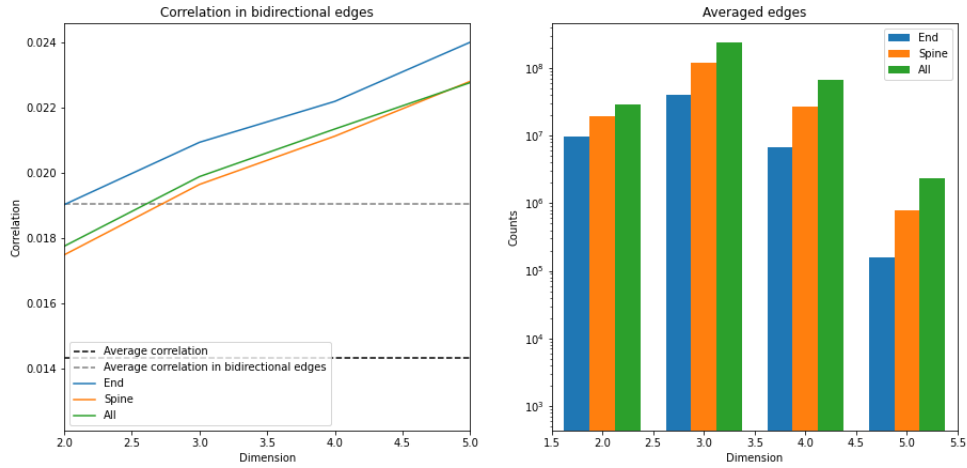


Figure 6.2 – Average correlation of spike trains of neurons in bidirectional edges in different positions of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the positions are described in 6.1. To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

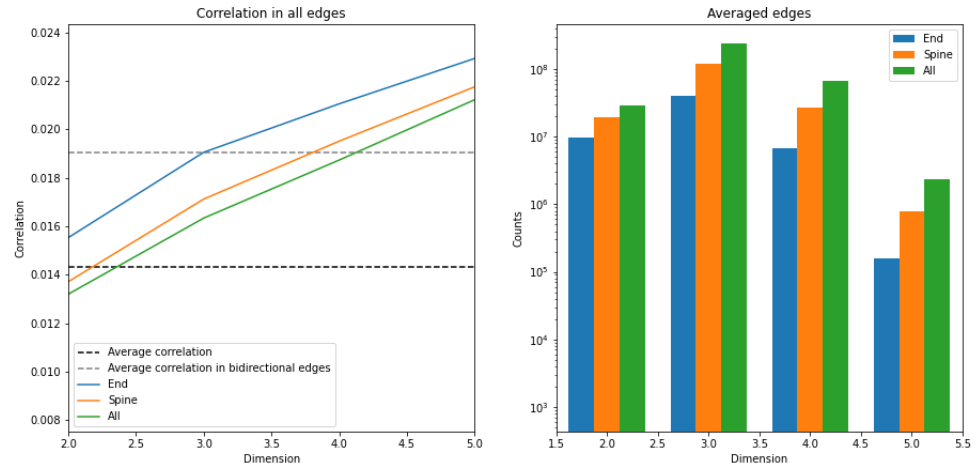


Figure 6.3 – Average correlation of spike trains of neurons in all edges in different positions of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the positions are described in 6.1. To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

6.1

7 Conclusions and perspectives

7.1 Conclusions

In the current work, we analyzed the presence and activity involvement of particular kinds of robust motifs in a very detailed somatosensory cortex computational model. The main challenge to the present work was to show that Robust Motif presence and role could not be explained by simplex overrepresentation found in Reimann et al., 2017.

Indeed we have shown in chapter 4 that bisimplices are overrepresented in the model, and their count by dimension cannot be explained by the mere presence of simplices. On the contrary, another special robust motif, the extended simplex, was underrepresented in the computational model, taking into account increased simplex counts. We have also shown that bisimplices tend to concentrate in specific m-types of the model, and this difference in concentration cannot be explained by presence of bidirectional edges alone.

Bisimplex overrepresentation was explained by an increase in bidirectional edge formation in simplices, as shown in 5. We didn't find preferential formation of reciprocal connections at the end of a clique. The distribution of bidirectional edges in cliques was non-random as it was different from the control models. This implies that other motifs, different from the bisimplex, should be overrepresented in the computational model as well.

Finally, we have shown that activity in bisimplices follows the same trend found in Reimann et al., 2017, with bidirectional edges having increased correlation at the end of the simplex, so that bisimplices increase even further neuron correlation. This suggests that bisimplices do play a role in the activity of the overall network.

7.2 Perspectives

The following are future areas of study that would strengthen our results but lie outside of the scope of this thesis.

7.2.1 Additional controls and other models

Additional controls can still be performed:

- The histograms of Section 4.5 can be generated controlling for simplex presence as well.
- The analysis of Section 6.1 can be re-run using spike trains more similar to those presented in Reimann et al., 2017, where low-activity time bins were not thresholded out.
- Correlation in Section 6.1 can be analyzed in terms of number of bisimplices of a given dimension a bidirectional edge belonged to, as done in Reimann et al., 2017.
- The results of Section 5.4 can be improved by increasing the number of control model instances used for the t-tests.
- The analysis of Section 5.4 can be repeated confronting the microcircuit with the bishuffled model as well.

Other models could be included in the analysis:

- The same analysis could be applied to other computational models, as for example *C. Elegans* or *Drosophila* connectomes, or the mouse cortex connectome developed by the Blue Brain Project.
- Results could be compared to more refined controls that better represent the various degree distributions of the computational model could be devised, following for example Section 2.2 of Spricer and Britton, 2015.

7.2.2 Future directions

While writing the present work, we thought of some directions for further research.

- Are robust motifs that do not contain a simplex also overrepresented in the model? While extended simplices are found in the model in higher number than controls, the ratio to simplex count was not as high as the bisimplex, so it might be that containing a simplex is not necessary for a motif to be overrepresented.

- Are networks with high directionality more likely to be represented in the model? Directed cliques and bisimplices have high directionality, and as shown in Curto et al., 2019, the fixed point of a DAG2 robust motif always lies in the 2-clique.
- Does the modified distribution of bidirectional edges in simplices influence activity? We have shown that a bidirectional edge at the end of the simplex increases correlation (section 6.1), and that some positions in the simplex have an increased chance of bidirectional edges (section 5.4). Maybe the changes in bidirectional edge formation chance in different positions drive a change in spike train correlation between involved neurons.

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A Supplementary figures for chapter 4

A.1 Absolute counts histograms for all dimensions

A.1.1 2D histogram of layer combinations in motifs

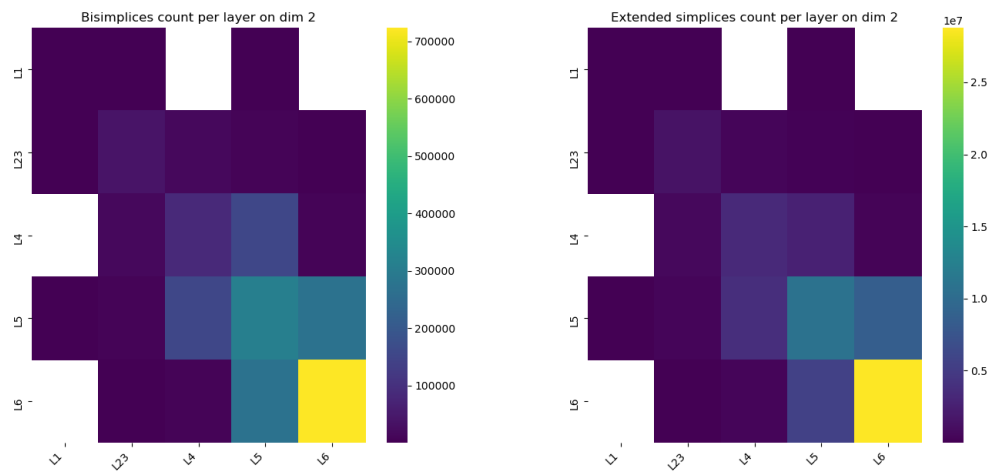


Figure A.1 – Count of Bisimplices and Extended Simplices per sink - extra edge layer combination in dimension 2.

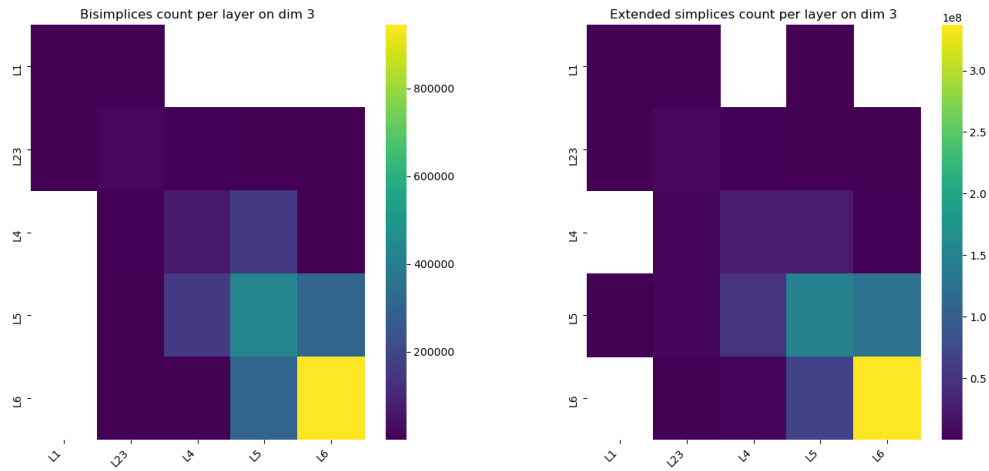


Figure A.2 – Count of Bisimplices and Extended Simplices per sink - extra edge layer combination in dimension 3.

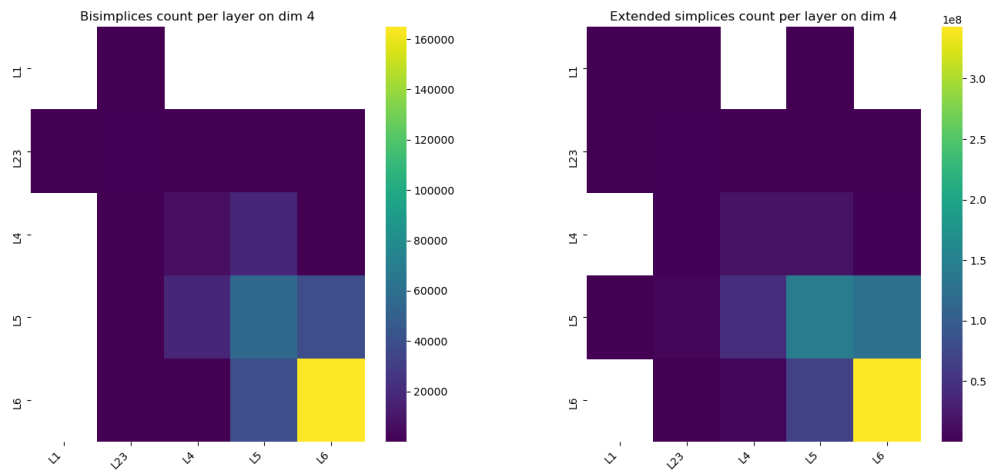


Figure A.3 – Count of Bisimplices and Extended Simplices per sink - extra edge layer combination in dimension 4.

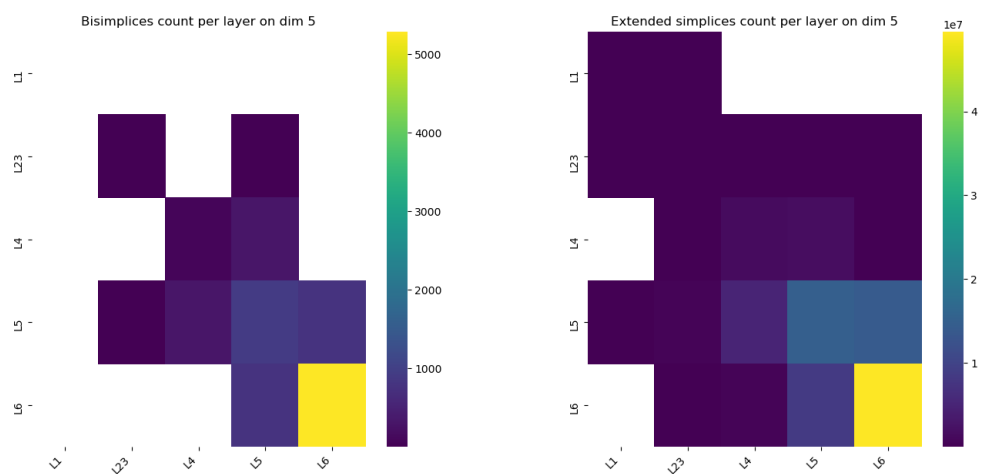


Figure A.4 – Count of Bisimplices and Extended Simplices per sink - extra edge layer combination in dimension 5.

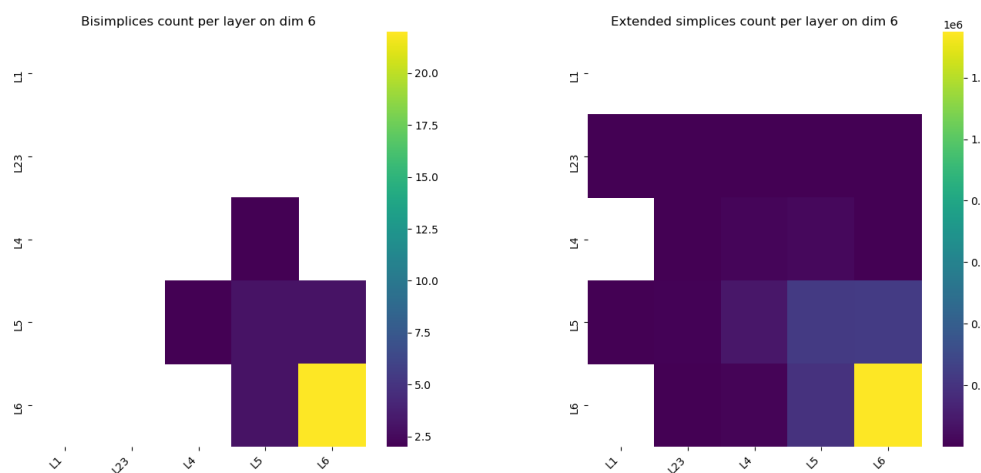


Figure A.5 – Count of Bisimplices and Extended Simplices per sink - extra edge layer combination in dimension 6.

A.1.2 Histogram of motif sink layer

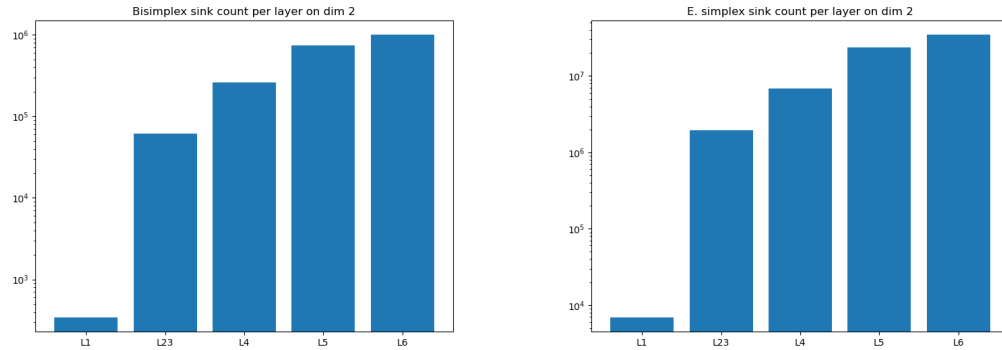


Figure A.6 – Count of Bisimplices and Extended Simplices sink layer in dimension 2.

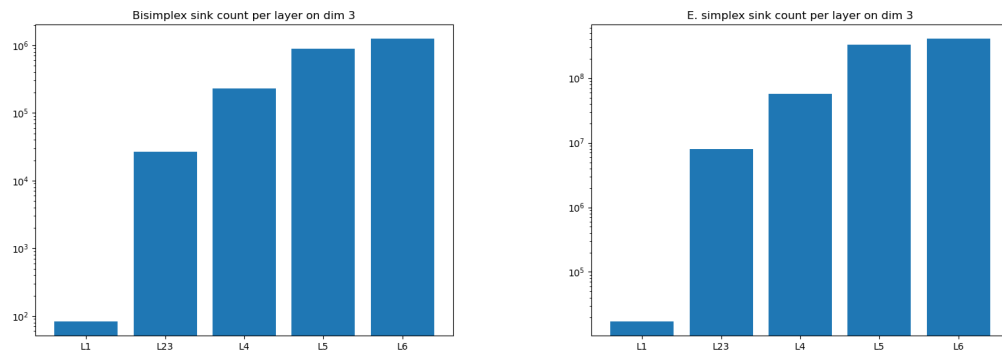


Figure A.7 – Count of Bisimplices and Extended Simplices sink layer in dimension 3.

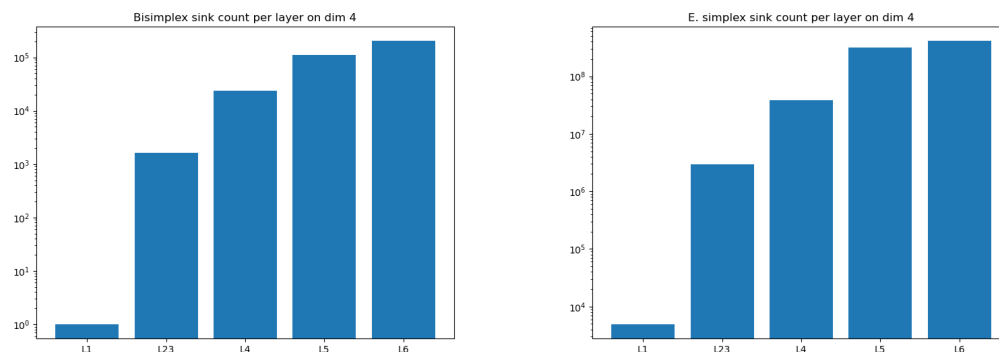


Figure A.8 – Count of Bisimplices and Extended Simplices sink layer in dimension 4.

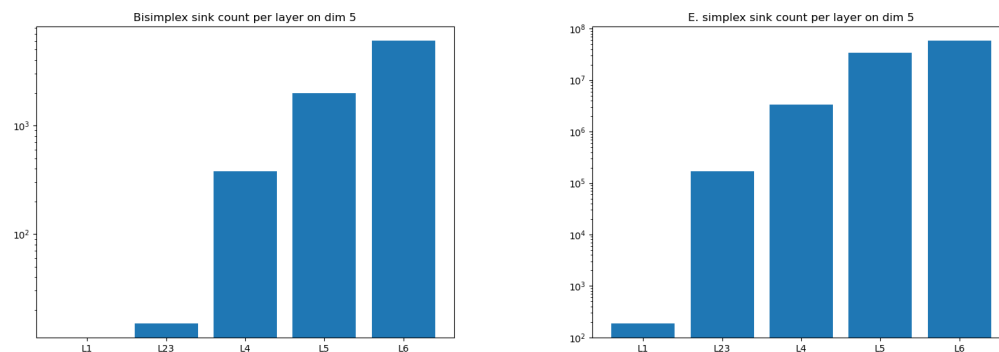


Figure A.9 – Count of Bisimplices and Extended Simplices sink layer in dimension 5.

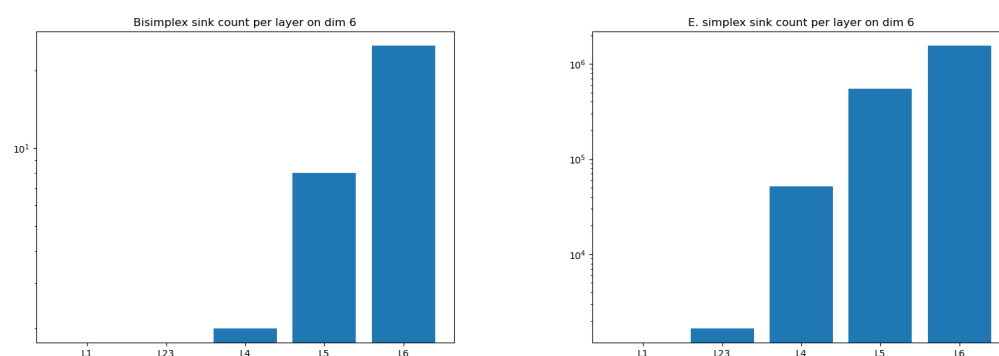


Figure A.10 – Count of Bisimplices and Extended Simplices sink layer in dimension 6.

A.1.3 2D Histogram of morphological type combinations in motifs

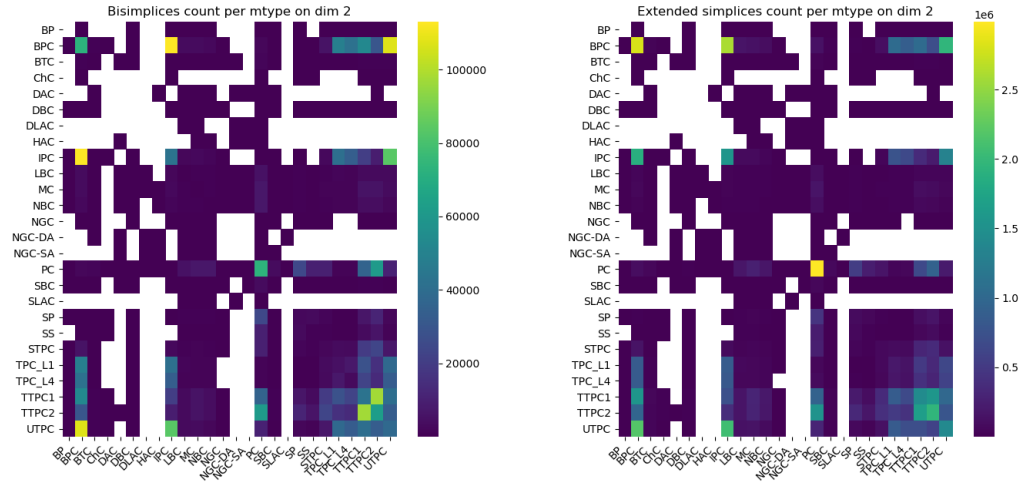


Figure A.11 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination in dimension 2.

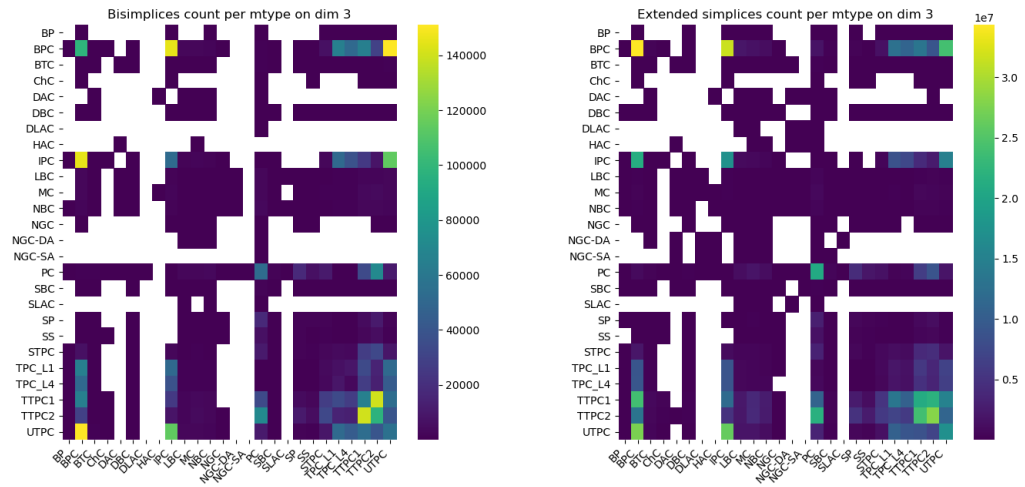
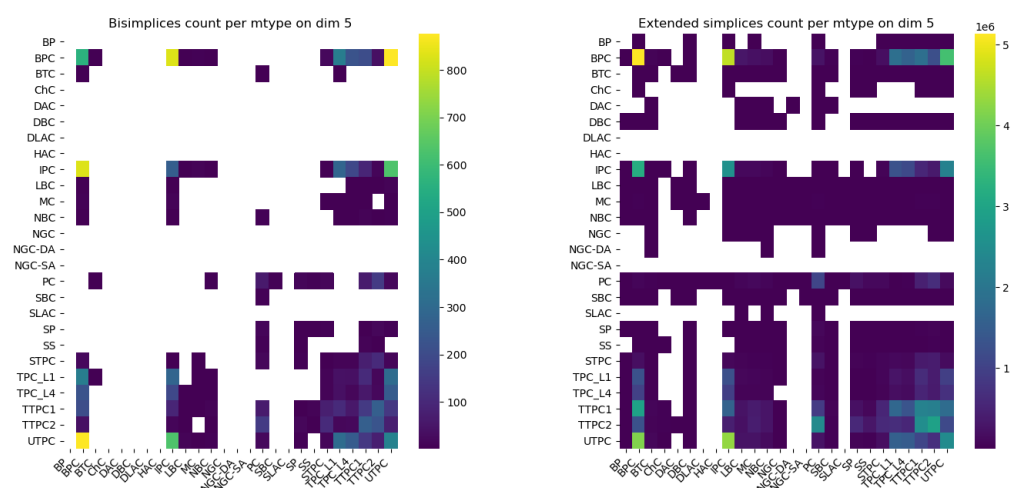
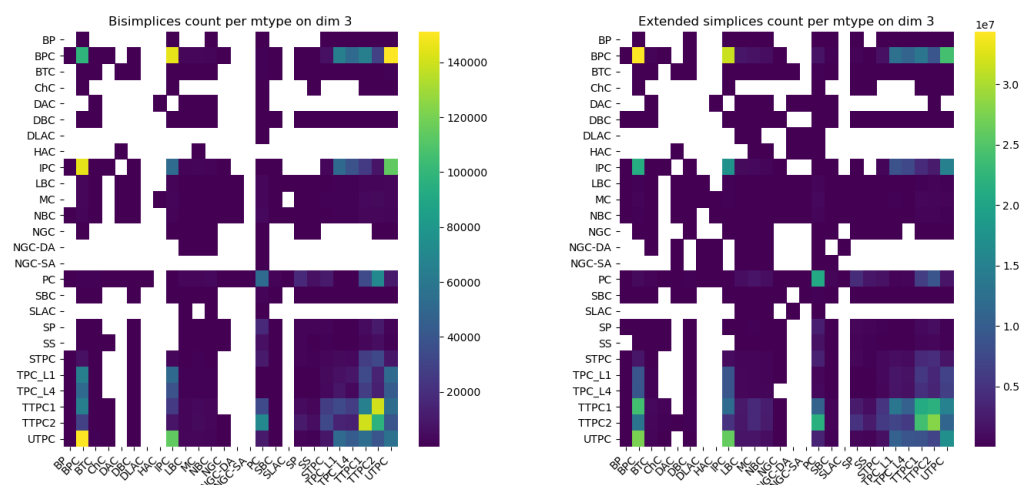


Figure A.12 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination in dimension 3.





A.1.4 Histogram of motif sink morphological type

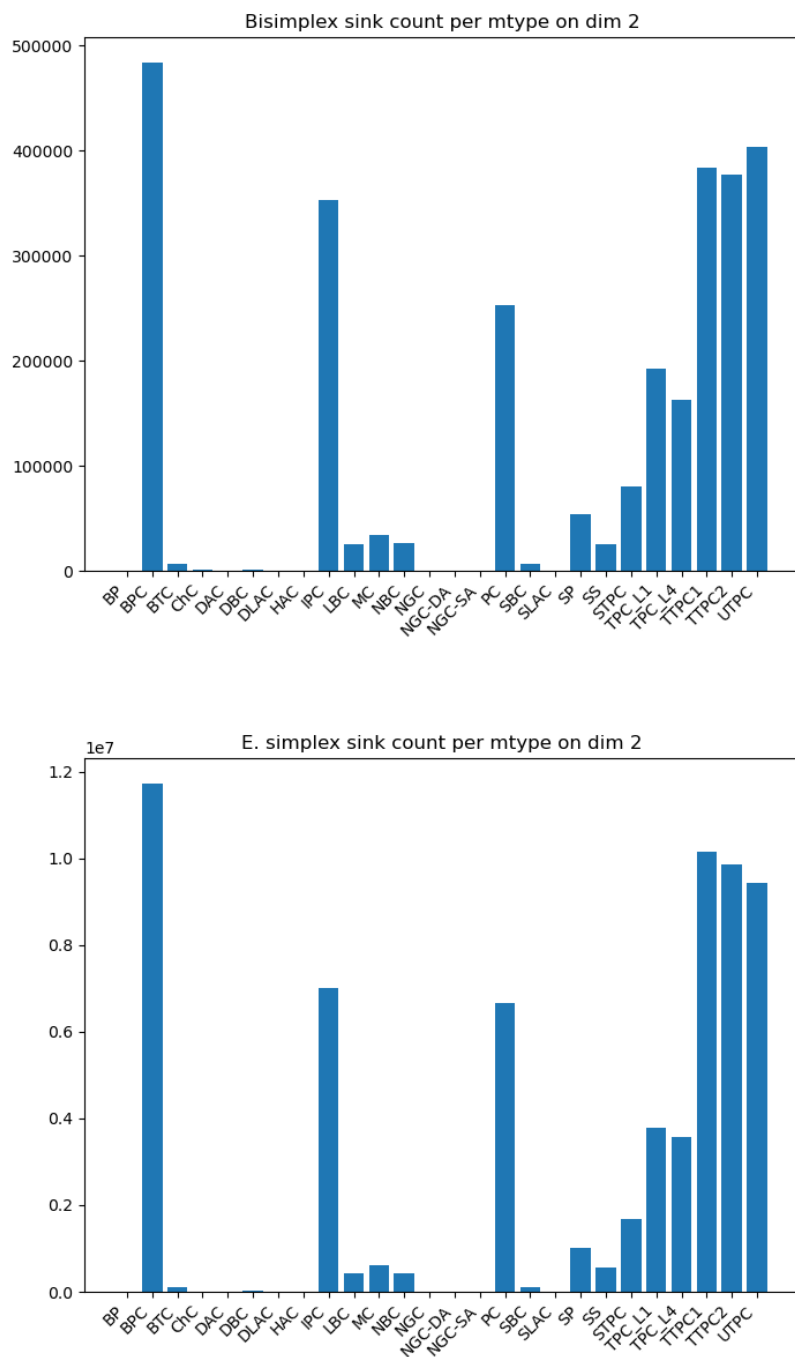


Figure A.16 – Count of Bisimplices and Extended Simplices sink mtype in dimension 2.

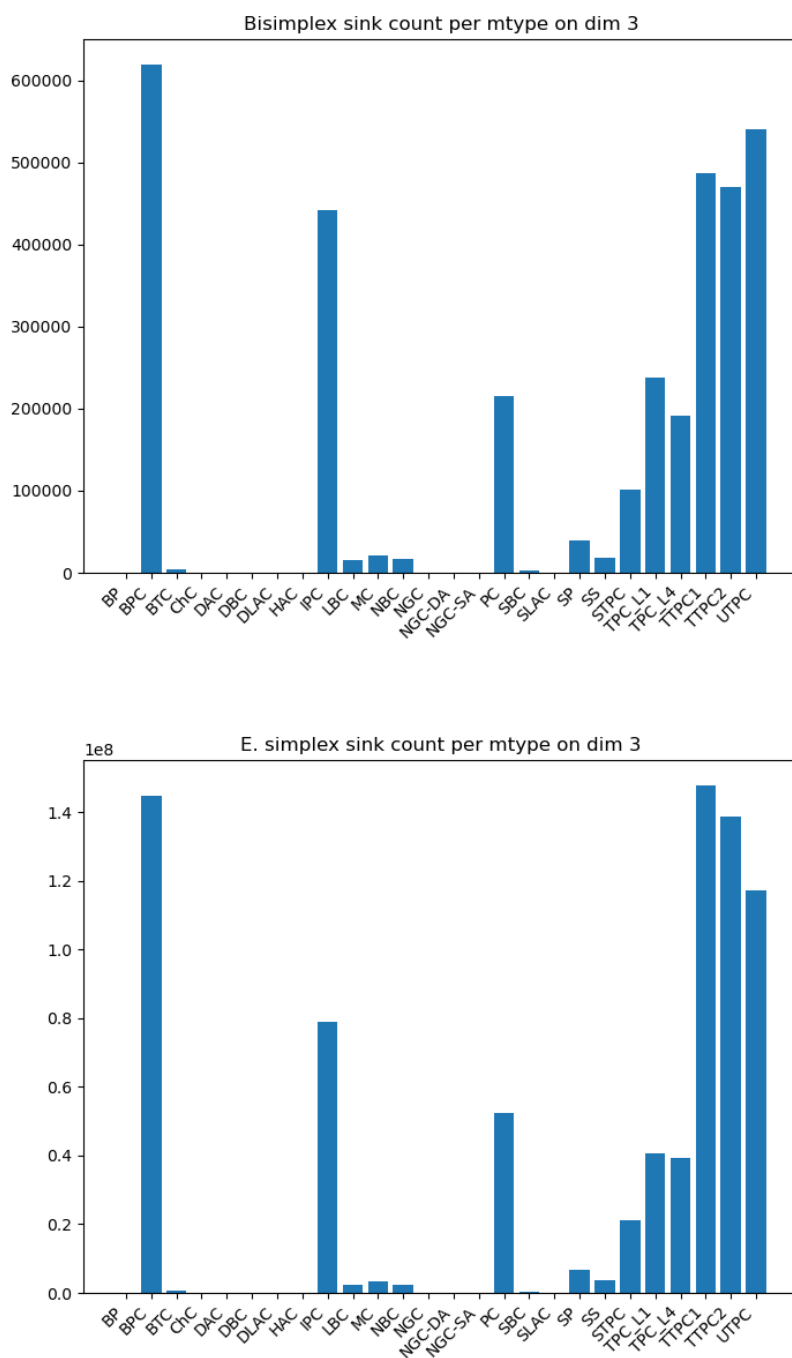


Figure A.17 – Count of Bisimplices and Extended Simplices sink mtype in dimension 3.

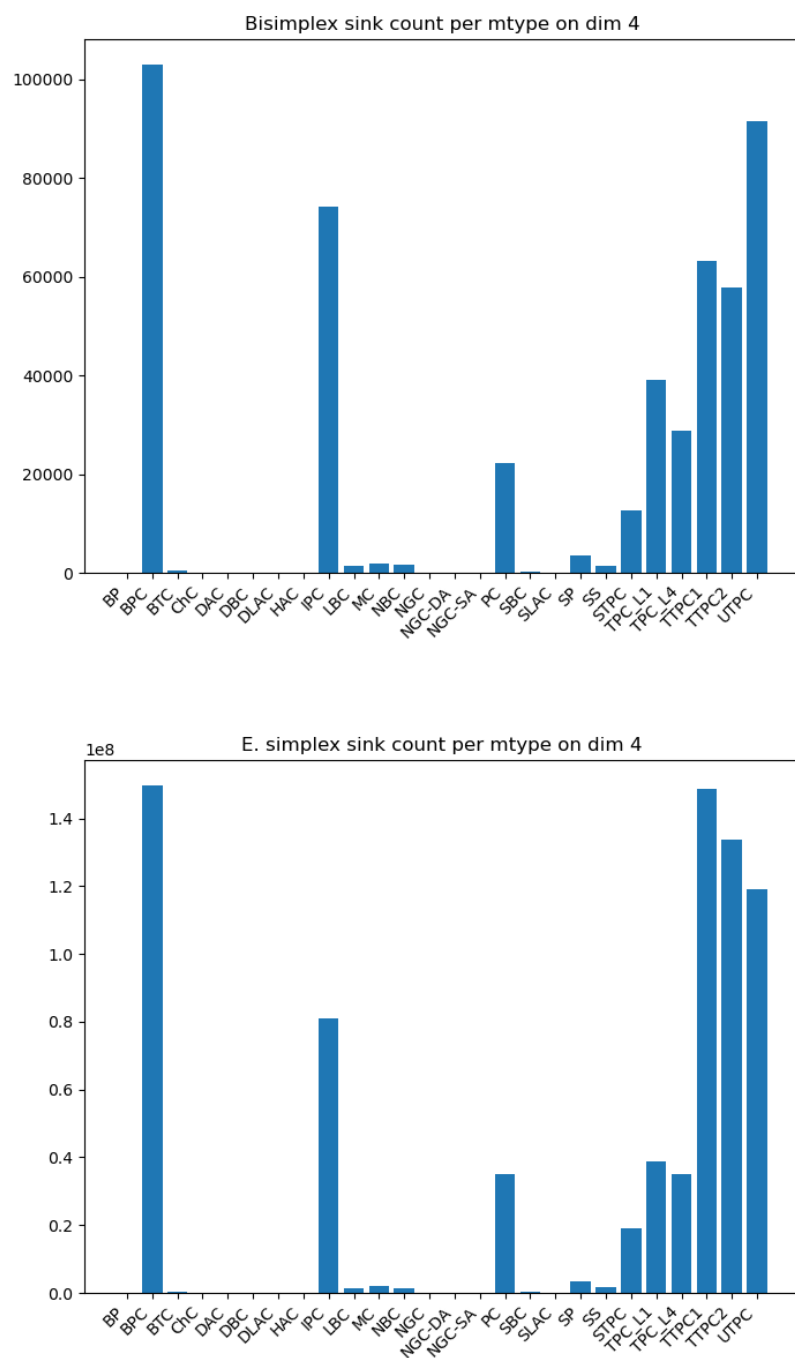


Figure A.18 – Count of Bisimplices and Extended Simplices sink mtype in dimension 4.

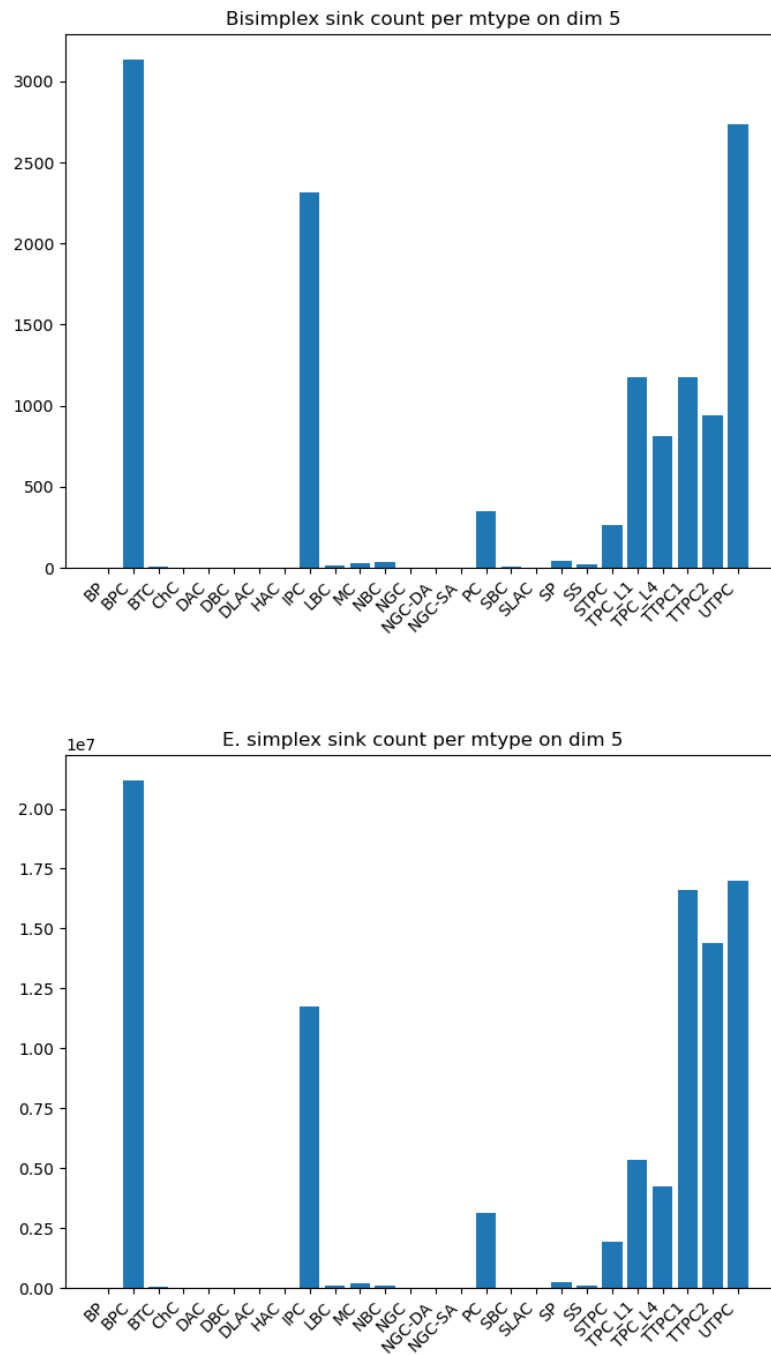


Figure A.19 – Count of Bisimplices and Extended Simplices sink mtype in dimension 5.

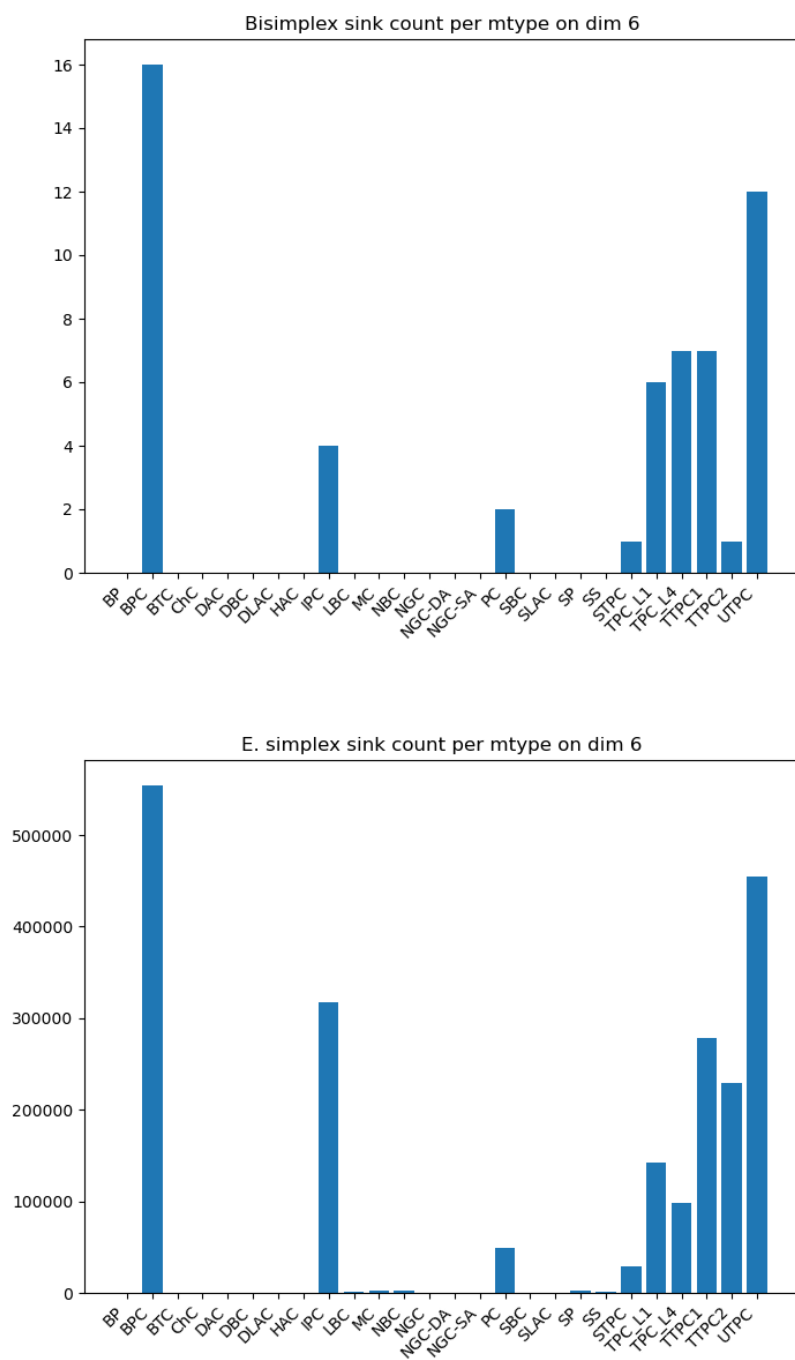


Figure A.20 – Count of Bisimplices and Extended Simplices sink mtype in dimension 6.

A.2 Normalized counts histograms for all dimensions

A.2.1 Normalized 2D histograms of layer combinations in motifs

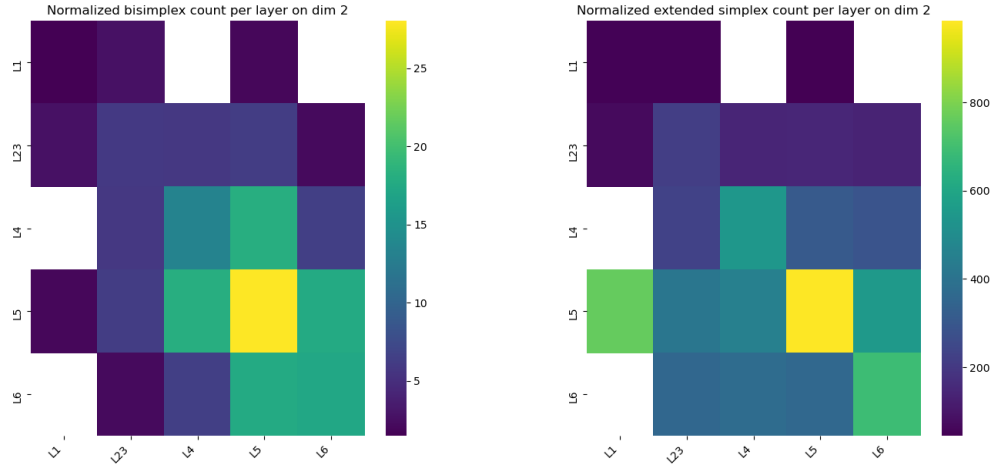


Figure A.21 – Count of bisimplices and extended simplices per sink - extra edge layer combination over bidirectional edges on neurons with the same combination in dimension 2.

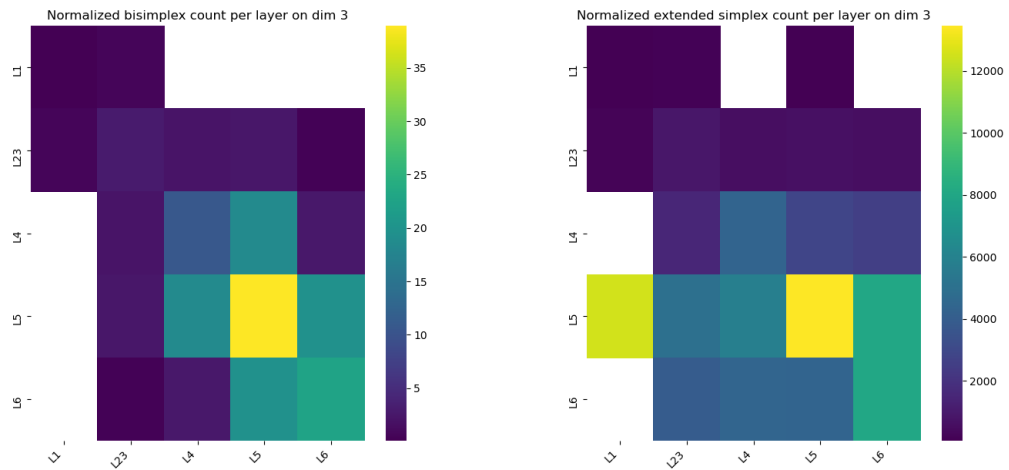


Figure A.22 – Count of bisimplices and extended simplices per sink - extra edge layer combination over bidirectional edges on neurons with the same combination in dimension 3.

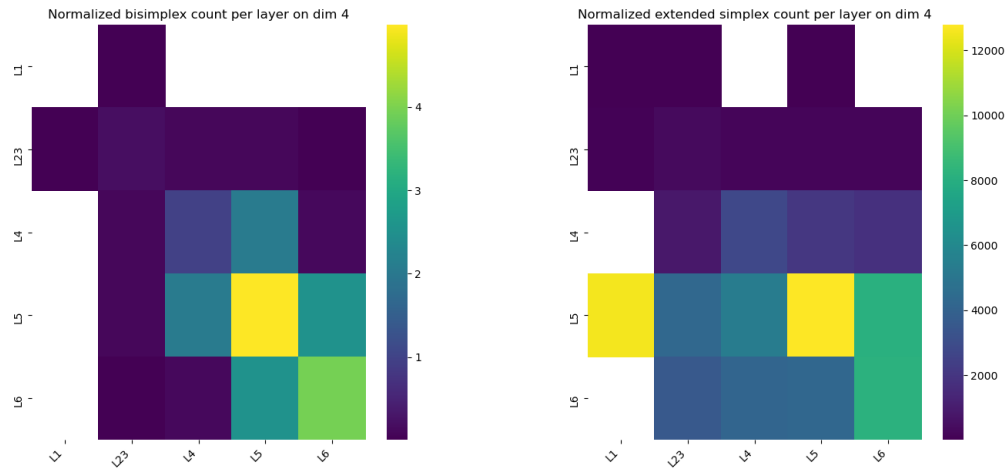


Figure A.23 – Count of bisimplices and extended simplices per sink - extra edge layer combination over bidirectional edges on neurons with the same combination in dimension 4.

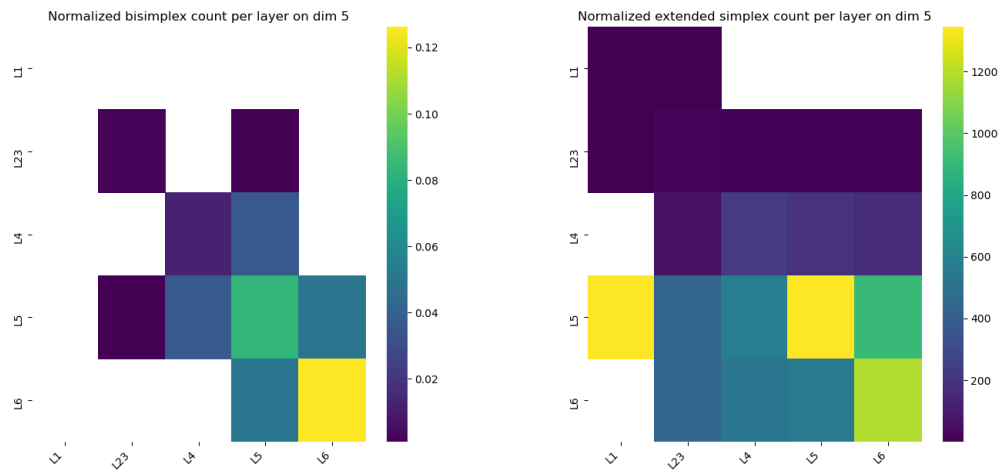


Figure A.24 – Count of bisimplices and extended simplices per sink - extra edge layer combination over bidirectional edges on neurons with the same combination in dimension 5.

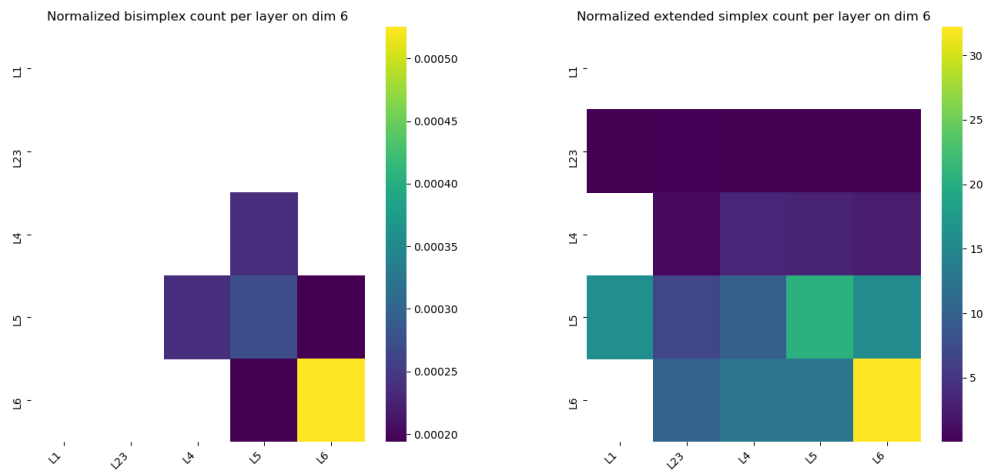


Figure A.25 – Count of bisimplices and extended simplices per sink - extra edge layer combination over bidirectional edges on neurons with the same combination in dimension 6.

A.2.2 Normalized histogram of motif sink layer

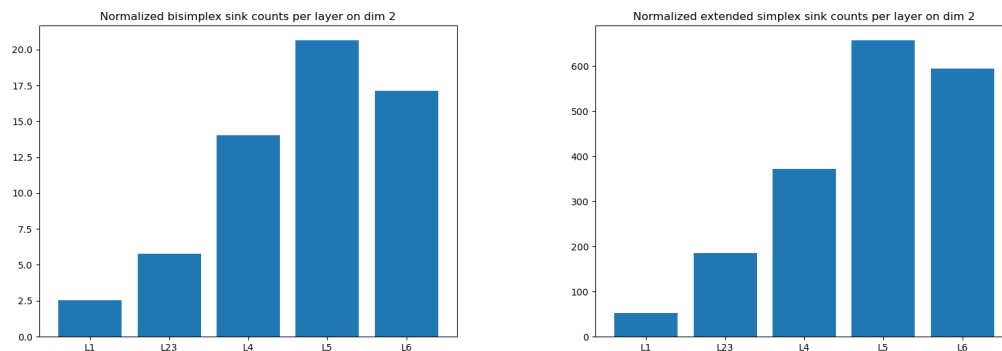


Figure A.26 – The motif count per layer type is compared to the number of bidirectional edges that include at least one neuron of that layer in dimension 2.

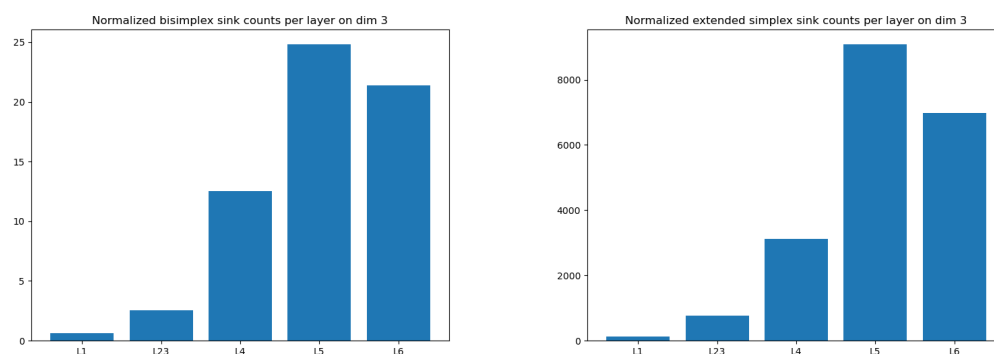


Figure A.27 – The motif count per layer type is compared to the number of bidirectional edges that include at least one neuron of that layer in dimension 3.

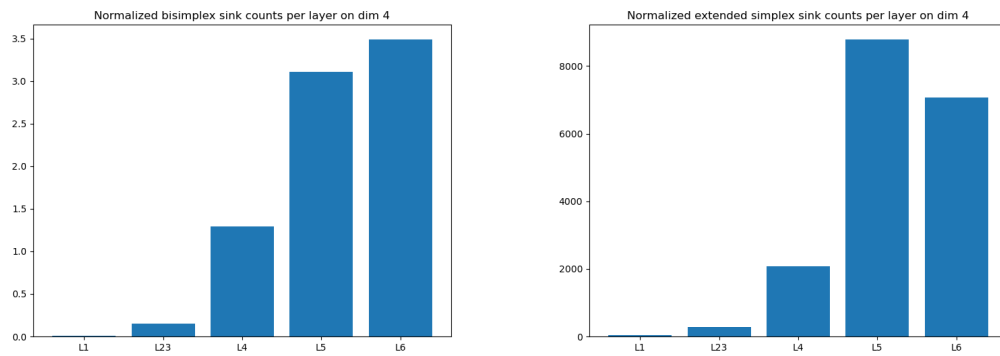


Figure A.28 – The motif count per layer type is compared to the number of bidirectional edges that include at least one neuron of that layer in dimension 4.

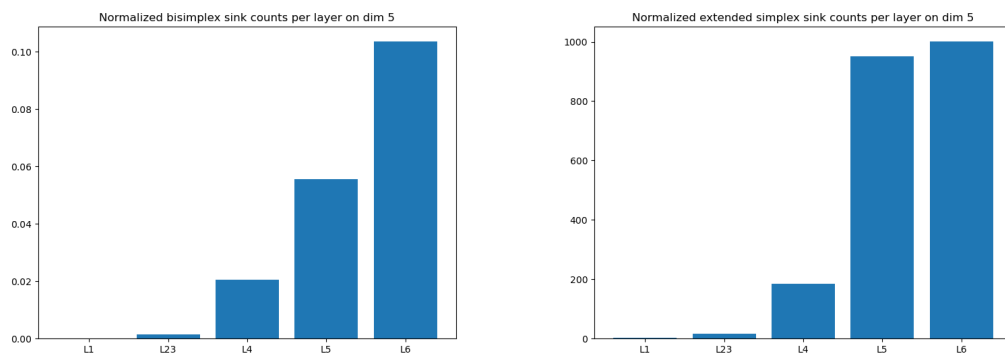


Figure A.29 – The motif count per layer type is compared to the number of bidirectional edges that include at least one neuron of that layer in dimension 5.

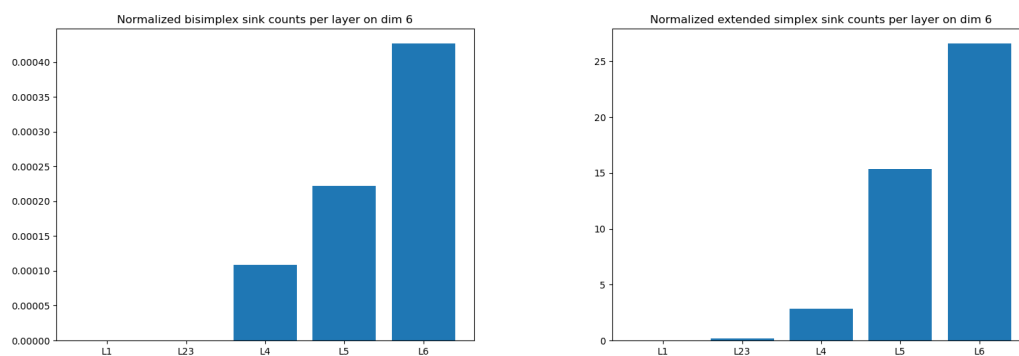


Figure A.30 – The motif count per layer type is compared to the number of bidirectional edges that include at least one neuron of that layer in dimension 6.

A.2.3 Normalized 2D histograms of morphological type combinations in motifs

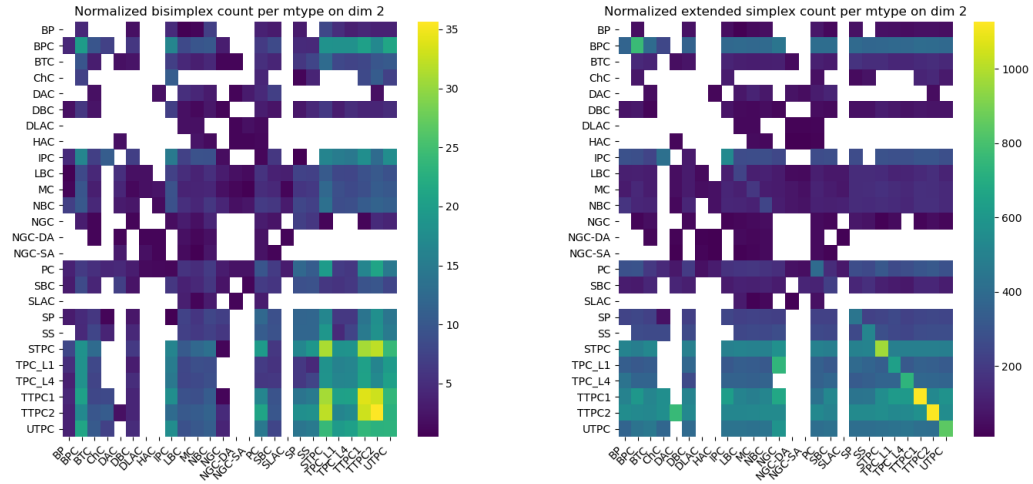


Figure A.31 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination over the number of bidirectional edges with that combination in dimension 2

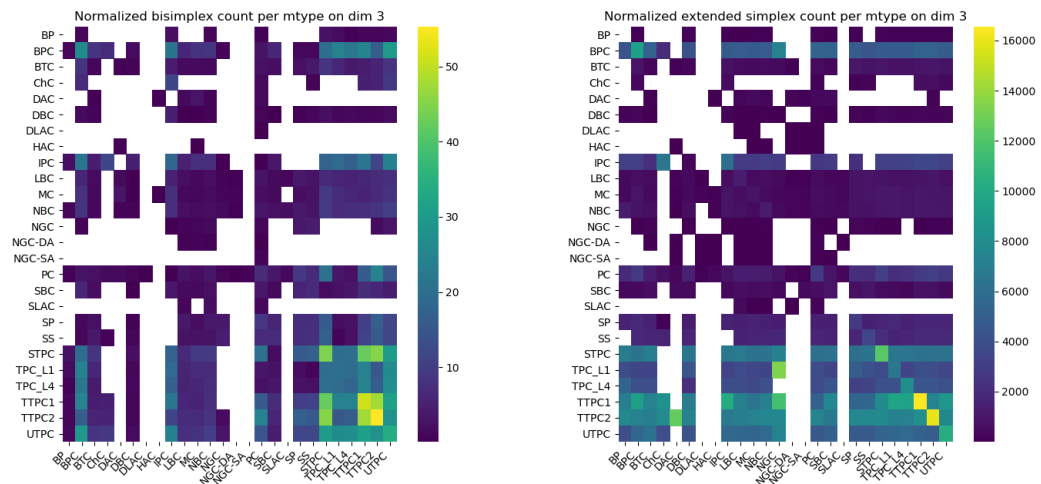


Figure A.32 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination over the number of bidirectional edges with that combination

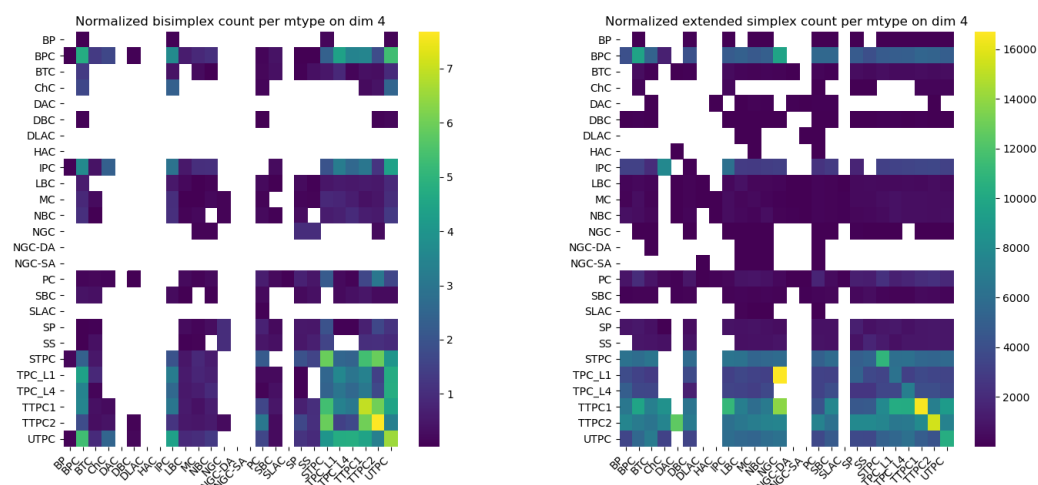


Figure A.33 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination over the number of bidirectional edges with that combination in dimension 4

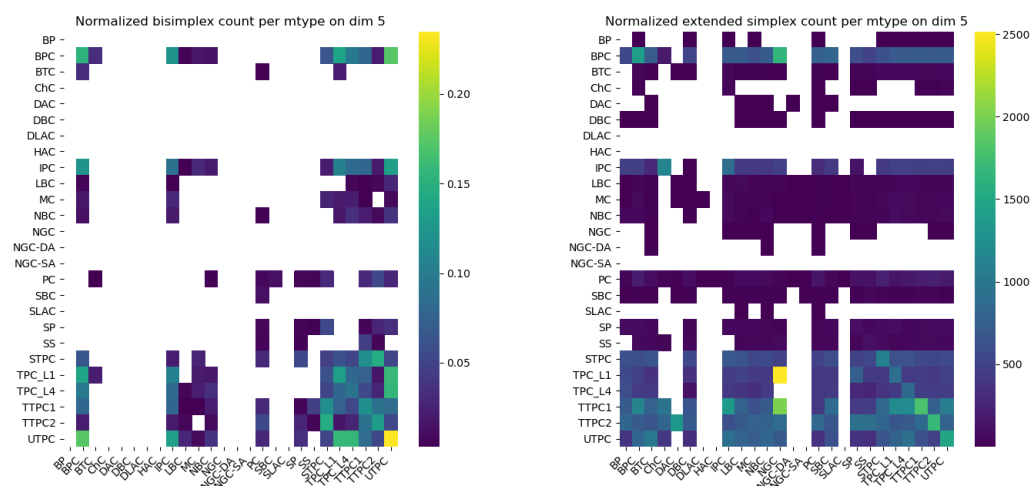


Figure A.34 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination over the number of bidirectional edges with that combination in dimension 5

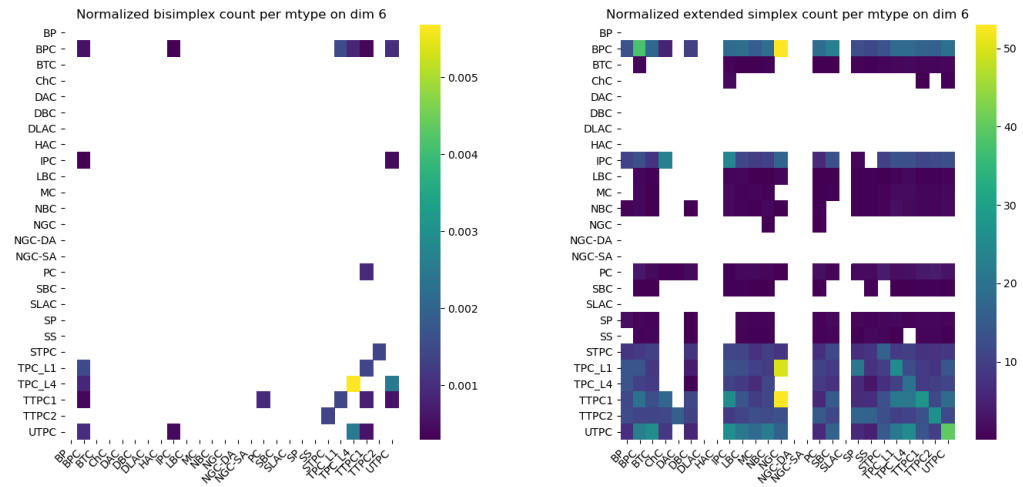


Figure A.35 – Count of Bisimplices and Extended Simplices per sink - extra edge mtype combination over the number of bidirectional edges with that combination in dimension 6

A.2.4 Normalized histogram of motif sink morphological type

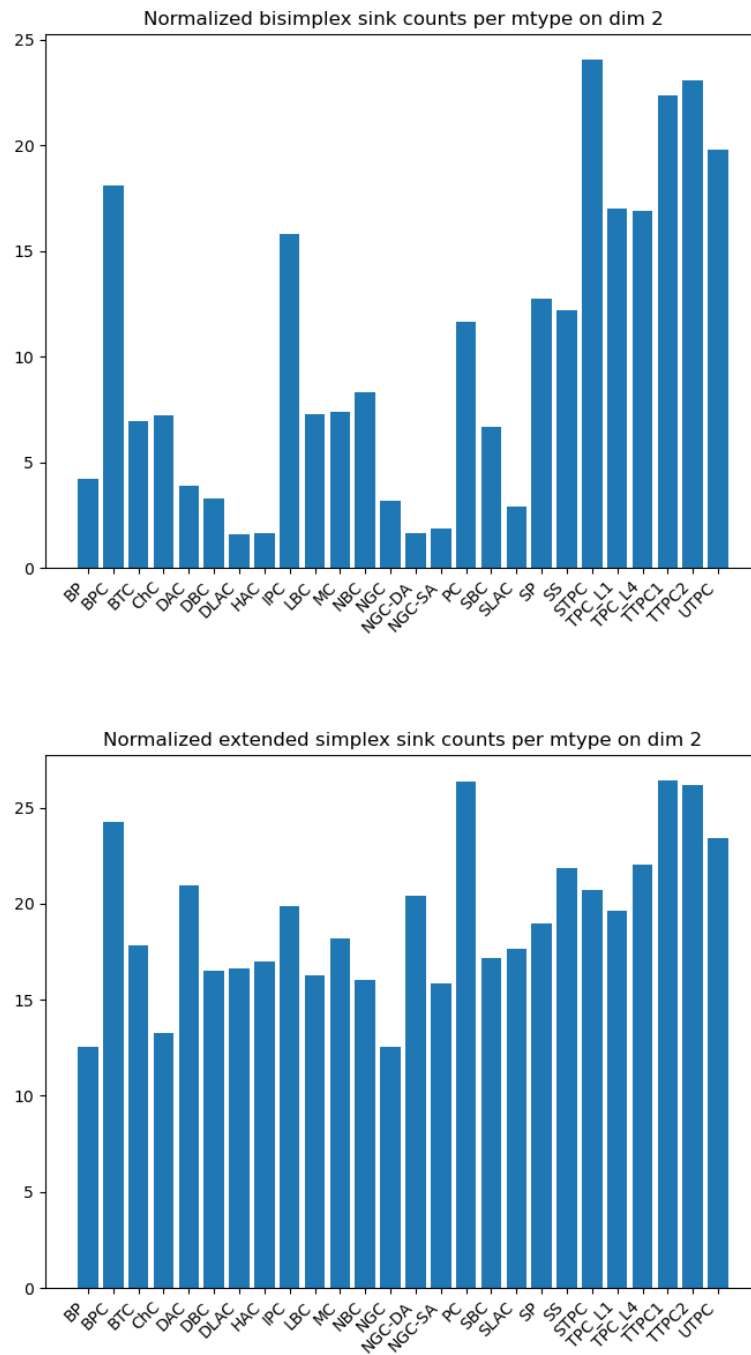


Figure A.36 – The motif count per m-type is compared to the number of bidirectional edges that include at least one neuron of m-type in dimension 2.

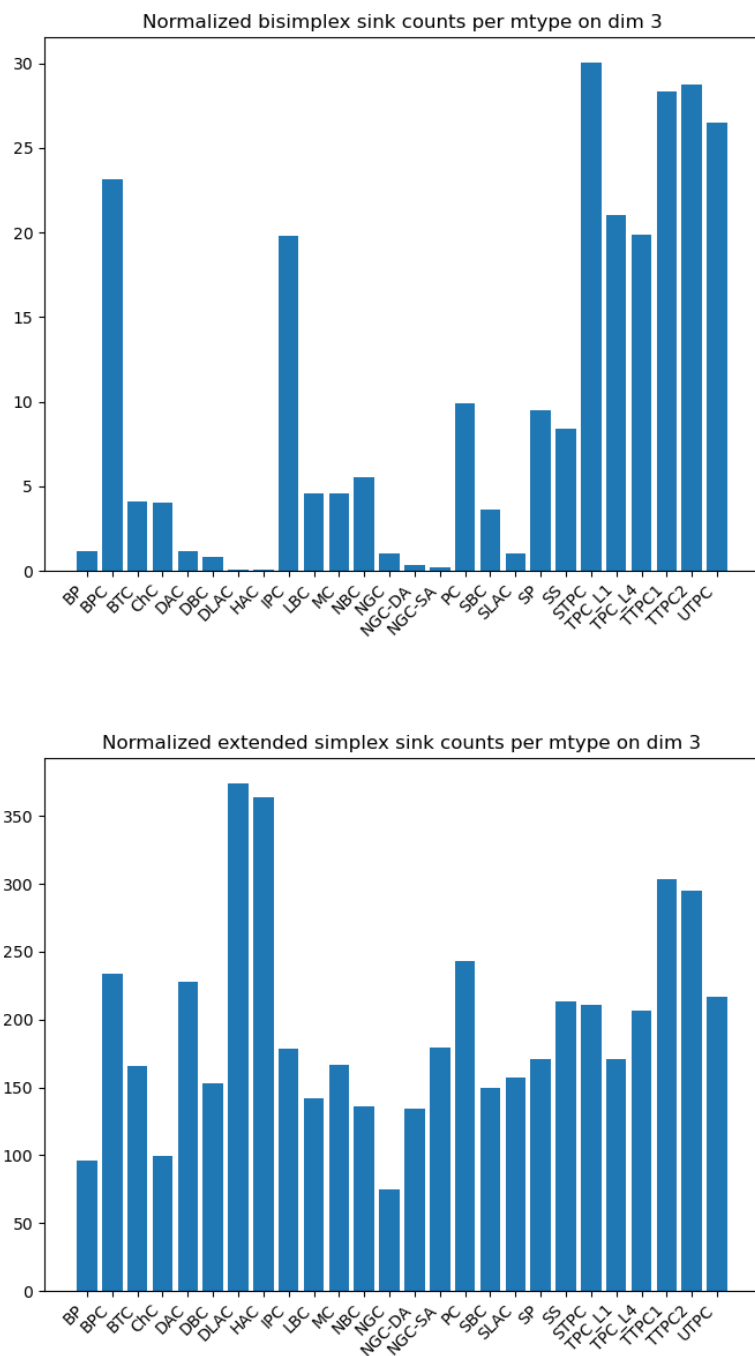


Figure A.37 – The motif count per m-type is compared to the number of bidirectional edges that include at least one neuron of m-type in dimension 3.

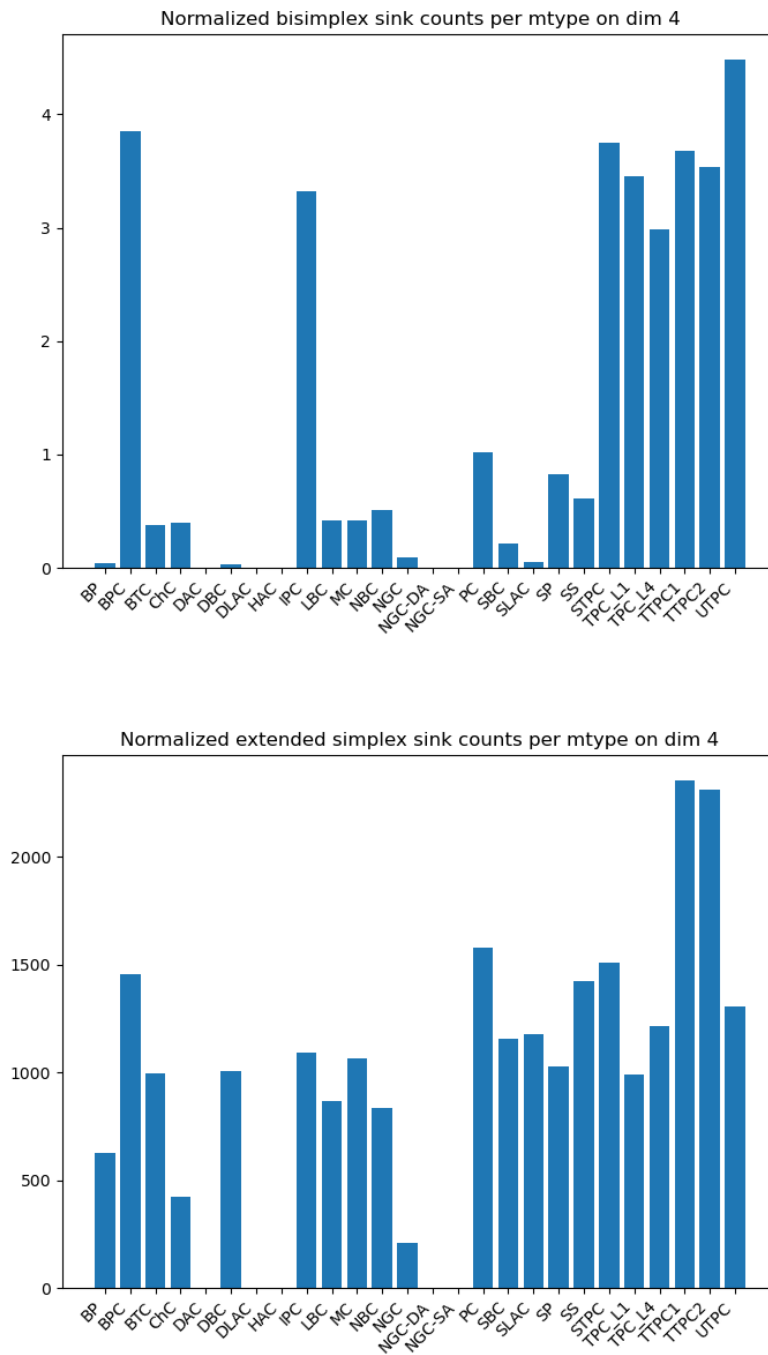


Figure A.38 – The motif count per m-type is compared to the number of bidirectional edges that include at least one neuron of m-type in dimension 4.

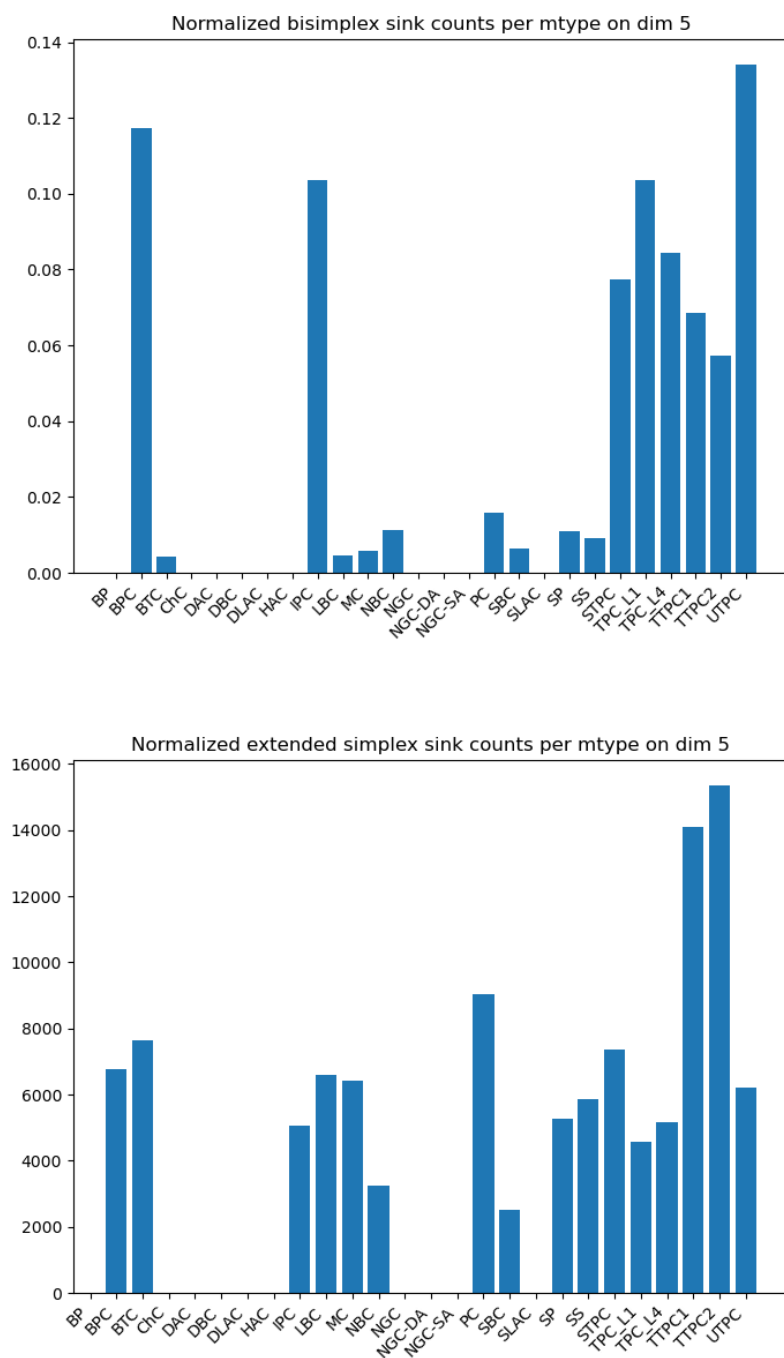


Figure A.39 – The motif count per m-type is compared to the number of bidirectional edges that include at least one neuron of m-type in dimension 5.

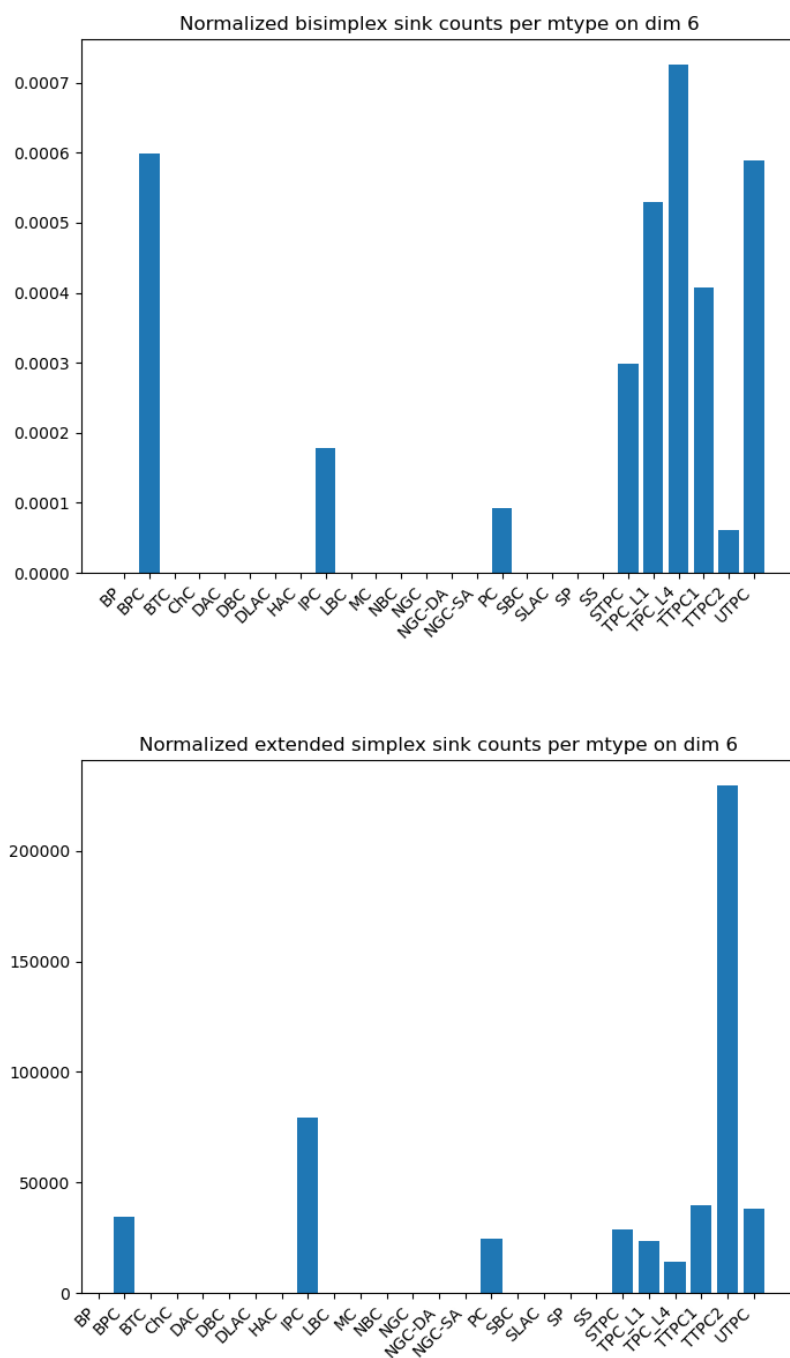


Figure A.40 – The motif count per m-type is compared to the number of bidirectional edges that include at least one neuron of m-type in dimension 6.

B Supplementary figures for chapter 5

B.1 Absolute count bidirectional edge matrices

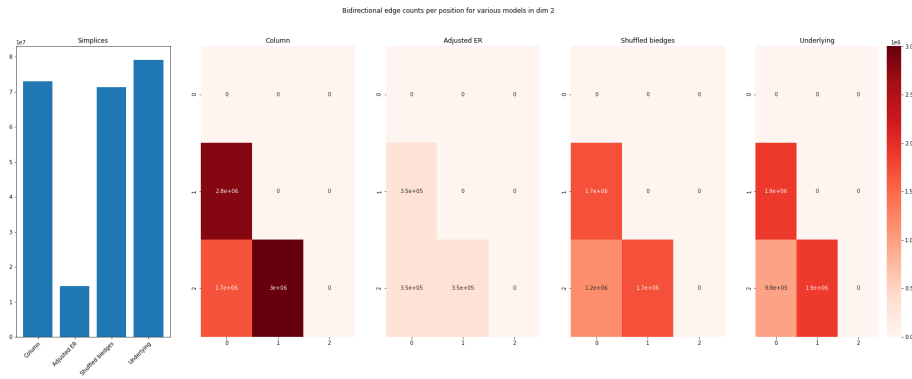


Figure B.1 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 2. To the left, a simplex count histogram for dimension 2. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

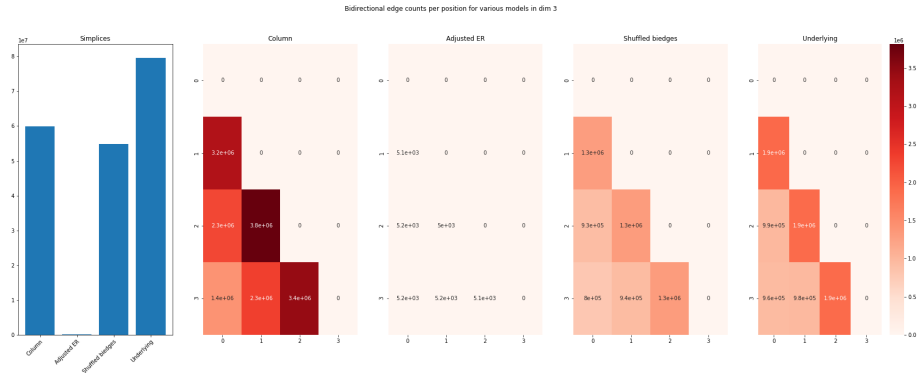


Figure B.2 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 3. To the left, a simplex count histogram for dimension 3. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

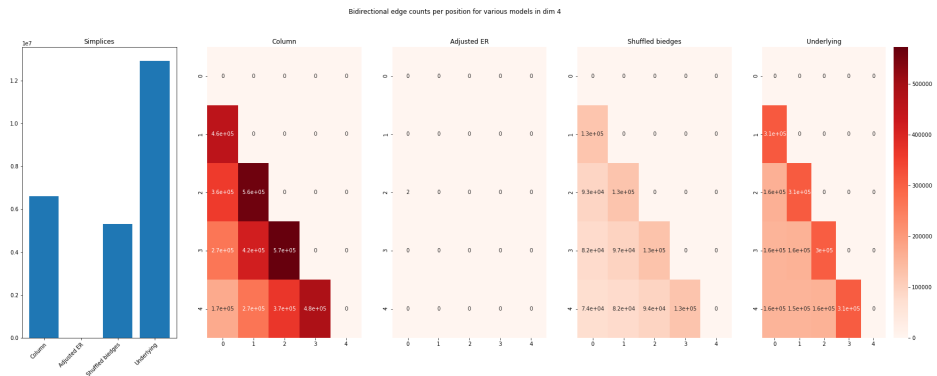


Figure B.3 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 4. To the left, a simplex count histogram for dimension 4. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

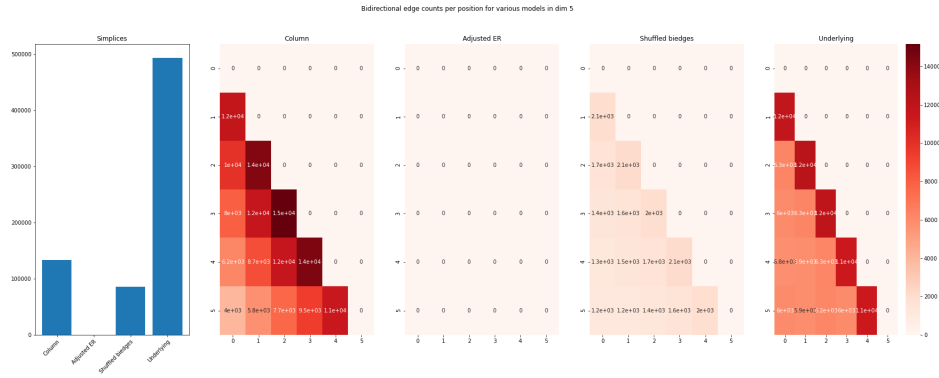


Figure B.4 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 5. To the left, a simplex count histogram for dimension 5. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

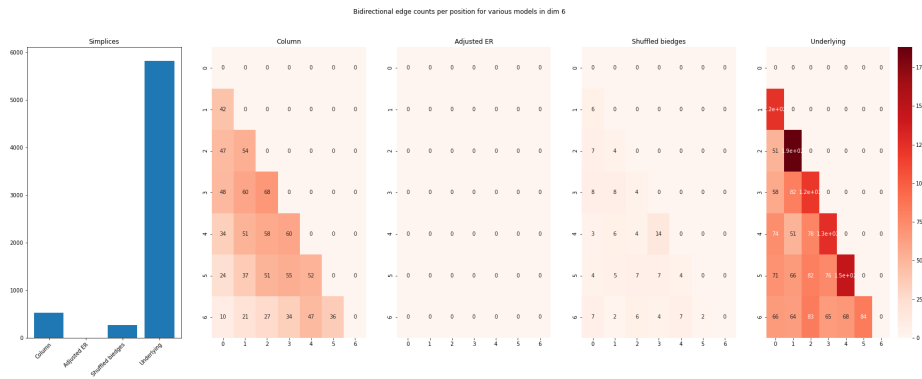


Figure B.5 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 6. To the left, a simplex count histogram for dimension 6. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

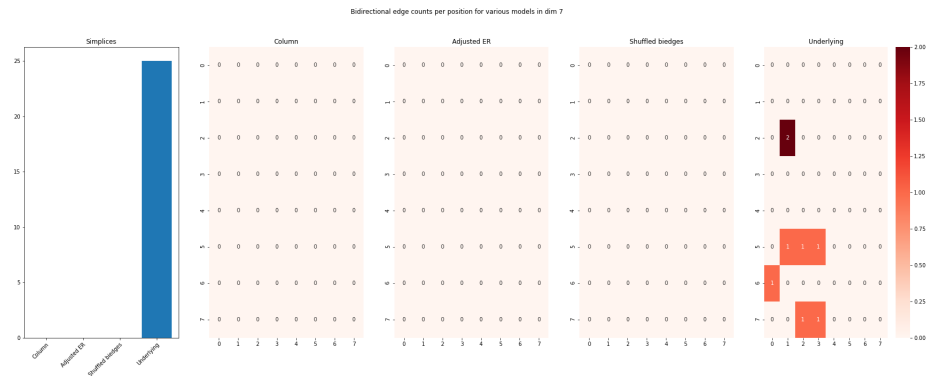


Figure B.6 – Absolute bidirectional edge count for the microcircuit and the controls in dimension 7. To the left, a simplex count histogram for dimension 7. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

B.2 Normalized count bidirectional edge matrices

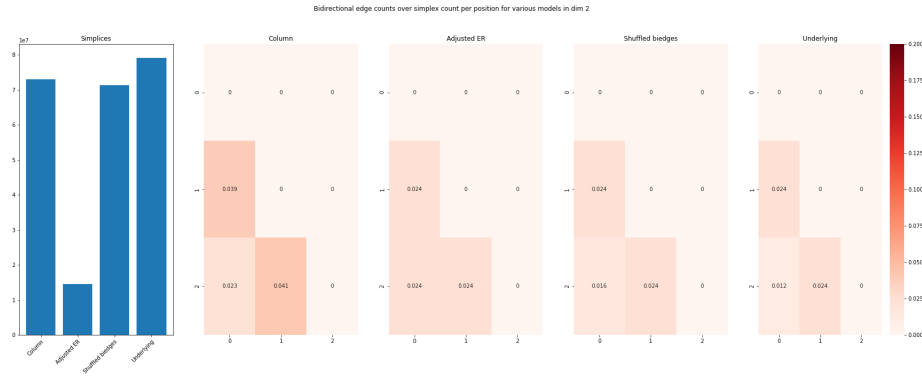


Figure B.7 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 2. To the left, a simplex count histogram for dimension 2. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

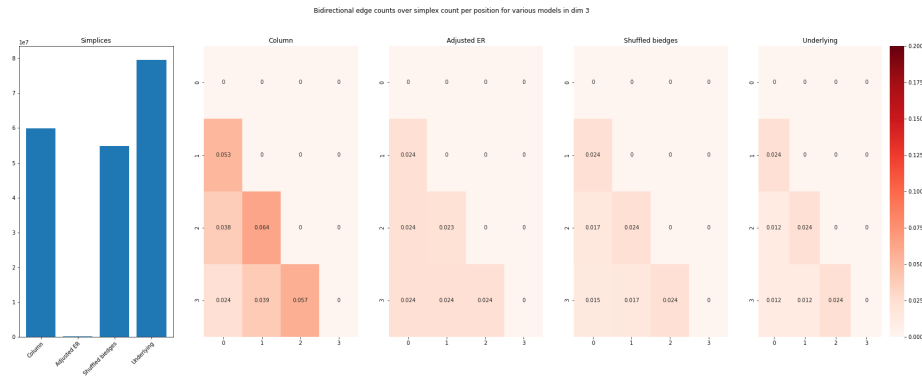


Figure B.8 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 3. To the left, a simplex count histogram for dimension 3. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

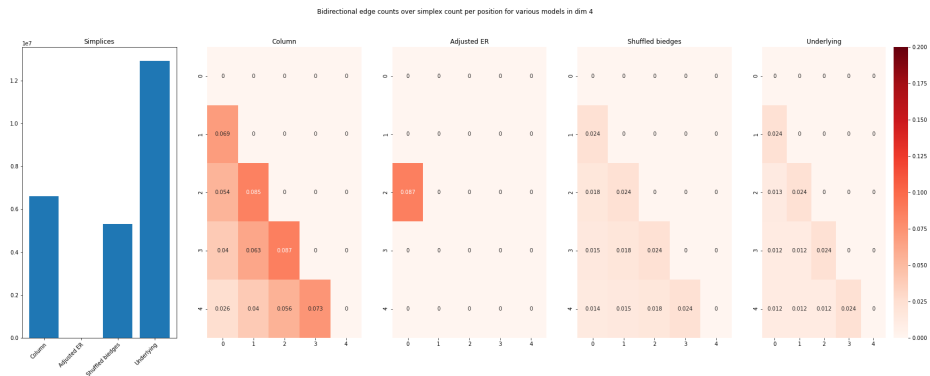


Figure B.9 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 4. To the left, a simplex count histogram for dimension 4. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

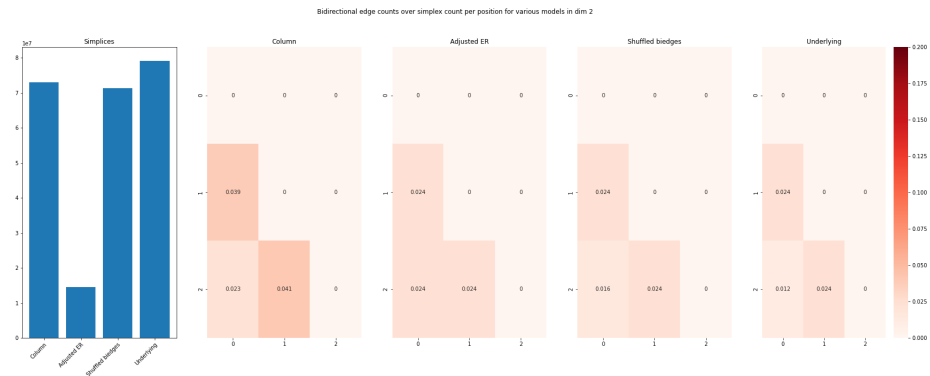


Figure B.10 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 5. To the left, a simplex count histogram for dimension 5. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

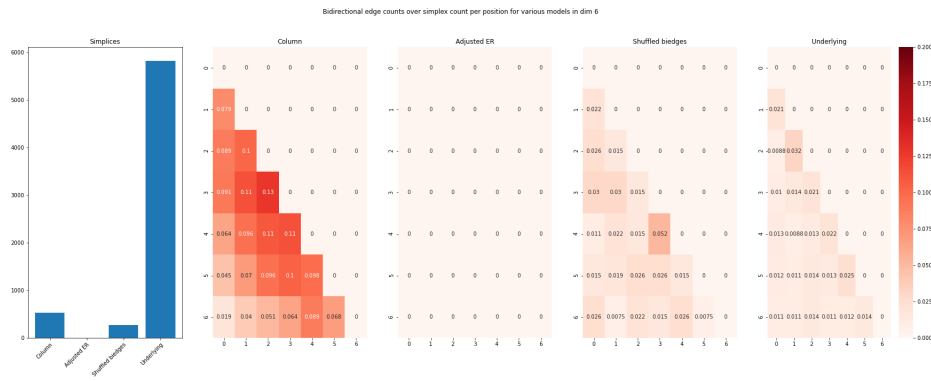


Figure B.11 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 6. To the left, a simplex count histogram for dimension 6. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

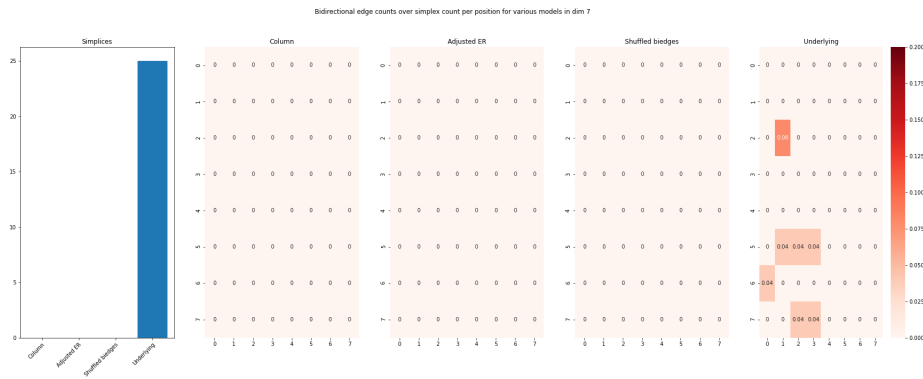


Figure B.12 – Normalized bidirectional edge count for the microcircuit and the controls in dimension 7. To the left, a simplex count histogram for dimension 7. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

B.3 Percent count bidirectional edge matrices

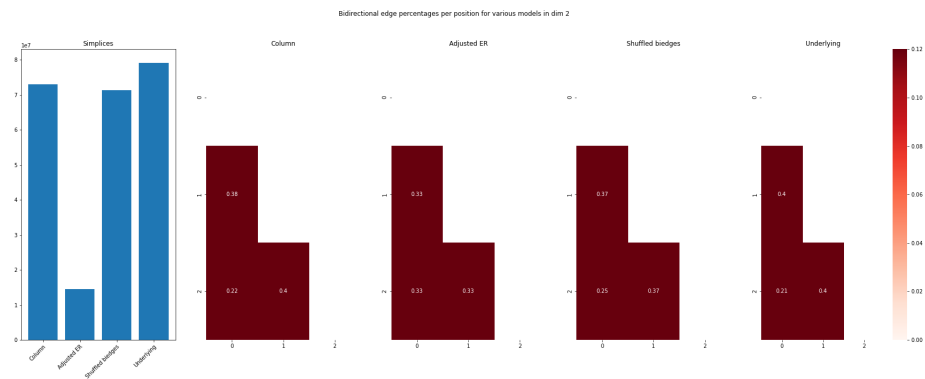


Figure B.13 – Percent bidirectional edge count for the model and the controls in dimension 2. To the left, a simplex count histogram for dimension 2. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

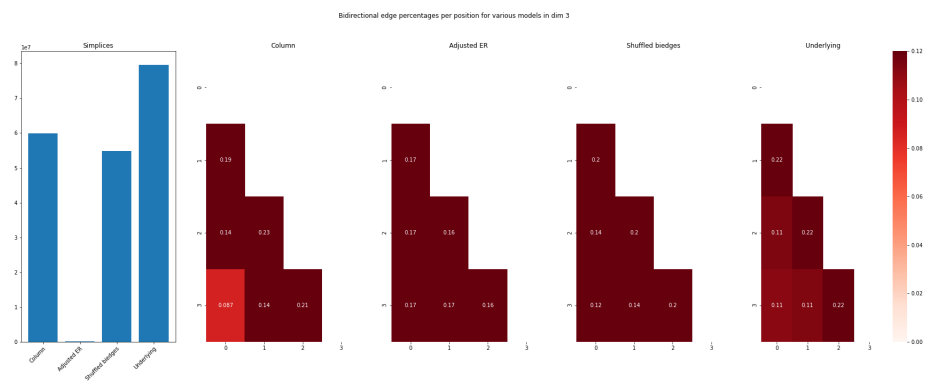


Figure B.14 – Percent bidirectional edge count for the model and the controls in dimension 3. To the left, a simplex count histogram for dimension 4. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

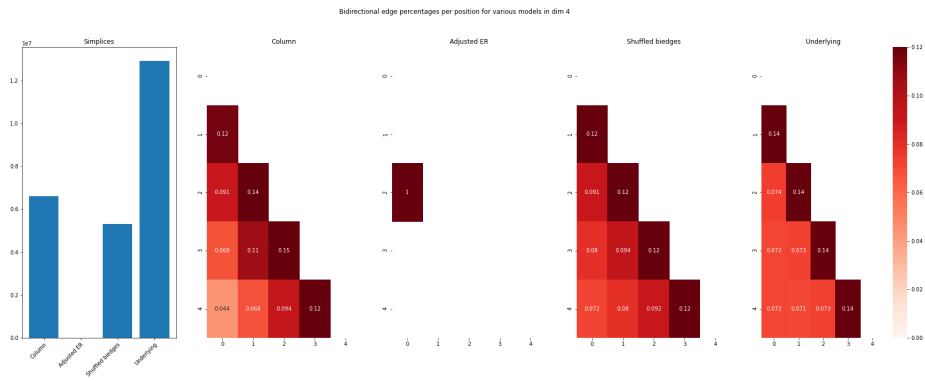


Figure B.15 – Percent bidirectional edge count for the model and the controls in dimension 4. To the left, a simplex count histogram for dimension 4. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

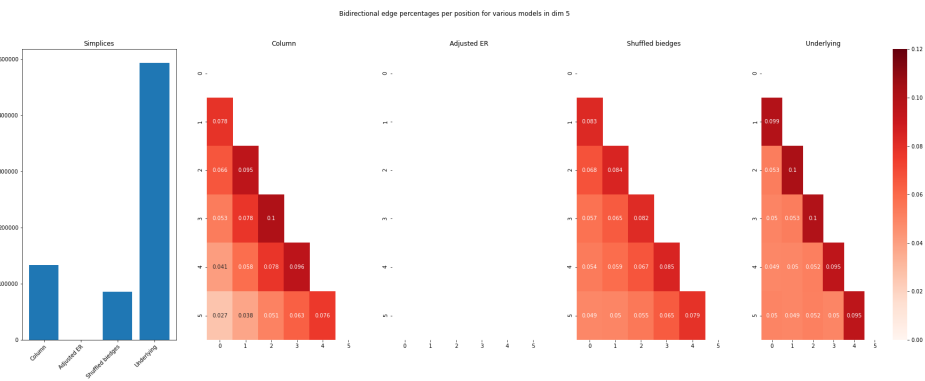


Figure B.16 – Percent bidirectional edge count for the model and the controls in dimension 5. To the left, a simplex count histogram for dimension 5. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

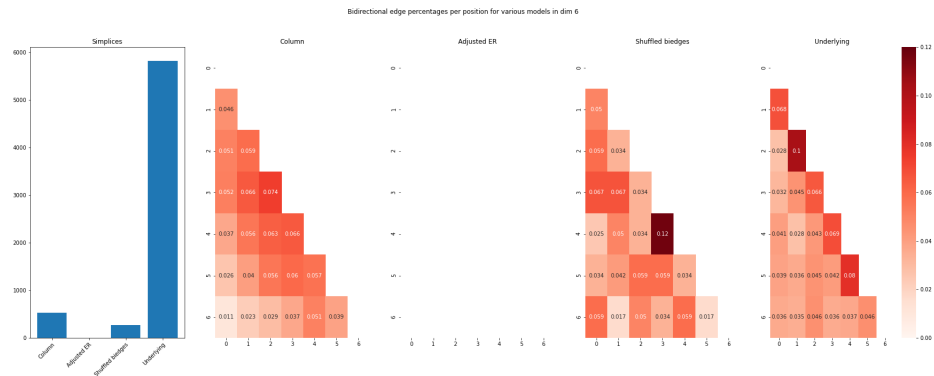


Figure B.17 – Percent bidirectional edge count for the model and the controls in dimension 6. To the left, a simplex count histogram for dimension 6. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

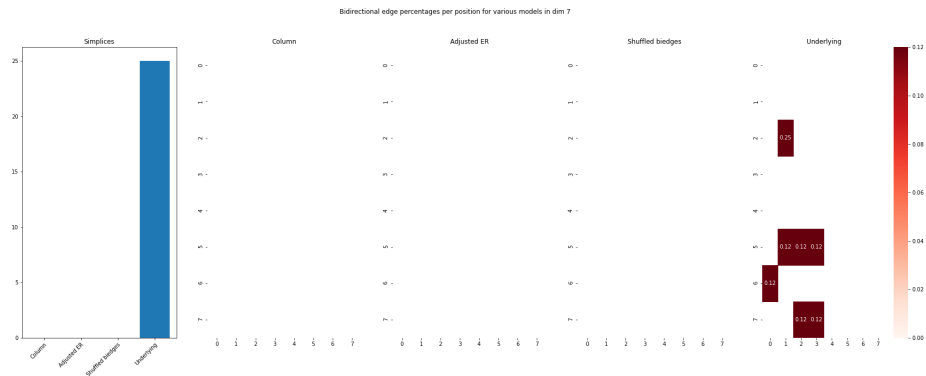


Figure B.18 – Percent bidirectional edge count for the model and the controls in dimension 7. To the left, a simplex count histogram for dimension 7. In order, from left to right, the microcircuit, Adjusted ER, Bishuffled and underlying models. A single instance for each matrix.

B.4 Difference between percent counts of models

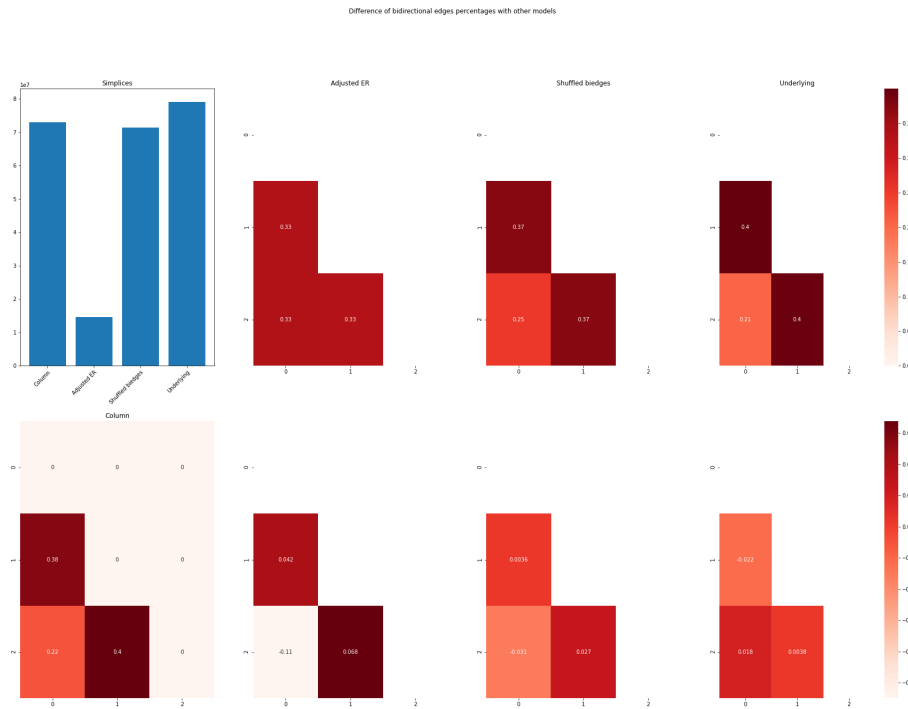


Figure B.19 – Percent bidirectional edge count difference for the model and the controls in dimension 2. Upper row is controls, the microcircuit counts are at the left. Difference in counts between microcircuit and each model is shown in the lower row. A single instance for each matrix.

B.5 Percentage t-tests for all dimensions

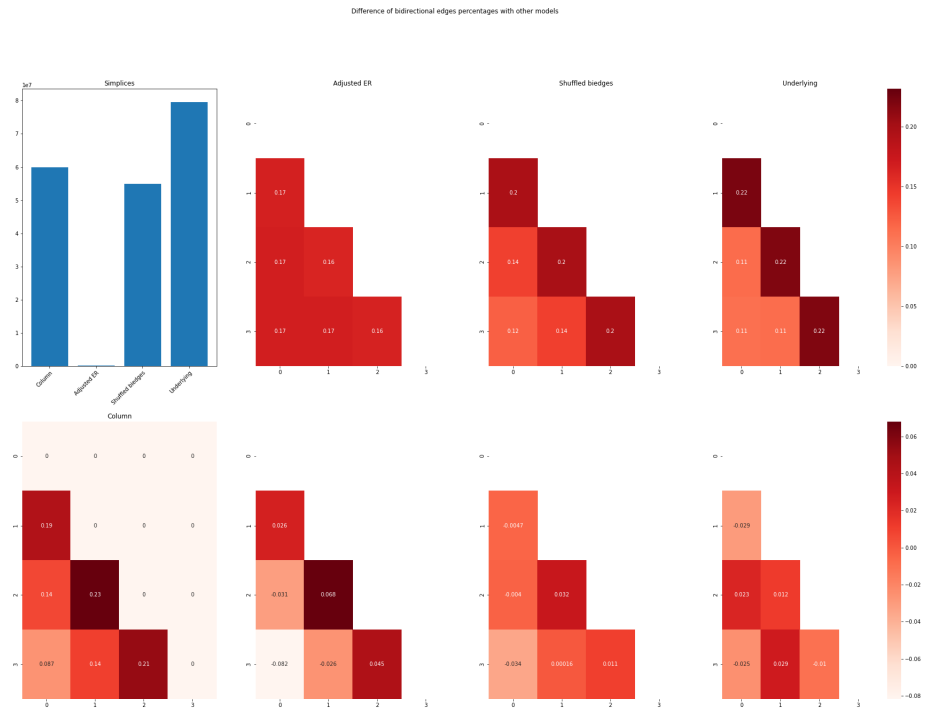


Figure B.20 – Percent bidirectional edge count difference for the model and the controls in dimension 3. Upper row is controls, the microcircuit counts are at the left. Difference in counts between microcircuit and each model is shown in the lower row. A single instance for each matrix.

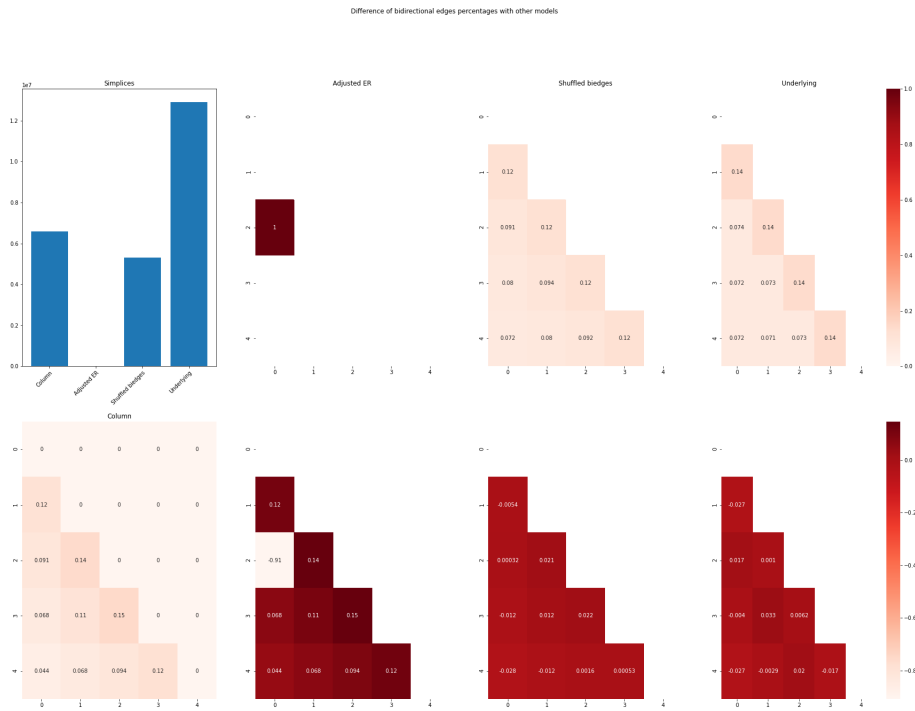


Figure B.21 – Percent bidirectional edge count difference for the model and the controls in dimension 4. Upper row is controls, the microcircuit counts are at the left. Difference in counts between microcircuit and each model is shown in the lower row. A single instance for each matrix.

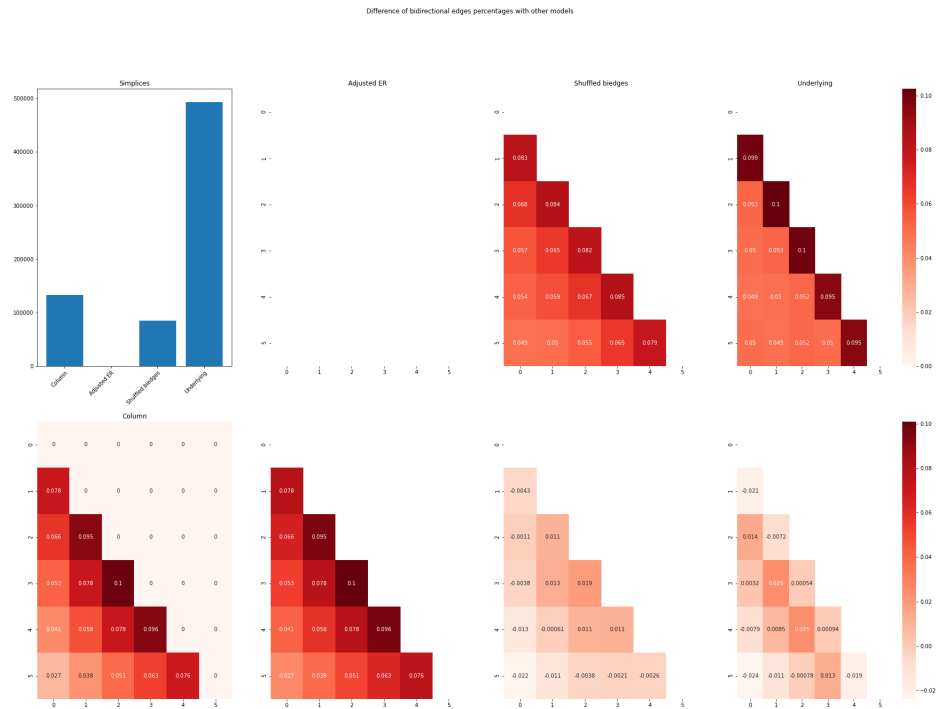


Figure B.22 – Percent bidirectional edge count difference for the model and the controls in dimension 5. Upper row is controls, the microcircuit counts are at the left. Difference in counts between microcircuit and each model is shown in the lower row. A single instance for each matrix.

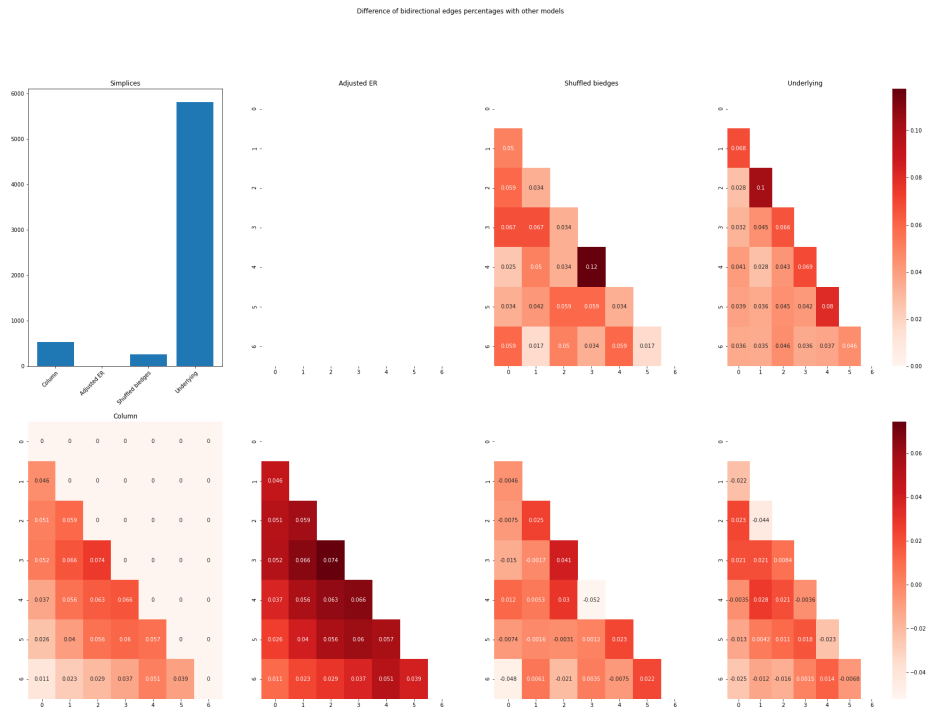


Figure B.23 – Percent bidirectional edge count difference for the model and the controls in dimension 6. Upper row is controls, the microcircuit counts are at the left. Difference in counts between microcircuit and each model is shown in the lower row. A single instance for each matrix.

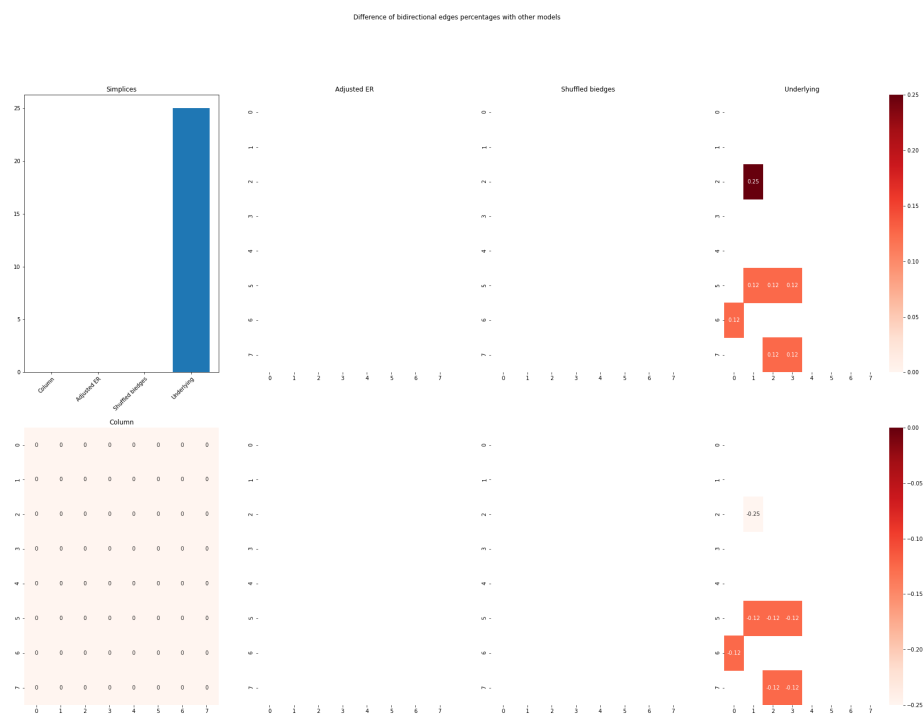


Figure B.24 – Percent bidirectional edge count difference for the model and the controls in dimension 7. Upper row is controls, the microcircuit counts are at the left. Difference in counts between microcircuit and each model is shown in the lower row. A single instance for each matrix.

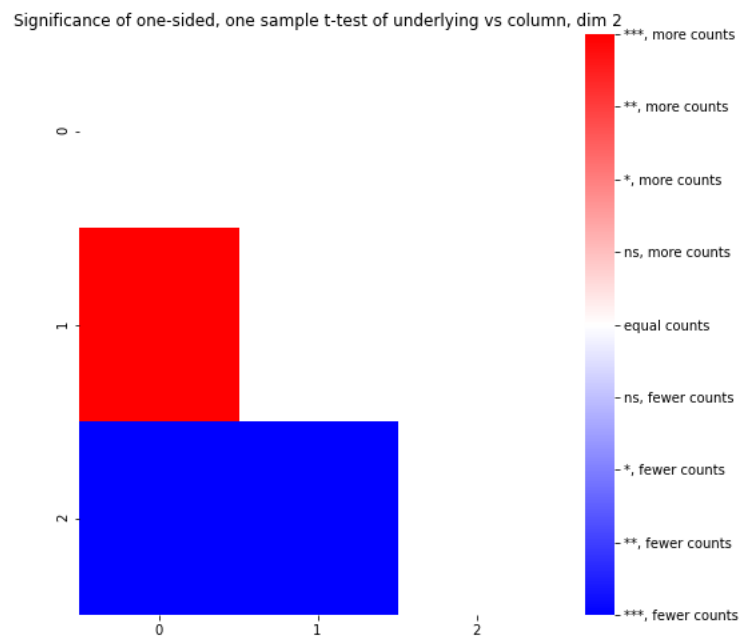


Figure B.25 – Significance of 1-sided t-tests of bidirectional edge counts in different position in computational model versus the underlying model in dimension 2. Blue squares mean that the original model has more bidirectional edges in percentage, red squares mean that the underlying model has more bidirectional edges in percentage. 30 instances of the control model.

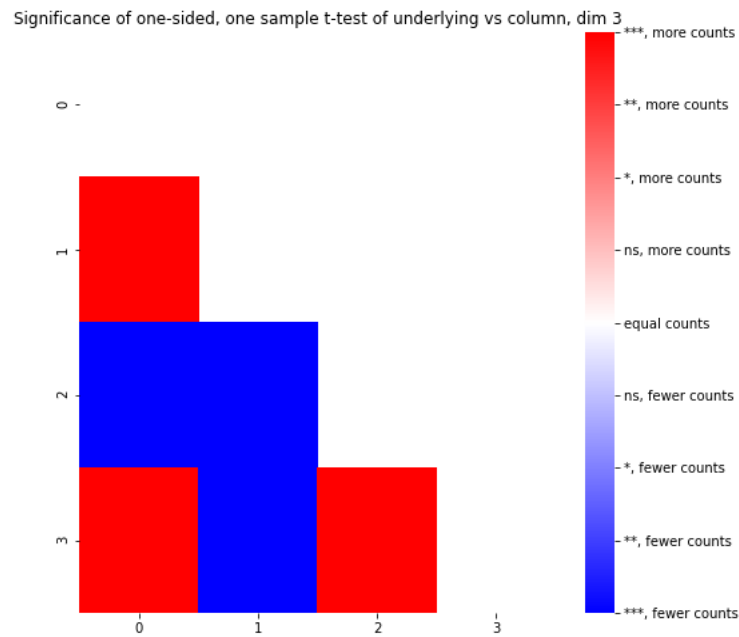


Figure B.26 – Significance of 1-sided t-tests of bidirectional edge counts in different position in computational model versus the underlying model in dimension 3. Blue squares mean that the original model has more bidirectional edges in percentage, red squares mean that the underlying model has more bidirectional edges in percentage. 30 instances of the control model.

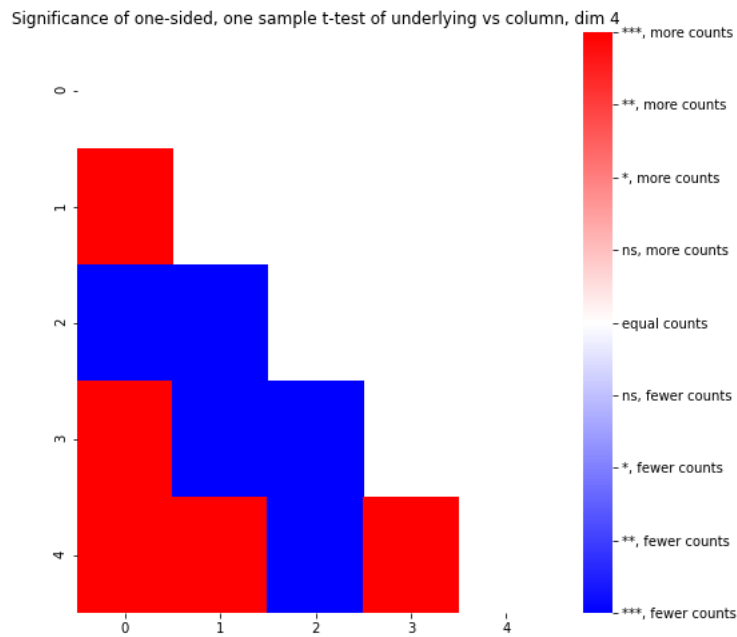


Figure B.27 – Significance of 1-sided t-tests of bidirectional edge counts in different position in computational model versus the underlying model in dimension 4. Blue squares mean that the original model has more bidirectional edges in percentage, red squares mean that the underlying model has more bidirectional edges in percentage. 30 instances of the control model.

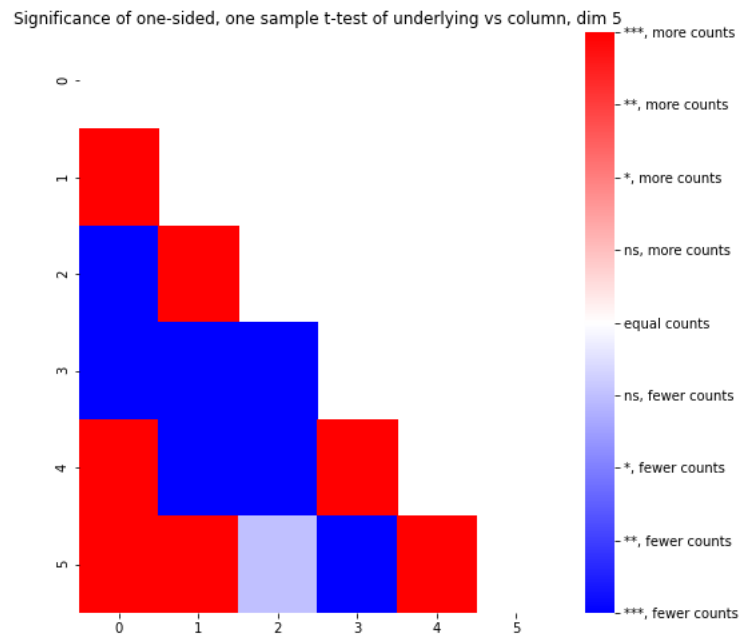


Figure B.28 – Significance of 1-sided t-tests of bidirectional edge counts in different position in computational model versus the underlying model in dimension 5. Blue squares mean that the original model has more bidirectional edges in percentage, red squares mean that the underlying model has more bidirectional edges in percentage. 30 instances of the control model.

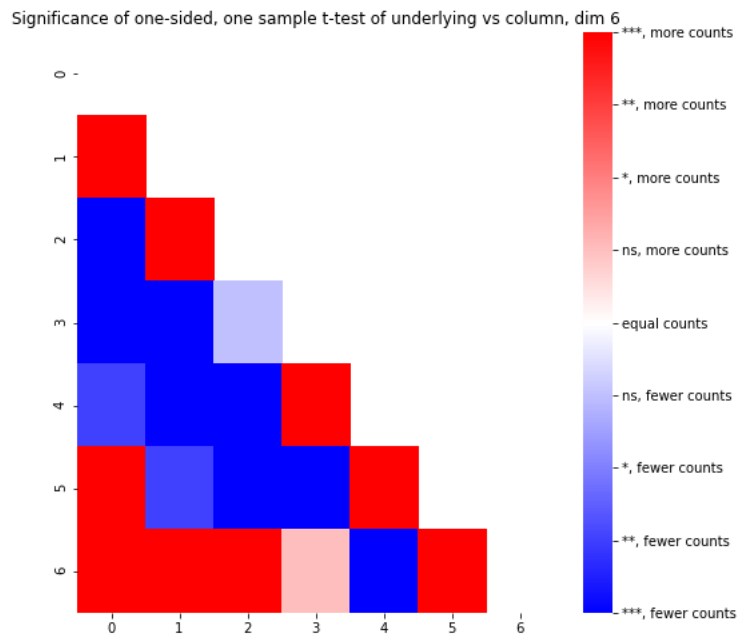


Figure B.29 – Significance of 1-sided t-tests of bidirectional edge counts in different position in computational model versus the underlying model in dimension 6. Blue squares mean that the original model has more bidirectional edges in percentage, red squares mean that the underlying model has more bidirectional edges in percentage. 30 instances of the control model.

C Supplementary figures for chapter 6

C.1 Correlation of edges per type

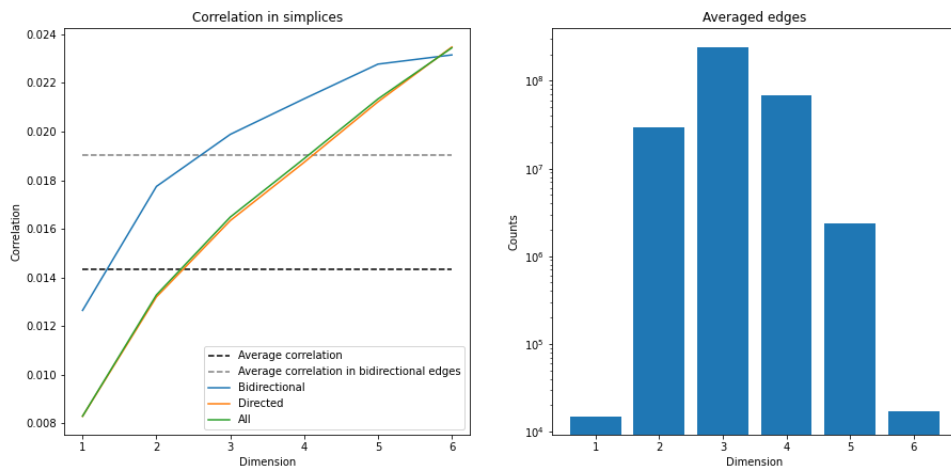


Figure C.1 – Average correlation of spike trains of neurons in edges in any position of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the edge type - blue for bidirectional, orange for non-bidirectional edges and green for all edges. To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

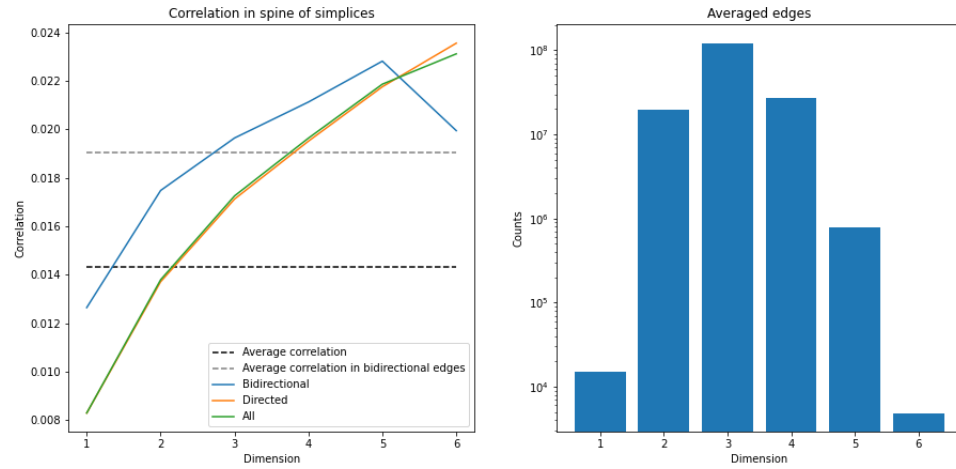


Figure C.2 – Average correlation of spike trains of neurons in edges in the spine of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the edge type - blue for bidirectional, orange for non-bidirectional edges and green for all edges. To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

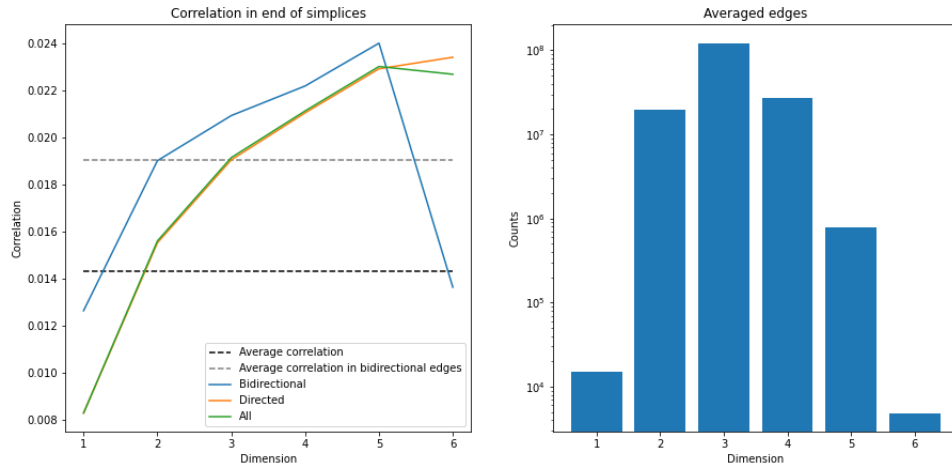


Figure C.3 – Average correlation of spike trains of neurons in edges in the end (position $(n, n - 1)$) of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the edge type - blue for bidirectional, orange for non-bidirectional edges and green for all edges. To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

C.2 Correlation of edges per position

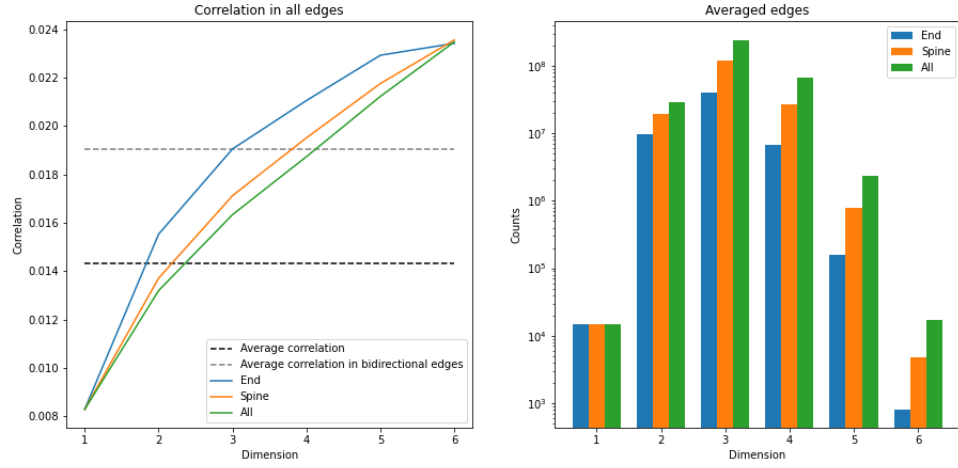


Figure C.4 – Average correlation of spike trains of neurons in edges of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the different groups - the whole simplex, the spine, or the end (position $(n, n - 1)$). To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

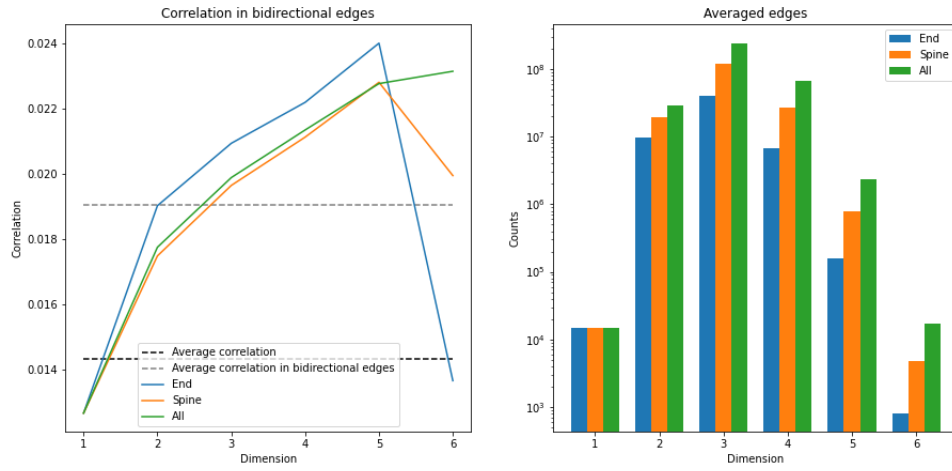


Figure C.5 – Average correlation of spike trains of neurons in bidirectional edges of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the different groups - the whole simplex, the spine, or the end (position $(n, n - 1)$). To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.

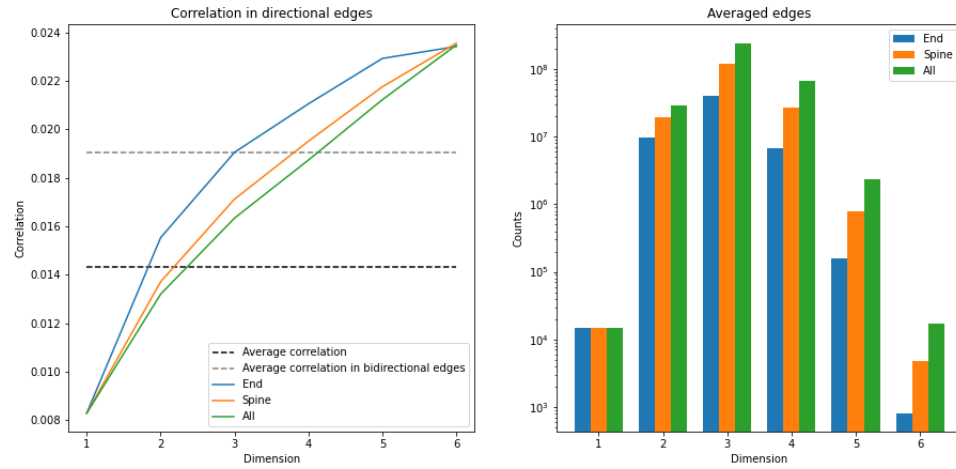


Figure C.6 – Average correlation of spike trains of neurons in non-bidirectional edges of the simplex. Horizontal lines are the average correlation of all neurons in edges of the graph (gray) and the average correlation of all neurons in bidirectional edges of the graph (black). Colors are the different groups - the whole simplex, the spine, or the end (position $(n, n - 1)$). To the right we show the amount of edges we averaged the correlation of. Each data point is the average of 20 spike trains.